

Project Title:

Automated SOC Lab using Suricata, Wazuh, Shuffle, TheHive, and Cortex for Real-Time Threat Detection and Response

Prepared By:

Adan Asghar

Project Architecture:

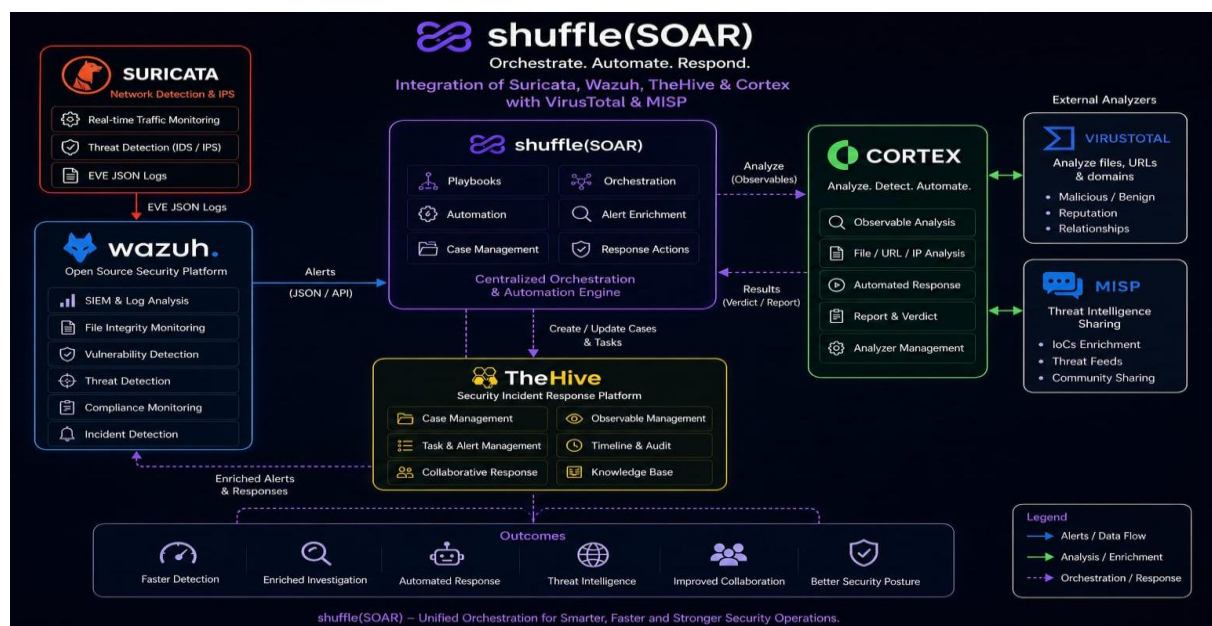


TABLE OF CONTENTS

- 1. Introduction**
- 2. Objectives**
- 3. SOC Architecture Overview**
 - 3.1 Core Components
- 4. Integration of Suricata with Wazuh**
 - 4.1 Overview of Suricata
 - 4.2 Installation of Suricata
 - 4.3 Configuration of Suricata
 - 4.4 Update Rules
 - 4.5 Test Configuration
 - 4.6 Start Suricata
 - 4.7 Log Files
- 5. Custom Rule Creation**
- 6. Wazuh Setup**
 - 6.1 Overview of Wazuh
 - 6.2 Service Management
 - 6.3 Accessing Wazuh Dashboard
- 7. Wazuh Agent Installation (Kali Linux)**
- 8. Integration of Suricata Logs with Wazuh**
- 9. Shuffle (SOAR) Implementation**
 - 9.1 Overview of Shuffle
 - 9.2 Docker Installation
 - 9.3 Deployment of Shuffle
 - 9.4 Accessing Shuffle
- 10. TheHive Installation**
 - 10.1 Dependency Installation
 - 10.2 Java Installation
 - 10.3 Cassandra Setup
 - 10.4 Elasticsearch Installation
 - 10.5 TheHive Configuration
 - 10.6 Accessing TheHive
- 11. Cortex Installation**
 - 11.1 Overview of Cortex
 - 11.2 Docker Deployment
 - 11.3 API Key Generation
- 12. Shuffle Workflow Integration (SOC Automation Flow)**
 - 12.1 Wazuh SIEM Flow
 - 12.2 Data Processing in Shuffle
 - 12.3 Automatic TheHive Case Creation
 - 12.4 Observable Enrichment
 - 12.5 Cortex Analysis (VirusTotal & AbuseIPDB)
 - 12.6 MISP Threat Intelligence Integration
 - 12.7 VirusTotal Validation

- 12.8 Final Case Update in TheHive
- 12.9 Email Notifications (Gmail Integration)

13. End-to-End SOC Automation Flow

- 13.1 Complete Attack Simulation Scenario
- 13.2 Alert Lifecycle in SOC
- 13.3 Incident Response Workflow

14. Results and Observations

- 14.1 System Performance
- 14.2 Detection Accuracy
- 14.3 Automation Efficiency

15. Conclusion

- 15.1 Project Summary
 - 15.2 Achievements
 - 15.3 Future Improvements
-

Introduction

This report presents the design and deployment of a **Security Operations Center (SOC) Automation Lab** using a modular Docker-based architecture. The objective is to simulate a real-world SOC where threat detection, analysis, automation, and response are fully integrated across two systems.

The lab demonstrates how alerts flow automatically from detection tools to case management and enrichment platforms, enabling efficient incident response.

Objectives

- Build a **fully automated SOC pipeline**
 - Distribute workload across **two systems** for performance optimization
 - Integrate detection, automation, and response tools
 - Simulate real-time threat detection and response
 - Maintain a **scalable and modular architecture**
-

SOC Architecture Overview

Core Components

Component	Function
Wazuh	SIEM (log collection, correlation, alerting)
Suricata	Network intrusion detection system
Shuffle	Security automation (SOAR)
TheHive	Incident response & case management
Cortex	Threat enrichment engine
MISP	Threat intelligence platform
VirusTotal	External malware scanning

2. Integration of Suricata with Wazuh

2.1 What is Suricata?

Suricata is a powerful open-source network security engine used for:

- Intrusion Detection System (IDS)

- Intrusion Prevention System (IPS)
- Network Security Monitoring (NSM)

It performs deep packet inspection, automatically detects protocols, and identifies malicious activities based on predefined rules. Suricata can be effectively integrated with platforms like Wazuh to enhance threat detection and centralized monitoring.

Installing Suricata

Step 1: Update System

The system packages were updated using the following command:

```
sudo apt update && sudo apt upgrade -y
```

[illegible]

Step 2: Install Suricata

Suricata was installed using:

```
sudo apt install suricata -y
```

```

--(kali@kali)-[~]
└─$ sudo apt install suricata -y
[sudo] password for kali:
The following packages were automatically installed and are no longer required:
curlftpfs libconfig-inifiles-perl libmpeg2encpp-2.1.0t64 libpostproc58 libswscale8 pocketsphinx-en-us python3-wapiti-arsenic
libaudio2 libbrotli1 libbrotli2-2.1.0t64 librubberband2 libvdpau-vl-gli python3-aiocache python3-awsfp
libavfilter10 libgav1-1 libmpdf75-1 libsmpeg4t64 linux-image-6.16.8-kali-amd64 python3-aiomcache ruby-unf-ext
libavformat61 libmpegutils-2.1.0t64 libpocketsphinx3 libspnbase3t64 mesa-vdpau-drivers python3-fs vdpau-driver-all
Use 'sudo apt autoremove' to remove them.

Upgrading:
  libhwloc-plugins libhwloc15 suricata

Summary:
  Upgrading: 3, Installing: 0, Removing: 0, Not Upgrading: 1146
  Download size: 0 B / 4,472 kB
  Freed space: 119 kB

(Reading database. 445694 files and directories currently installed.)
Preparing to unpack ./libhwloc-plugins_2.13.0-2_amd64.deb...
Unpacking libhwloc-plugins:amd64 (2.13.0-2) over (2.12.2-1+b1)...
Preparing to unpack ./libhwloc15_2.13.0-2_amd64.deb...
Unpacking libhwloc15:amd64 (2.13.0-2) over (2.12.2-1+b1)...
Preparing to unpack ./suricata_1k3a6.0.4-1_amd64.deb...
Unpacking suricata (1:8.0.4-1) over (1:8.0.3-1)...
Setting up libhwloc15:amd64 (2.13.0-2)...
Setting up suricata (1:8.0.4-1)...
Installing new version of config file /etc/logrotate.d/suricata...
Installing new version of config file /etc/suricata/rules/http2-events.rules...

Configuration file «/etc/suricata/suricata.yaml»
⇒ Modified (by you or by a script) since installation.
⇒ Package distributor has shipped an updated version.
What would you like to do about it ? Your options are:
Y or I : install the package maintainer's version
N or O : keep your currently-installed version
D      : show the differences between the versions
Z      : start a shell to examine the situation
The default action is to keep your current version.
**+ suricata.yaml (Y/I/N/O/D/Z) [default=N] ? y
Installing new version of config file /etc/suricata/suricata.yaml...
Setting up libhwloc-plugins:amd64 (2.13.0-2)...
Processing triggers for kali-menu (2026.1.13)...
Processing triggers for libc-bin (2.42-5)...

```

Step 3: Enable and Check Service

sudo systemctl enable suricata.service

sudo systemctl status suricata

```

--(kali@kali)-[~]
└─$ sudo systemctl enable suricata.service

--(kali@kali)-[~]
└─$ sudo systemctl status suricata
* suricata.service - Suricata IDS/IDP daemon
   Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-04-23 15:18:16 EDT; 5min ago
  Invocation: 9a4c8c13b6f54217954bc6f15c094037
     Docs: man:suricata(8)
           man:suricata-sc(8)
           https://suricata.io/documentation/
    Main PID: 71220 (Suricata-Main)
      Tasks: 10 (limit: 7035)
    Memory: 49.2M (peak: 50.6M)
       CPU: 5.92ms
    CGroup: /system.slice/suricata.service
            └─71220 /usr/bin/suricata -D -af-packet -c /etc/suricata/suricata.yaml --pidfile /var/run/suricata/suricata.pid

Apr 23 15:18:15 kali systemd[1]: Starting suricata.service - Suricata IDS/IDP daemon...
Apr 23 15:18:15 kali suricata[71221]: i: suricata: This is Suricata version 8.0.4 RELEASE running in SYSTEM mode
Apr 23 15:18:16 kali systemd[1]: Started suricata.service - Suricata IDS/IDP daemon.
Apr 23 15:22:05 kali systemd[1]: /usr/lib/systemd/system/suricata.service:11: PIDFile= references a path below legacy directory /var/run/, updating /var/run/suricata/suricata.pid + /run/suricata/suricata.pid;
lines 1-18/18 (END)

```

Step 4: Stop Service for Configuration

The Suricata service was stopped to perform configuration:

sudo systemctl stop suricata

```

--(kali@kali)-[~]
└─$ sudo systemctl status suricata
* suricata.service - Suricata IDS/IDP daemon
   Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-04-23 15:18:16 EDT; 5min ago
  Invocation: 9a4c8c13b6f54217954bc6f15c094037
     Docs: man:suricata(8)
           man:suricata-sc(8)
           https://suricata.io/documentation/
    Main PID: 71220 (Suricata-Main)
      Tasks: 10 (limit: 7035)
    Memory: 49.2M (peak: 50.6M)
       CPU: 5.92ms
    CGroup: /system.slice/suricata.service
            └─71220 /usr/bin/suricata -D -af-packet -c /etc/suricata/suricata.yaml --pidfile /var/run/suricata/suricata.pid

Apr 23 15:18:15 kali systemd[1]: Starting suricata.service - Suricata IDS/IDP daemon...
Apr 23 15:18:15 kali suricata[71221]: i: suricata: This is Suricata version 8.0.4 RELEASE running in SYSTEM mode
Apr 23 15:18:16 kali systemd[1]: Started suricata.service - Suricata IDS/IDP daemon.
Apr 23 15:22:05 kali systemd[1]: /usr/lib/systemd/system/suricata.service:11: PIDFile= references a path below legacy directory /var/run/, updating /var/run/suricata/suricata.pid + /run/suricata/suricata.pid;
lines 1-18/18 (END)

--(kali@kali)-[~]
└─$ sudo systemctl stop suricata

```

4. Configuration of Suricata

4.1 Check System IP Address

The IP address of the system was identified using:

ip a

```
(kali@kali)-[~]
$ ip a
: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
  link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
  inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
  inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
  link/ether 00:0c:29:8b:14:93 brd ff:ff:ff:ff:ff:ff
  inet 192.168.178.128/24 brd 192.168.178.255 scope global dynamic noprefixroute eth0
    valid_lft 1038sec preferred_lft 1038sec
  inet6 fe80::12c3:41b:c31c:ecb2/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
  link/ether 02:42:b9:be:a9:23 brd ff:ff:ff:ff:ff:ff
  inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
    valid_lft forever preferred_lft forever
```

4.2 Access Configuration File

The main configuration file is located at:

`/etc/suricata/suricata.yaml`

This file was opened and modified according to the system requirements.

```
(kali@kali)-[~]
$ ls -al /etc/suricata
total 216
drwxr-xr-x  3 root root  4096 Apr 23 15:18 .
drwxr-xr-x 190 root root 12288 Apr 23 15:18 ..
-rw-r--r--  1 root root  3327 Jan  9 10:27 classification.config
-rw-r--r--  1 root root  1701 Jan  9 10:27 reference.config
drwxr-xr-x  2 root root  4096 Apr 23 15:18 rules
-rw-r--r--  1 root root 93438 Mar 17 10:23 suricata.yaml
-rw-r--r--  1 root root 93060 Feb 28 02:25 suricata.yaml.dpkg-old
-rw-r--r--  1 root root  1643 Jan  9 10:27 threshold.config
```

```
(kali@kali)-[~]
$ ls -al /etc/suricata/rules
total 184
drwxr-xr-x 2 root root 4096 Apr 23 15:18 .
drwxr-xr-x 3 root root 4096 Apr 23 15:18 ..
-rw-r--r-- 1 root root 1858 Jan 9 10:27 app-layer-events.rules
-rw-r--r-- 1 root root 21287 Jan 9 10:27 decoder-events.rules
-rw-r--r-- 1 root root 468 Jan 9 10:27 dhcp-events.rules
-rw-r--r-- 1 root root 1786 Jan 9 10:27 dnp3-events.rules
-rw-r--r-- 1 root root 2202 Jan 9 10:27 dns-events.rules
-rw-r--r-- 1 root root 524 Jan 9 10:27 enip-events.rules
-rw-r--r-- 1 root root 4005 Jan 9 10:27 files.rules
-rw-r--r-- 1 root root 446 Jan 9 10:27 ftp-events.rules
-rw-r--r-- 1 root root 4188 Mar 12 18:36 http2-events.rules
-rw-r--r-- 1 root root 15088 Jan 9 10:27 http-events.rules
-rw-r--r-- 1 root root 2834 Jan 9 10:27 ipsec-events.rules
-rw-r--r-- 1 root root 585 Jan 9 10:27 kerberos-events.rules
-rw-r--r-- 1 root root 565 Mar 12 18:36 ldap-events.rules
-rw-r--r-- 1 root root 2576 Jan 9 10:27 mdns-events.rules
-rw-r--r-- 1 root root 1856 Jan 9 10:27 modbus-events.rules
-rw-r--r-- 1 root root 2187 Jan 9 10:27 mqtt-events.rules
-rw-r--r-- 1 root root 729 Jan 9 10:27 nfs-events.rules
-rw-r--r-- 1 root root 558 Jan 9 10:27 ntp-events.rules
-rw-r--r-- 1 root root 658 Jan 9 10:27 postgres-events.rules
-rw-r--r-- 1 root root 882 Jan 9 10:27 pop3-events.rules
-rw-r--r-- 1 root root 721 Jan 9 10:27 quic-events.rules
-rw-r--r-- 1 root root 926 Jan 9 10:27 rfb-events.rules
-rw-r--r-- 1 root root 4607 Jan 9 10:27 smb-events.rules
-rw-r--r-- 1 root root 5433 Jan 9 10:27 smtp-events.rules
-rw-r--r-- 1 root root 719 Jan 9 10:27 ssh-events.rules
-rw-r--r-- 1 root root 17569 Jan 9 10:27 stream-events.rules
-rw-r--r-- 1 root root 7081 Jan 9 10:27 tls-events.rules
-rw-r--r-- 1 root root 614 Jan 9 10:27 websocket-events.rules
```

Open yaml file to make changes

Sudo vim /etc/suricata/suricata.yaml

```
(kali@kali)-[~]
$ sudo vim /etc/suricata/suricata.yaml
```

4.3 Configure Network Variables

The system IP address was added in the configuration file to monitor network traffic effectively.

```

GNU nano 0.7 /etc/suricata/suricata.yaml *
YAML 1.1

# Suricata configuration file. In addition to the comments describing all
# options in this file, full documentation can be found at:
# https://docs.suricata.io/en/latest/configuration/suricata-yaml.html

# This configuration file was generated by Suricata 0.0.4.
suricata-version: "0.0"

#
# Step 1: Inform Suricata about your network
#

vars:
  # more specific is better for alert accuracy and performance
  address-groups:
    HOME_NET: "[192.168.178.128/24]"
    #HOME_NET: "[192.168.0.0/16]"
    #HOME_NET: "[10.0.0.0/8]"
    #HOME_NET: "[172.16.0.0/12]"
    #HOME_NET: "any"

    EXTERNAL_NET: "!$HOME_NET"
    #EXTERNAL_NET: "any"

  HTTP_SERVERS: "$HOME_NET"
  SMTP_SERVERS: "$HOME_NET"
  SQL_SERVERS: "$HOME_NET"
  DNS_SERVERS: "$HOME_NET"
  TELNET_SERVERS: "$HOME_NET"
  AIM_SERVERS: "$EXTERNAL_NET"
  DC_SERVERS: "$HOME_NET"
  DNP3_SERVER: "$HOME_NET"
  DNP3_CLIENT: "$HOME_NET"
  MODBUS_CLIENT: "$HOME_NET"
  MODBUS_SERVER: "$HOME_NET"
  ENIP_CLIENT: "$HOME_NET"
  ENIP_SERVER: "$HOME_NET"

port-groups:
  HTTP_PORTS: "80"
  SHELLCODE_PORTS: "!80"
  ORACLE_PORTS: 1521

```

4.4 Set Network Interface

The network interface (e.g., eth0) was configured in the YAML file to capture live traffic.

```

Linux high speed capture support (requires Linux kernel > 3.1)
# packet:
# - interface: eth0
#   # Number of receive threads. "auto" uses the number of cores
#   #threads: auto
#   # Default clusterid. AF_PACKET will load balance packets based on flow.
#   cluster-id: 99
#   # Default AF_PACKET cluster type. AF_PACKET can load balance per flow or per hash.
#   # This is only supported for Linux kernel > 3.1
#   # possible value are:
#   # * cluster_flow: all packets of a given flow are sent to the same socket
#   # * cluster_cpu: all packets treated in kernel by a CPU are sent to the same socket
#   # * cluster_qm: all packets linked by network card to a RSS queue are sent to the same
#   # socket. Requires at least Linux 3.14.
#   # * cluster_ebpf: eBPF file load balancing. See doc/userguide/capture-hardware/ebpf-xdp.rst for
#   # more info.
#   # Recommended modes are cluster_flow on most boxes and cluster_cpu or cluster_qm on system
#   # with capture card using RSS (requires cpu affinity tuning and system IRQ tuning)
#   # cluster_rollover has been deprecated; if used, it'll be replaced with cluster_flow.
#   cluster-type: cluster_flow
#   # In some fragmentation cases, the hash can not be computed. If "defrag" is set

```

```

Cross platform libpcap capture support (requires libpcap > 1.0.0)
# cap:
# - interface: eth0
#   # On Linux, pcap will try to use mmap'ed capture and will use "buffer-size"
#   # as total memory used by the ring. So set this to something bigger
#   # than 1% of your bandwidth.
#   #buffer-size: 16777216
#   #bpf-filter: "tcp and port 25"
#   # Choose checksum verification mode for the interface. At the moment
#   # of the capture, some packets may have an invalid checksum due to
#   # the checksum computation being offloaded to the network card.
#   # Possible values are:
#   # - yes: checksum validation is forced
#   # - no: checksum validation is disabled
#   # - auto: Suricata uses a statistical approach to detect when
#   # checksum off-loading is used. (default)
#   # Warning: 'capture.checksum-validation' must be set to yes to have any validation
#   #checksum-checks: auto
#   # With some accelerator cards using a modified libpcap (like Myricom), you
#   # may want to have the same number of capture threads as the number of capture
#   # rings. In this case, set up the threads variable to N to start N threads
#   # listening on the same interface.
#   #threads: 16
#   # set to no to disable promiscuous mode:
#   #promisc: no
#   # set snaplen, if not set it defaults to MTU if MTU can be known
#   # via ioctl call and to full capture if not.
#   #snaplen: 1518
#   # Put default values here
#   - interface: default
#   #checksum-checks: auto

```

4.5 Enable Community ID

The Community ID feature was enabled to improve event correlation:

community-id: **true**

```
# pipelining:
# enabled: yes ## set enable to yes to enable query pipelining
# batch-size: 10 ## number of entries to keep in buffer

# Include top level metadata. Default yes.
#metadata: no
# Include suricata version. Default no.
#suricata-version: yes

# Include the name of the input pcap file in pcap file processing mode
pcap-file: false

# Community Flow ID
# Adds a 'community_id' field to EVE records. These are meant to give
# records a predictable flow ID that can be used to match records to
# output of other tools such as Zeek (Bro).
#
# Takes a 'seed' that needs to be same across sensors and tools
# to make the id less predictable.

# enable/disable the community id feature.
community-id: true
# Seed value for the ID output. Valid values are 0-65535.
community-id-seed: 0

# HTTP X-Forwarded-For support by adding an extra field or overwriting
# the source or destination IP address (depending on flow direction)
# with the one reported in the X-Forwarded-For HTTP header. This is
# helpful when reviewing alerts for traffic that is being reverse
# or forward proxied.
xff:
  enabled: no
  # Two operation modes are available: "extra-data" and "overwrite".
  mode: extra-data
  # Two proxy deployments are supported: "reverse" and "forward". In
  # a "reverse" deployment the IP address used is the last one, in a
  # "forward" deployment the first IP address is used.
  deployment: reverse
  # Header name where the actual IP address will be reported. If more
  # than one IP address is present, the last IP address will be the
  # one taken into consideration.
  header: X-Forwarded-For

types:
```

4.6 Update Suricata Rules

Rules were updated using:

```
sudo suricata-update
```

Additional rule sources were enabled:

```
sudo suricata-update enable-source et/open
sudo suricata-update
```

```
[kali@kali:~]$ sudo suricata-update
23/4/2026 -- 16:13:52 -- <info> -- Using data-directory /var/lib/suricata.
23/4/2026 -- 16:13:52 -- <info> -- Using Suricata configuration /etc/suricata/suricata.yaml
23/4/2026 -- 16:13:52 -- <info> -- Using /etc/suricata/rules for Suricata provided rules.
23/4/2026 -- 16:13:52 -- <info> -- Found Suricata version 8.0.4 at /usr/bin/suricata.
23/4/2026 -- 16:13:52 -- <info> -- Loading /etc/suricata/suricata.yaml
23/4/2026 -- 16:13:52 -- <info> -- Disabling rules for protocol pgsq
23/4/2026 -- 16:13:52 -- <info> -- Disabling rules for protocol modbus
23/4/2026 -- 16:13:52 -- <info> -- Disabling rules for protocol dnp3
23/4/2026 -- 16:13:52 -- <info> -- Disabling rules for protocol enb
23/4/2026 -- 16:13:52 -- <info> -- No sources configured, will use Emerging Threats Open
23/4/2026 -- 16:13:52 -- <info> -- Fetching https://rules.emergingthreats.net/open/suricata-8.0.4/emerging.rules.tar.gz.
100% -- 5439551/5439551 -- Done.
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/app-layer-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/decoder-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/dhcp-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/ipssec-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/dns-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/http2-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/http-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/iptables-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/kerberos-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/modbus-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/mqtt-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/nfs-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/ntp-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/quic-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/rfb-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/smb-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/smtp-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/ssh-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/stream-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Loading distribution rule file /etc/suricata/rules/tls-events.rules
23/4/2026 -- 16:14:25 -- <info> -- Ignoring file e8e18dbadbc07eebb54ecdb5c78f603/rules/emerging-deleted.rules
23/4/2026 -- 16:14:30 -- <info> -- Loaded 65682 rules.
23/4/2026 -- 16:14:30 -- <info> -- Disabled 15 rules.
23/4/2026 -- 16:14:30 -- <info> -- Enabled 0 rules.
23/4/2026 -- 16:14:30 -- <info> -- Modified 0 rules.
23/4/2026 -- 16:14:30 -- <info> -- Dropped 0 rules.
23/4/2026 -- 16:14:30 -- <info> -- Enabled 126 rules for Flowbit dependencies.
23/4/2026 -- 16:14:30 -- <info> -- Backing up current rules.
23/4/2026 -- 16:14:30 -- <info> -- Writing rules to /var/lib/suricata/rules/suricata.rules: total: 65682; enabled: 49815; added: 65682; removed 0; modified: 0
23/4/2026 -- 16:14:31 -- <info> -- Writing /var/lib/suricata/rules/classification.config
23/4/2026 -- 16:14:32 -- <info> -- Testing with suricata -T.
```

```
sudo ls -al /var/lib/suricata/rules/
```

This command lists all Suricata rule files with detailed information to verify installed rules

```
(kali㉿kali)-[~]
└─$ sudo ls -al /var/lib/suricata/rules/
total 42352
drwxr-xr-x 2 root root 4096 Apr 23 16:14 . (Ethernet)
drwxr-xr-x 5 root root 4096 Apr 23 16:13 ..
-rw-r--r-- 1 root root 3228 Apr 23 16:14 classification.config
-rw-r--r-- 1 root root 43353030 Apr 23 16:14 suricata.rules
```

```
Session Actions Edit View Help
Subscription: https://www.secureworks.com/contact/ (Please reference CTU Countermeasures)
Name: stamus/lateral
Vendor: Stamus Networks
Summary: Lateral movement rules
License: GPL-3.0-only
Name: stamus/nrd-14-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 14 day list, complete
License: Commercial
Parameters: secret-code
Subscription: https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed
Name: stamus/nrd-30-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 30 day list, complete
License: Commercial
Parameters: secret-code
Subscription: https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed
Name: stamus/nrd-entropy-14-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 14 day list, high entropy
License: Commercial
Parameters: secret-code
Subscription: https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed
Name: stamus/nrd-entropy-30-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 30 day list, high entropy
License: Commercial
Parameters: secret-code
Subscription: https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed
Name: stamus/nrd-phishing-14-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 14 day list, phishing
License: Commercial
Parameters: secret-code
Subscription: https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed
Name: stamus/nrd-phishing-30-open
Vendor: Stamus Networks
Summary: Newly Registered Domains Open only - 30 day list, phishing
License: Commercial
Parameters: secret-code
Subscription: https://www.stamus-networks.com/stamus-labs/subscribe-to-threat-intel-feed
Name: tgreen/hunting
Vendor: tgreen
Summary: Threat hunting rules
License: GPLv3
```

```
(kali㉿kali)-[~]
└─$ sudo suricata-update enable-source et/open
23/4/2026 -- 16:28:58 - <Info> -- Using data-directory /var/lib/suricata.
23/4/2026 -- 16:28:58 - <Info> -- Using Suricata configuration /etc/suricata/suricata.yaml
23/4/2026 -- 16:28:58 - <Info> -- Using /etc/suricata/rules for Suricata provided rules.
23/4/2026 -- 16:28:58 - <Info> -- Found Suricata version 8.0.4 at /usr/bin/suricata.
23/4/2026 -- 16:28:58 - <Warning> -- Source index does not exist, will use bundled one.
23/4/2026 -- 16:28:58 - <Warning> -- Please run suricata-update update-sources.
23/4/2026 -- 16:28:58 - <Info> -- Creating directory /var/lib/suricata/update/sources
23/4/2026 -- 16:28:58 - <Info> -- Source et/open enabled
```

sudo suricata-update enable-source et/open

then sudo suricata-update

```
3/4/2026 -- 16:28:58 - <Info> -- Using data-directory /var/lib/suricata.
3/4/2026 -- 16:28:58 - <Info> -- Using Suricata configuration /etc/suricata/suricata.yaml
3/4/2026 -- 16:28:58 - <Info> -- Using /etc/suricata/rules for Suricata provided rules.
3/4/2026 -- 16:28:58 - <Info> -- Found Suricata version 8.0.4 at /usr/bin/suricata.
3/4/2026 -- 16:28:58 - <Warning> -- Source index does not exist, will use bundled one.
3/4/2026 -- 16:28:58 - <Warning> -- Please run suricata-update update-sources.
3/4/2026 -- 16:28:58 - <Info> -- Creating directory /var/lib/suricata/update/sources
3/4/2026 -- 16:28:58 - <Info> -- Source et/open enabled

--(kali@kali)-[~]
$ sudo suricata-update
3/4/2026 -- 16:30:02 - <Info> -- Using data-directory /var/lib/suricata.
3/4/2026 -- 16:30:02 - <Info> -- Using Suricata configuration /etc/suricata/suricata.yaml
3/4/2026 -- 16:30:02 - <Info> -- Using /etc/suricata/rules for Suricata provided rules.
3/4/2026 -- 16:30:02 - <Info> -- Found Suricata version 8.0.4 at /usr/bin/suricata.
3/4/2026 -- 16:30:02 - <Info> -- Loading /etc/suricata/suricata.yaml
3/4/2026 -- 16:30:02 - <Info> -- Disabling rules for protocol pgsq
3/4/2026 -- 16:30:02 - <Info> -- Disabling rules for protocol modbus
3/4/2026 -- 16:30:02 - <Info> -- Disabling rules for protocol dnp3
3/4/2026 -- 16:30:02 - <Info> -- Disabling rules for protocol enip
3/4/2026 -- 16:30:02 - <Warning> -- No index exists, will use bundled index.
3/4/2026 -- 16:30:02 - <Warning> -- Please run suricata-update update-sources.
3/4/2026 -- 16:30:02 - <Info> -- Checking https://rules.emergingthreats.net/open/suricata-8.0.4/emerging.rules.tar.gz.md5.
3/4/2026 -- 16:30:04 - <Info> -- Remote checksum has not changed. Not fetching.
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/app-layer-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/decoder-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/dhcp-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/dnp3-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/dns-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/files.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/http2-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/http-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/ipsec-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/kerberos-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/modbus-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/mqtt-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/nfs-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/ntp-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/quic-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/rfb-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/smb-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/smtp-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/ssh-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/stream-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Loading distribution rule file /etc/suricata/rules/tls-events.rules
3/4/2026 -- 16:30:04 - <Info> -- Ignoring file e8e18dbaadbcd7eebb54ecdb5c78f603/rules/emerging-deleted.rules
```

Suricata was kept inactive during the initial configuration phase to avoid interference with system setup and integration testing of other security components

```
---(kali@kali)-[~]
$ sudo systemctl stop suricata

---(kali@kali)-[~]
$ sudo systemctl status suricata.service
suricata.service - Suricata IDS/IDP daemon
Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: disabled)
Active: inactive (dead) since Thu 2026-04-23 15:25:56 EDT; 1h 7min ago
Duration: 7min 37.810s
Invocation: 9a4c8c13b6f54217954bc6f13c694037
Docs: man:suricata(8)
      https://suricata.io/documentation/
Process: 71220 ExecStart=/usr/bin/suricata -D -af-packet -c /etc/suricata/suricata.yaml --pidfile /var/run/suricata/suricata.pid (code=exited, status=0/SUCCESS)
Process: 75138 ExecStop=/usr/bin/suricatasc -c shutdown (code=exited, status=0/SUCCESS)
Main PID: 71220 (code=exited, status=0/SUCCESS)
Mem peak: 51.1M
CPU: 9.539s

Apr 23 15:18:15 kali systemd[1]: Starting suricata.service - Suricata IDS/IDP daemon...
Apr 23 15:18:15 kali suricata[71211]: i: suricata: This is Suricata version 8.0.4 RELEASE running in SYSTEM mode
Apr 23 15:18:16 kali systemd[1]: Started suricata.service - Suricata IDS/IDP daemon.
Apr 23 15:22:05 kali systemd[1]: /usr/lib/systemd/system/suricata.service:11: PIDFile= references a path below legacy directory /var/run/, updating /var/run/suricata/suricata.pid → /run/suricata/suricata.pid;
Apr 23 15:25:54 kali systemd[1]: Stopping suricata.service - Suricata IDS/IDP daemon...
Apr 23 15:25:54 kali suricatasc[75138]: {"message":"Closing Suricata","return":"OK"}
Apr 23 15:25:56 kali systemd[1]: suricata.service: Deactivated successfully.
Apr 23 15:25:56 kali systemd[1]: Stopped suricata.service - Suricata IDS/IDP daemon.
Apr 23 15:25:56 kali systemd[1]: suricata.service: Consumed 9.539s CPU time over 7min 40.911s wall clock time, 51.1M memory peak.
```

4.7 Test Configuration

The configuration was tested using:

sudo suricata -T -c /etc/suricata/suricata.yaml -v

```
(kali@kali)-[~]
$ sudo suricata -T -c /etc/suricata/suricata.yaml -v
Notice: suricata: This is Suricata version 8.0.4 RELEASE running in SYSTEM mode
Info: cpu: CPUs/cores online: 4
Info: suricata: Running suricata under test mode
Info: suricata: Setting engine mode to IDS mode by default
Info: exception-policy: master exception-policy set to: auto
Info: suricata: Preparing unexpected signal handling
Info: logopenfile: fast output device (regular) initialized: fast.log
Info: logopenfile: eve-log output device (regular) initialized: eve.json
Info: logopenfile: stats output device (regular) initialized: stats.log
^VInfo: detect: 1 rule files processed, 49815 rules successfully loaded, 0 rules failed, 0 rules skipped
Info: threshold-config: Threshold config parsed: 0 rule(s) found
Info: detect: 49820 signatures processed, 1289 are IP-only rules, 4485 are inspecting packet payload, 43810 inspect application layer, 110 are decoder event only
Notice: suricata: Configuration provided was successfully loaded. Exiting.
```

4.8 Start Suricata Service

sudo systemctl start suricata.service

```
(kali@kali)-[~]
$ sudo systemctl start suricata.service
(kali@kali)-[~]
$ sudo systemctl status suricata.service
● suricata.service - Suricata IDS/IDP daemon
   Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-04-23 16:42:40 EDT; 1min 17s ago
 Invocation: 6169020922c44ed2a9acfd16ff071249
    Docs: man:suricata(8)
          man:suricatasc(8)
          https://suricata.io/documentation/
 Main PID: 112582 (Suricata-Main)
   Tasks: 10 (limit: 7035)
  Memory: 410.9M (peak: 410.9M)
    CPU: 16.800s
   CGroup: /system.slice/suricata.service
           └─112582 /usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.yaml --pidfile /var/run/suricata/suricata.pid

Apr 23 16:42:23 kali systemd[1]: Starting suricata.service - Suricata IDS/IDP daemon...
Apr 23 16:42:24 kali suricata[112581]: i: suricata: This is Suricata version 8.0.4 RELEASE running in SYSTEM mode
Apr 23 16:42:40 kali systemd[1]: Started suricata.service - Suricata IDS/IDP daemon.
```

4.9 Log Files

Suricata generates logs such as:

- fast.log (alerts summary)
- eve.json (detailed JSON logs)

These logs are located in:

/var/log/suricata/

```
(kali@kali)-[~]
$ ls -al /var/log/suricata
total 42184
drwxr-xr-x  2 root root    4096 Apr 15 14:16 .
drwxr-xr-x 22 root root    4096 Apr 23 11:04 ..
-rw-r--r--  1 root root 22927660 Apr 23 16:45 eve.json
-rw-r--r--  1 root root 337261   Apr 15 14:16 eve.json.1.gz
-rw-r--r--  1 root root 1927607   Apr  5 08:29 eve.json.2.gz
-rw-r--r--  1 root root    0      Apr 15 14:16 fast.log
-rw-r--r--  1 root root    20     Apr  5 08:29 fast.log.1.gz
-rw-r--r--  1 root root    20     Feb 28 01:56 fast.log.2.gz
-rw-r--r--  1 root root 17001923 Apr 23 16:45 stats.log
-rw-r--r--  1 root root 151862   Apr 15 14:16 stats.log.1.gz
-rw-r--r--  1 root root 792709   Apr  5 08:29 stats.log.2.gz
-rw-r--r--  1 root root 11445    Apr 23 16:42 suricata.log
-rw-r--r--  1 root root 1206     Apr 15 14:06 suricata.log.1.gz
-rw-r--r--  1 root root 2130     Apr  4 20:23 suricata.log.2.gz
```

Curl <http://testmyids.org/uid/index.html>

```
(kali@kali)-[~]
$ curl http://testmyids.org/uid/index.html
uid=0(root) gid=0(root) groups=0(root)

(kali@kali)-[~]
$
```

Sudo cat /var/log/suricata/fast.log

This command is used to **view Suricata alert logs** in real time or after detection.

```
(kali@kali)-[~]
$ sudo cat /var/log/suricata/fast.log
4/23/2026-16:46:58.829402  [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 192.168.178.128:68 → 92.168.178.254:67
4/23/2026-16:50:06.882765  [**] [1:2100498:7] GPL ATTACK_RESPONSE id check returned root [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 143.204.106.69:80 → 192.168.178.128:44390

(kali@kali)-[~]
$
```

Stop the suricata Service

sudo systemctl stop suricata.service

```
(kali@kali)-[~]
$ sudo systemctl stop suricata.service

(kali@kali)-[~]
$
```

5. Custom Rule Creation

5.1 Open Local Rules File

`sudo nano /etc/suricata/rules/local.rules`

```
kali@kali:~$  
$ sudo vim /etc/suricata/rules/local.rules  
kali@kali:~$  
$
```

5.2 Add Custom Rules

Custom rules were added to detect specific activities

```
Session Actions Edit View Help  
alert icmp any any → any any (msg:"Ping Detected"; sid:1000001; rev:1;)  
alert http any any → any any (msg:"HTTP Traffic Detected"; sid:1000002; rev:1;)  
alert http any any → any any (msg:"Suspicious User-Agent"; content:'curl'; http_user_agent; sid:1000003; rev:1;)  
alert tcp any any → any any (msg:"Possible HTTP Flood"; threshold:type both, track_by_src, count 50, seconds 10; sid:1000004; rev:1;)  
alert tcp any any → any any (msg:"Possible Port Scan"; flags:SYN; threshold:per both, track_by_src, count 70, seconds 5; sid:1000005; rev:1;)  
alert http any any → any any (msg:"SQL Injection Attempt"; content:" OR 1=1"; nocase; sid:1000006; rev:1;)  
alert http any any → any any (msg:"XSS Attempt"; content:"<script>"; nocase; sid:1000007; rev:1;)  
alert http any any → any any (msg:"Possible Malware Download EXE"; content:".exe"; http_uri; sid:1000008; rev:1;)  
alert http any any → any any (msg:"HTTP POST Upload Detected"; flow:to_server,established; content:"POST"; http_method; sid:1000009; rev:1;)
```

5.3 Verify Rules

Rules were verified to ensure correct syntax and functionality.

```

(kali@kali)-[~]
$ cat /etc/suricata/rules/local.rules
alert icmp any any → any any (msg:"Ping Detected"; sid:1000001; rev:1;)
alert http any any → any any (msg:"HTTP Traffic Detected"; sid:1000002; rev:1;)
alert http any any → any any (msg:"Suspicious User-Agent"; content:"curl"; http_user_agent; sid:1000003; rev:1;)
alert tcp any any → any 80 (msg:"Possible HTTP Flood"; threshold:type both, track by_src, count 50, seconds 10; sid:1000004; rev:1;)
alert tcp any any → any any (msg:"Possible Port Scan"; flags:S; threshold:type both, track by_src, count 20, seconds 5; sid:1000002; rev:1;)
alert http any any → any any (msg:"SQL Injection Attempt"; content:"' OR 1=1"; nocase; sid:1000005; rev:1;)
alert http any any → any any (msg:"XSS Attempt"; content:"<script>"; nocase; sid:1000006; rev:1;)
alert http any any → any any (msg:"Possible Malware Download EXE"; content:".exe"; http_url; sid:1000009; rev:1;)
alert tcp any any → any any (msg:"Too many connections"; threshold:type threshold, track by_src, count 10, seconds 5; sid:1000001;)

(kali@kali)-[~]
$ sudo suricata -T -c /etc/suricata/suricata.yaml -v
Notice: suricata: This is Suricata version 8.0.4 RELEASE running in SYSTEM mode
Info: cpu: CPUs/cores online: 4
Info: suricata: Running suricata under test mode
Info: suricata: Setting engine mode to IDS mode by default
Info: exception-policy: master exception-policy set to: auto
Info: suricata: Preparing unexpected signal handling
Info: logopenfile: fast output device (regular) initialized: fast.log
Info: logopenfile: eve-log output device (regular) initialized: eve.json
Info: logopenfile: stats output device (regular) initialized: stats.log
Info: detect: 2 rule files processed. 49815 rules successfully loaded, 0 rules failed, 0 rules skipped
Info: threshold-config: Threshold config parsed: 0 rule(s) found
Info: detect: 49820 signatures processed. 1289 are IP-only rules, 4485 are inspecting packet payload, 43810 inspect application layer, 110 are decoder event only
Notice: suricata: Configuration provided was successfully loaded. Exiting.

```

5.4 Update Configuration

The local rules file was included in:

suricata.yaml

```

(kali@kali)-[~]
$ sudo vim /etc/suricata/suricata.yaml
(kali@kali)-[~]
$

```

Adding /etc/suricata/rules/local.rules in rule-file

```

Session Actions Edit View Help
#
# The most common hashmode commands are: hash2tuple, hash2tuplesorted,
# hash5tuple, hash5tuplesorted and roundrobin.
#
# See Nmaptech NTPL documentation other hashmodes and details on their use.
#
# This parameter has no effect if auto-config is disabled.
#
hashmode: hash5tuplesorted

#
# Configure Suricata to load Suricata-Update managed rules.
#
default-rule-path: /var/lib/suricata/rules
rule-files:
- suricata.rules
- /etc/suricata/rules/local.rules
#
# Auxiliary configuration files.
#
classification-file: /etc/suricata/classification.config
reference-config-file: /etc/suricata/reference.config
threshold-file: /etc/suricata/threshold.config

#
# Suricata as a Firewall options (experimental)
#
firewall:
# toggle to enable firewall mode
Enabled: no
# Firewall rule file are in their own path and are not managed
# by Suricata-Update.
#rule-path: /etc/suricata/firewall/
# List of files with firewall rules. Order matters, files are loaded
# in order and rules are applied in that order (per state, see docs)
#rule-files:
# - firewall.rules

#
# Include other configs

```

Sudo suricata -T -c /etc/suricata/suricata.yaml -v

This command tests and validates the Suricata configuration file for errors before running the service.

```
(kali@kali)-[~]
└─$ sudo suricata -T -c /etc/suricata/suricata.yaml -v
notice: suricata: This is Suricata version 8.0.4 RELEASE running in SYSTEM mode
info: cpu: CPUs/cores online: 4
info: suricata: Running suricata under test mode
info: suricata: Setting engine mode to IDS mode by default
info: exception-policy: master exception-policy set to: auto
info: suricata: Preparing unexpected signal handling
info: logopenfile: fast output device (regular) initialized: fast.log
info: logopenfile: eve-log output device (regular) initialized: eve.json
info: logopenfile: stats output device (regular) initialized: stats.log
info: detect: 2 rule files processed. 49815 rules successfully loaded, 0 rules failed, 0 rules skipped
info: threshold-config: Threshold config parsed: 0 rule(s) found
info: detect: 49820 signatures processed. 1289 are IP-only rules, 4485 are inspecting packet payload, 43810 inspect application layer, 110 are decoder event only
notice: suricata: Configuration provided was successfully loaded. Exiting.
```

Start Suricata Service

Sudo systemctl start suricata.service

```
(kali@kali)-[~]
└─$ sudo systemctl start suricata.service
(kali@kali)-[~]
└─$
```

suricata cat /var/log/suricata/fast.log

This command is intended to display Suricata's fast.log file to view generated alerts.

```
(kali@kali)-[~]
└─$ sudo systemctl start suricata.service
(kali@kali)-[~]
└─$ sudo cat /var/log/suricata/fast.log
04/23/2026-16:46:58.829402  [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 192.168.178.128:68 → 192.168.178.254:67
04/23/2026-16:50:06.882765  [**] [1:2100498:7] GPL ATTACK_RESPONSE id check returned root [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 143.204.106.69:80 → 192.168.178.128:44390
(kali@kali)-[~]
└─$ sudo systemctl status suricata.service
● suricata.service - Suricata IDS/IDP daemon
   Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-04-23 17:20:46 EDT; 2min 52s ago
     Invocation: b9d965633c82402db2bcbad8e1328637
       Docs: man:suricata(8)
             man:suricata-sc(8)
             https://suricata.io/documentation/
    Main PID: 131285 (Suricata-Main)
      Tasks: 10 (limit: 7035)
     Memory: 411.2M (peak: 413.2M)
        CPU: 17.817s
    CGroup: /system.slice/suricata.service
            └─131285 /usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.yaml --pidfile /var/run/suricata/suricata.pid

Apr 23 17:20:30 kali systemd[1]: Starting suricata.service - Suricata IDS/IDP daemon...
Apr 23 17:20:31 kali suricata[131276]: i: suricata: This is Suricata version 8.0.4 RELEASE running in SYSTEM mode
Apr 23 17:20:46 kali systemd[1]: Started suricata.service - Suricata IDS/IDP daemon.
```

sudo apt-get install jq

This command installs jq, a tool used to parse and read JSON-formatted Suricata logs easily.

```
(kali@kali)-[~]
└─$ sudo apt-get install jq
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following packages were automatically installed and are no longer required:
  curlftpfs libaudio2 libavfilter10 libavformat61 libconfig-inifiles-perl libfuse2t64 libgavi-1 libjpegutils-2.1-0t64 libmpeg2encpp-2.1-0t64 libmplx2-2.1-0t64 libmupdf25.1 libpocketsphinx3 libpostproc58
  librubberband2 libsnmp4t64 libspinnbase3t64 libswscale8 libvdpau-va-gli linux-image-6.16.8-kali-amd64 mesa-vdpau-drivers pocketsphinx-en-us python3-aiocache python3-aiomcache python3-fs
  python3-kapiti-arsenic python3-yaswfp ruby-unf-ext vdpau-driver-all
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  jq
0 upgraded, 1 newly installed, 0 to remove and 1146 not upgraded.
Need to get 85.7 kB of archives.
After this operation, 137 kB of additional disk space will be used.
Get:1 http://http.kali.org/kali kali-rolling/main amd64 jq amd64 1.8.1-4+b1 [85.7 kB]
Fetched 85.7 kB in 1s (74.8 kB/s)
Selecting previously unselected package jq.
(Reading database... 44993 files and directories currently installed.)
Preparing to unpack ./jq1.8.1-4+b1_amd64.deb...
Unpacking jq (1.8.1-4+b1)...
Setting up jq (1.8.1-4+b1)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.1.3)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

sudo tail -f /var/log/suricata/eve.json | jq 'select(.event_type= "alert")'

This command filters and displays real-time Suricata alert logs from eve.json in a readable JSON format using jq.

```
(kali@kali)-[~]
└─$ sudo tail -f /var/log/suricata/eve.json | jq 'select(.event_type="alert")'
[sudo] password for kali:
{
  "timestamp": "2026-04-23T17:46:58.852647-0400",
  "flow_id": 565869257650445,
  "in_iface": "eth0",
  "event_type": "alert",
  "src_ip": "192.168.178.128",
  "src_port": 68,
  "dest_ip": "192.168.178.254",
  "dest_port": 67,
  "proto": "UDP",
  "ip_v": 4,
  "pkt_src": "wire/pcap",
  "community_id": "1:U7HVgGKeNMWl3M+3Knf+3407pLs=",
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 2022973,
    "rev": 1,
    "signature": "ET INFO Possible Kali Linux hostname in DHCP Request Packet",
    "category": "Potential Corporate Privacy Violation",
    "severity": 1,
    "metadata": {
      "attack_target": [
        "Client_Endpoint"
      ],
      "confidence": [
        "Medium"
      ],
      "created_at": [
        "2016_07_18"
      ],
      "deployment": [
        "Perimeter"
      ],
      "performance_impact": [
        "Low"
      ],
      "reviewed_at": [
        "2024_04_19"
      ],
      "signature_severity": [
        "Informational"
      ]
    }
  }
},
```

6. Wazuh Setup

6.1 What is Wazuh?

Wazuh is an open-source security platform used for:

- Threat detection
- Log analysis
- File integrity monitoring
- Security event management

It provides centralized monitoring and visualization of security events.

Install and open wazuh



username : wazuh-user

password : wazuh

```

Welcome to the Wazuh OVA version
Wazuh - 4.12.0
Login credentials:
  User: wazuh-user
  Password: wazuh

wazuh-server login: █

```

Sudo systemctl start wazuh-manager

[illegible]

sudo systemctl status wazuh-manager to check status

```
■ wazuh-manager.service - Wazuh manager
Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; pr
Active: active (running) since Sun 2026-04-26 19:10:12 UTC; 2min 22s ago
Process: 3742 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (co
Tasks: 174 (limit: 9469)
Memory: 860.6M
CPU: 1min 47.235s
CGroup: /system.slice/wazuh-manager.service
├─3993 /var/ossec/framework/python/bin/python3 /var/ossec/api/scri
├─3994 /var/ossec/framework/python/bin/python3 /var/ossec/api/scri
├─3995 /var/ossec/framework/python/bin/python3 /var/ossec/api/scri
├─3997 /var/ossec/framework/python/bin/python3 /var/ossec/api/scri
├─4001 /var/ossec/framework/python/bin/python3 /var/ossec/api/scri
├─4049 /var/ossec/bin/wazuh-authd
├─4070 /var/ossec/bin/wazuh-db
├─4099 /var/ossec/bin/wazuh-execd
├─4118 /var/ossec/bin/wazuh-analysisd
├─4133 /var/ossec/bin/wazuh-syscheckd
├─4154 /var/ossec/bin/wazuh-remoted
├─4196 /var/ossec/bin/wazuh-logcollector
├─4274 /var/ossec/bin/wazuh-monitord
└─4301 /var/ossec/bin/wazuh-modulesd

Apr 26 19:10:07 wazuh-server env[3742]: Started wazuh-remoted...
Apr 26 19:10:07 wazuh-server env[3742]: wazuh-logcollector: Process 4205 not us
Apr 26 19:10:08 wazuh-server env[3742]: Started wazuh-logcollector...
Apr 26 19:10:08 wazuh-server env[3742]: wazuh-monitord: Process 4276 not used b
Apr 26 19:10:10 wazuh-server env[3742]: Started wazuh-monitord...
Apr 26 19:10:10 wazuh-server env[3742]: wazuh-modulesd: Process 4301 not used b
lwazuh-user@wazuh-server ~]$
```

Sudo systemctl start wazuh-indexer

This command checks the running status of the Wazuh Indexer service to ensure it is active and functioning.

```
Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; pr
Active: active (running) since Sun 2026-04-26 19:10:12 UTC; 2min 22s ago
Process: 3742 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (co
Tasks: 174 (limit: 9469)
Memory: 860.6M
CPU: 1min 47.235s
CGroup: /system.slice/wazuh-manager.service
├─3993 /var/ossec/framework/python/bin/python3 /var/ossec/api/scri
├─3994 /var/ossec/framework/python/bin/python3 /var/ossec/api/scri
├─3995 /var/ossec/framework/python/bin/python3 /var/ossec/api/scri
├─3997 /var/ossec/framework/python/bin/python3 /var/ossec/api/scri
├─4001 /var/ossec/framework/python/bin/python3 /var/ossec/api/scri
├─4049 /var/ossec/bin/wazuh-authd
├─4070 /var/ossec/bin/wazuh-db
├─4099 /var/ossec/bin/wazuh-execd
├─4118 /var/ossec/bin/wazuh-analysisd
├─4133 /var/ossec/bin/wazuh-syscheckd
├─4154 /var/ossec/bin/wazuh-remoted
├─4196 /var/ossec/bin/wazuh-logcollector
├─4274 /var/ossec/bin/wazuh-monitord
└─4301 /var/ossec/bin/wazuh-modulesd

Apr 26 19:10:07 wazuh-server env[3742]: Started wazuh-remoted...
Apr 26 19:10:07 wazuh-server env[3742]: wazuh-logcollector: Process 4205 not us
Apr 26 19:10:08 wazuh-server env[3742]: Started wazuh-logcollector...
Apr 26 19:10:08 wazuh-server env[3742]: wazuh-monitord: Process 4276 not used b
Apr 26 19:10:10 wazuh-server env[3742]: Started wazuh-monitord...
Apr 26 19:10:10 wazuh-server env[3742]: wazuh-modulesd: Process 4301 not used b
lwazuh-user@wazuh-server ~]$ sudo systemctl start wazuh-indexer
lwazuh-user@wazuh-server ~]$
```

```

Apr 26 19:10:07 wazuh-server env[3742]: Started wazuh-remoted...
Apr 26 19:10:07 wazuh-server env[3742]: wazuh-logcollector: Process 4205 not used b
Apr 26 19:10:08 wazuh-server env[3742]: Started wazuh-logcollector...
Apr 26 19:10:08 wazuh-server env[3742]: wazuh-monitord: Process 4276 not used b
Apr 26 19:10:10 wazuh-server env[3742]: Started wazuh-monitord...
Apr 26 19:10:10 wazuh-server env[3742]: wazuh-modulesd: Process 4301 not used b
[wazuh-user@wazuh-server ~]$ sudo systemctl start wazuh-indexer
[wazuh-user@wazuh-server ~]$ sudo systemctl status wazuh-indexer
■ wazuh-indexer.service - wazuh-indexer
   Loaded: loaded (/usr/lib/systemd/system/wazuh-indexer.service; enabled; pr
   Active: active (running) since Sun 2026-04-26 19:11:07 UTC; 6min ago
     Docs: https://documentation.wazuh.com
    Main PID: 3741 (java)
      Tasks: 88 (limit: 9469)
     Memory: 4.6G
        CPU: 3min 38.296s
    CGroup: /system.slice/wazuh-indexer.service
            └─3741 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -Dopense

Apr 26 19:10:16 wazuh-server systemd-entrypoint[3741]: WARNING: A terminally de
Apr 26 19:10:16 wazuh-server systemd-entrypoint[3741]: WARNING: System::setSecu
Apr 26 19:10:16 wazuh-server systemd-entrypoint[3741]: WARNING: Please consider
Apr 26 19:10:16 wazuh-server systemd-entrypoint[3741]: WARNING: System::setSecu
Apr 26 19:10:20 wazuh-server systemd-entrypoint[3741]: Apr 26, 2026 7:10:20 PM >
Apr 26 19:10:20 wazuh-server systemd-entrypoint[3741]: WARNING: COMPAT locale p
Apr 26 19:10:24 wazuh-server systemd-entrypoint[3741]: WARNING: A terminally de
Apr 26 19:10:24 wazuh-server systemd-entrypoint[3741]: WARNING: System::setSecu
Apr 26 19:10:24 wazuh-server systemd-entrypoint[3741]: WARNING: Please consider
Apr 26 19:10:24 wazuh-server systemd-entrypoint[3741]: WARNING: System::setSecu
lines 1-21/21 (END)

```

Sudo systemctl start wazuh-dashboard

This command is used to check whether the Wazuh Dashboard service is running and active.

```

Apr 26 19:10:08 wazuh-server env[3742]: Started wazuh-logcollector...
Apr 26 19:10:08 wazuh-server env[3742]: wazuh-monitord: Process 4276 not used b
Apr 26 19:10:10 wazuh-server env[3742]: Started wazuh-monitord...
Apr 26 19:10:10 wazuh-server env[3742]: wazuh-modulesd: Process 4301 not used b
[wazuh-user@wazuh-server ~]$ sudo systemctl start wazuh-indexer
[wazuh-user@wazuh-server ~]$ sudo systemctl status wazuh-indexer
■ wazuh-indexer.service - wazuh-indexer
   Loaded: loaded (/usr/lib/systemd/system/wazuh-indexer.service; enabled; pr
   Active: active (running) since Sun 2026-04-26 19:11:07 UTC; 6min ago
     Docs: https://documentation.wazuh.com
    Main PID: 3741 (java)
      Tasks: 88 (limit: 9469)
     Memory: 4.6G
        CPU: 3min 38.296s
    CGroup: /system.slice/wazuh-indexer.service
            └─3741 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -Dopense

Apr 26 19:10:16 wazuh-server systemd-entrypoint[3741]: WARNING: A terminally de
Apr 26 19:10:16 wazuh-server systemd-entrypoint[3741]: WARNING: System::setSecu
Apr 26 19:10:16 wazuh-server systemd-entrypoint[3741]: WARNING: Please consider
Apr 26 19:10:16 wazuh-server systemd-entrypoint[3741]: WARNING: System::setSecu
Apr 26 19:10:20 wazuh-server systemd-entrypoint[3741]: Apr 26, 2026 7:10:20 PM >
Apr 26 19:10:20 wazuh-server systemd-entrypoint[3741]: WARNING: COMPAT locale p
Apr 26 19:10:24 wazuh-server systemd-entrypoint[3741]: WARNING: A terminally de
Apr 26 19:10:24 wazuh-server systemd-entrypoint[3741]: WARNING: System::setSecu
Apr 26 19:10:24 wazuh-server systemd-entrypoint[3741]: WARNING: Please consider
Apr 26 19:10:24 wazuh-server systemd-entrypoint[3741]: WARNING: System::setSecu
[wazuh-user@wazuh-server ~]$
[wazuh-user@wazuh-server ~]$ sudo systemctl start wazuh-dashboard
[wazuh-user@wazuh-server ~]$

```

Check status by using sudo systemctl status wazuh-dashboard

```

[wazuh-user@wazuh-server ~]$ sudo systemctl start wazuh-dashboard
[wazuh-user@wazuh-server ~]$ sudo systemctl status wazuh-dashboard
■ wazuh-dashboard.service - wazuh-dashboard
   Loaded: loaded (/etc/systemd/system/wazuh-dashboard.service; enabled; pres
   Active: active (running) since Sun 2026-04-26 19:09:09 UTC; 10min ago
   Main PID: 3051 (node)
   Tasks: 11 (limit: 9469)
   Memory: 278.9M
   CPU: 24.244s
   CGroup: /system.slice/wazuh-dashboard.service
           └─3051 /usr/share/wazuh-dashboard/node/bin/node --no-warnings --ma

Apr 26 19:11:27 wazuh-server opensearch-dashboards[3051]: {"type":"log","@times
Apr 26 19:11:28 wazuh-server opensearch-dashboards[3051]: {"type":"log","@times
Apr 26 19:11:28 wazuh-server opensearch-dashboards[3051]: {"type":"log","@times
Apr 26 19:11:29 wazuh-server opensearch-dashboards[3051]: {"type":"log","@times
Apr 26 19:11:29 wazuh-server opensearch-dashboards[3051]: {"type":"log","@times
Apr 26 19:11:29 wazuh-server opensearch-dashboards[3051]: {"type":"log","@times
Apr 26 19:15:00 wazuh-server opensearch-dashboards[3051]: {"type":"log","@times
Apr 26 19:15:00 wazuh-server opensearch-dashboards[3051]: {"type":"log","@times
Apr 26 19:15:00 wazuh-server opensearch-dashboards[3051]: {"type":"log","@times
Apr 26 19:15:00 wazuh-server opensearch-dashboards[3051]: {"type":"log","@times
lines 1-20/20 (END)

```

6.3 Access Wazuh Dashboard

1. Check system IP:

ip a

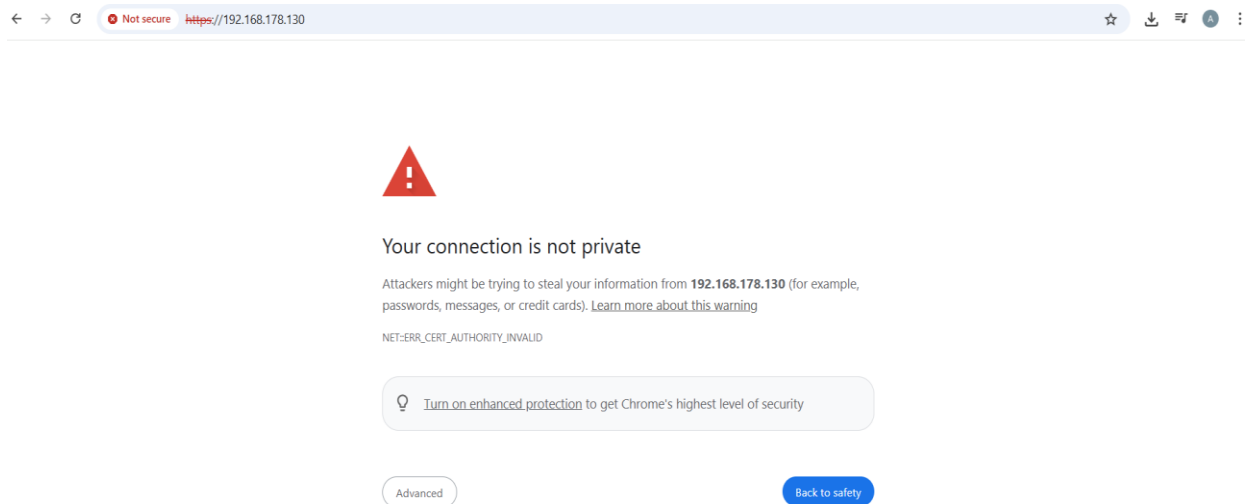
```

[wazuh-user@wazuh-server ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group
    p default qlen 1000
    link/ether 00:0c:29:21:57:6b brd ff:ff:ff:ff:ff:ff
    altname enp2s0
    altname ens32
    inet 192.168.178.130/24 metric 1024 brd 192.168.178.255 scope global dynamic
    eth0
        valid_lft 1117sec preferred_lft 1117sec
    inet6 fe80::20c:29ff:fe21:576b/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[wazuh-user@wazuh-server ~]$

```

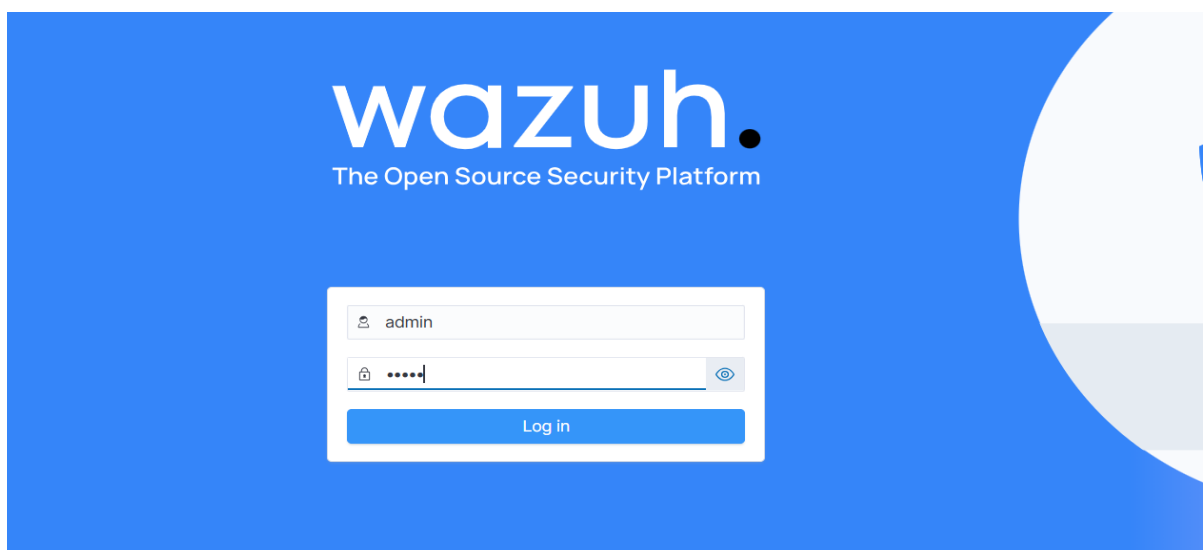
2. Open browser and enter:

https://<your-ip-address>



3. Login credentials:

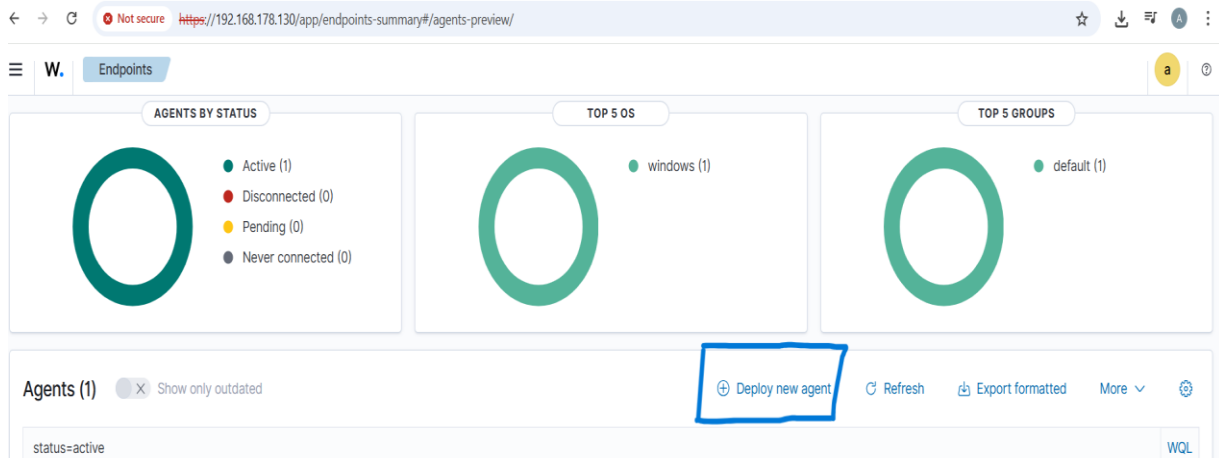
- Username: admin
- Password: admin



7. Wazuh Agent Installation (Kali Linux)

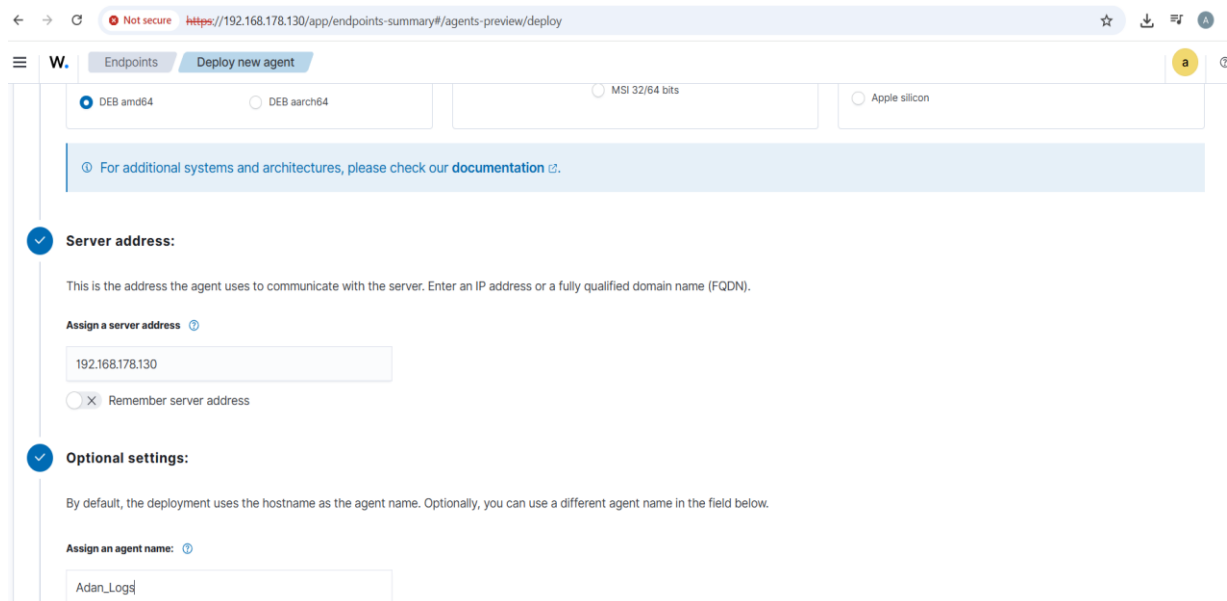
7.1 Deploy Agent

- Navigate to **Deploy New Agent** in Wazuh dashboard
- Select **DEB AMD64**



7.2 Configure Agent

- Enter server IP address
- Assign hostname



7.3 Install Agent

Copy the generated command and run it on Kali Linux.

Select one or more existing groups: [?](#)

Default

4 Run the following commands to download and install the agent:

```
wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.12.0-1_amd64.deb
```

Copy command

Requirements

- You will need administrator privileges to perform this installation.
- Shell Bash is required.

Keep in mind you need to run this command in a Shell Bash terminal.

5 Start the agent

Run it on kali linux

```
kali@kali: ~  
Session Actions Edit View Help  
--(kali@kali)-[~]  
-$ sudo wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.12.0-1_amd64.deb 66 sudo WAZUH_MANAGER='192.168.178.130' WAZUH_AGENT_NAME='Adan_Logs' dpkg -i ./wazuh-agent_4.12.0-1_amd64.deb  
sudo: password for kali:  
2026-04-26 15:55:49 -- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.12.0-1_amd64.deb  
resolving packages.wazuh.com (packages.wazuh.com) ... 18.64.141.68, 18.64.141.80, 18.64.141.53, ...  
connecting to packages.wazuh.com (packages.wazuh.com)[18.64.141.68]:443 ... connected.  
HTTP request sent, awaiting response ... 200 OK  
length: 11963008 (11M) [application/vnd.debian.binary-package]  
saving to: 'wazuh-agent_4.12.0-1_amd64.deb'  
wazuh-agent_4.12.0-1_amd64.deb 100%[=====] 11.41M 97.2KB/s in 68s  
2026-04-26 15:56:59 (172 KB/s) - 'wazuh-agent_4.12.0-1_amd64.deb' saved [11963008/11963008]  
Selecting previously unselected package wazuh-agent.  
Reading database... 445703 files and directories currently installed.)  
Preparing to unpack ./wazuh-agent_4.12.0-1_amd64.deb...  
Unpacking wazuh-agent (4.12.0-1)...  
Setting up wazuh-agent (4.12.0-1)...  
--(kali@kali)-[~]  
-$
```

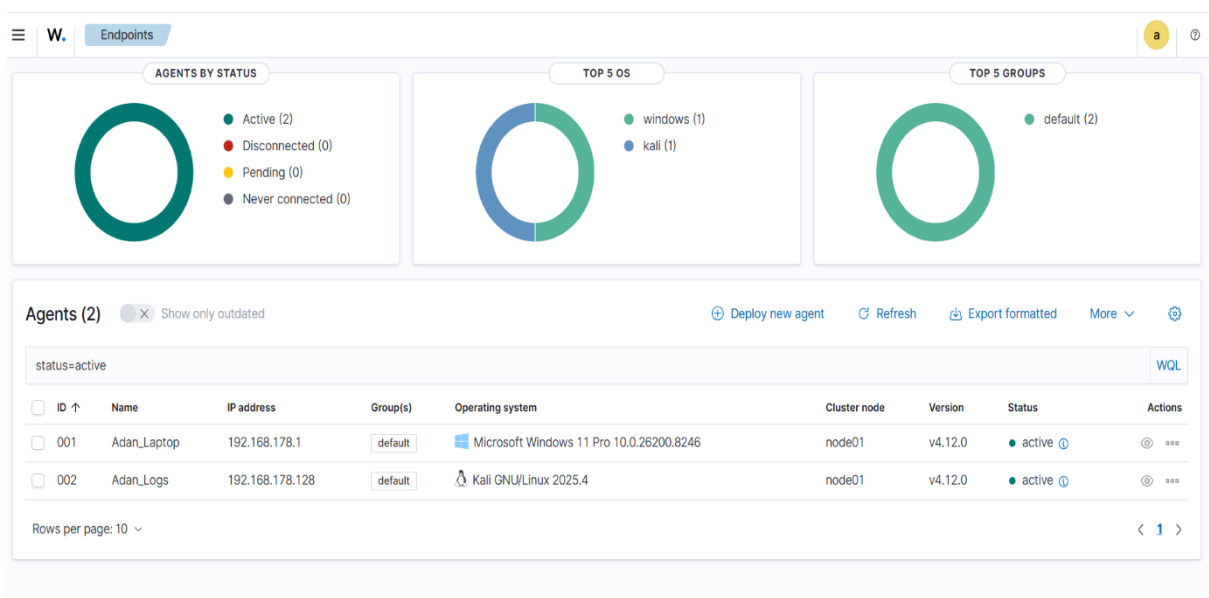
Then enter:

```
sudo systemctl daemon-reload  
sudo systemctl enable wazuh-agent  
sudo systemctl start wazuh-agent
```

```
(kali@kali)~$ sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
Created symlink '/etc/systemd/system/multi-user.target.wants/wazuh-agent.service' → '/usr/lib/systemd/system/wazuh-agent.service'.

(kali@kali)~$
```

A new log source named ‘Adan_Logs’ was successfully created and registered for monitoring and analysis within the security monitoring system



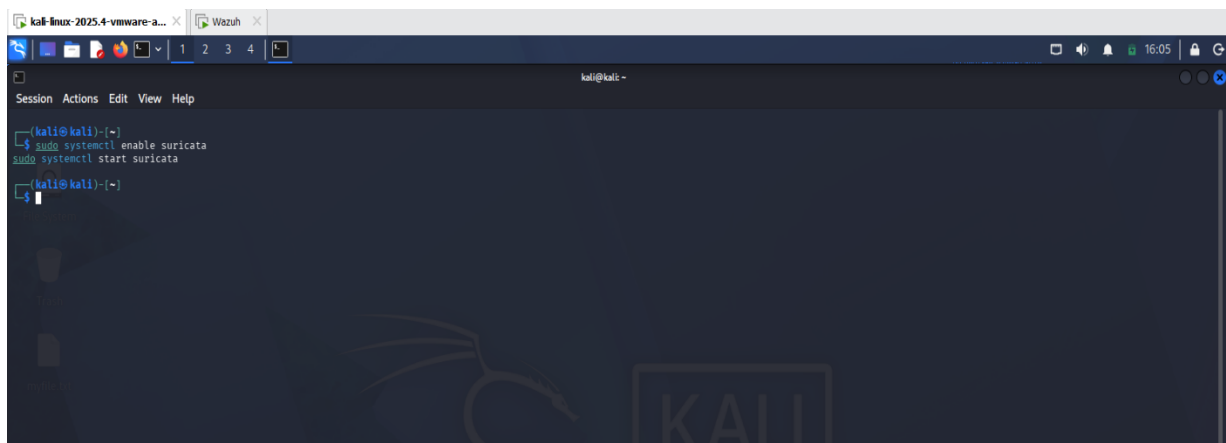
8. Integration of Suricata Logs with Wazuh

8.1 Configure Wazuh to Read Suricata Logs

Enable and Start Suricata

```
sudo systemctl enable suricata
```

```
sudo systemctl start suricata
```



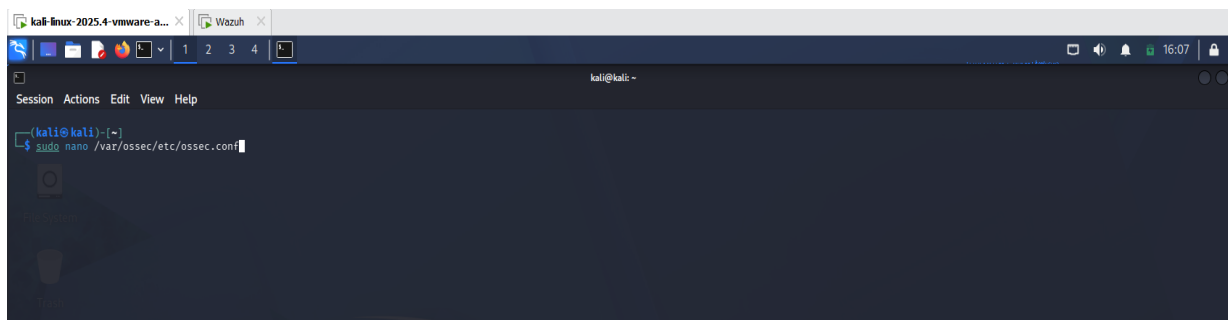
Check logs `sudo tail -f /var/log/suricata/eve.json`

This command monitors Suricata's eve.json log file in real time to observe detected events.



Configure “`sudo nano /var/ossec/etc/ossec.conf`”

This command opens the Wazuh configuration file (ossec.conf) for editing.



Add

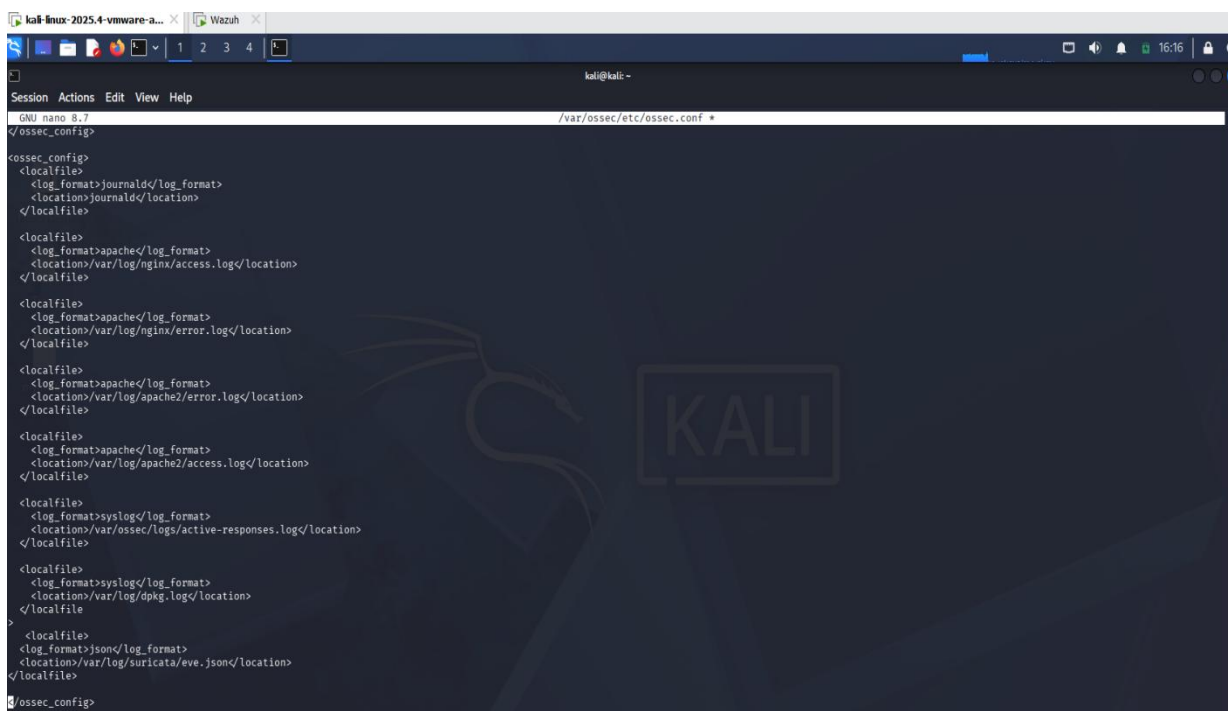
```
<localfile>
```

```
<log_format>json</log_format>
```

```
<location>/var/log/suricata/eve.json</location>
```

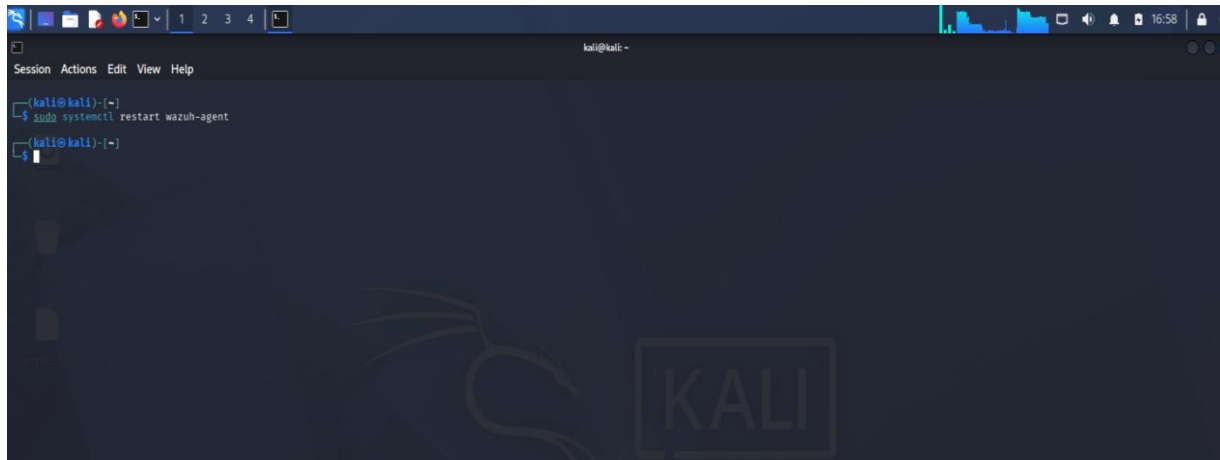
```
</localfile>
```

This configuration adds Suricata's eve.json file to Wazuh for monitoring JSON logs.



Then Restart Wazuh Agent

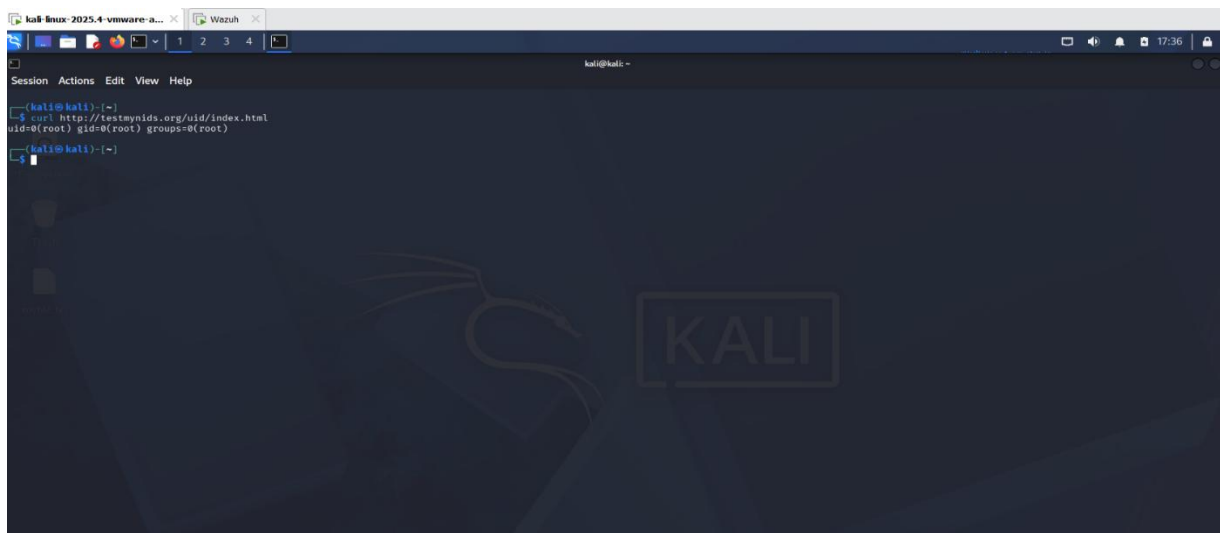
```
sudo systemctl restart wazuh-agent
```



```
kali@kali: ~  
$ sudo systemctl restart wazuh-agent  
$
```

Curl <http://testmyids.org/uid/index.html>

this command sends a test request to trigger IDS alerts and verify that Suricata is detecting threats correctly.



```
kali@kali: ~  
$ curl http://testmyids.org/uid/index.html  
uid=0(root) gid=0(root) groups=0(root)  
$
```

Suricata logs can be viewed in Wazuh Dashboard under the **Security Events / Alerts** section.

Export Formatted 730 available fields Columns Density 1 fields sorted Full screen				
timestamp	agent.name	rule.description	rule.level	rule.id
Apr 26, 2026 @ 14:36:15.443	Adan_Logs	Suricata: Alert - GPL ATTACK_RESPONSE id check returned root	3	86601

Ping -c 192.168.178.128

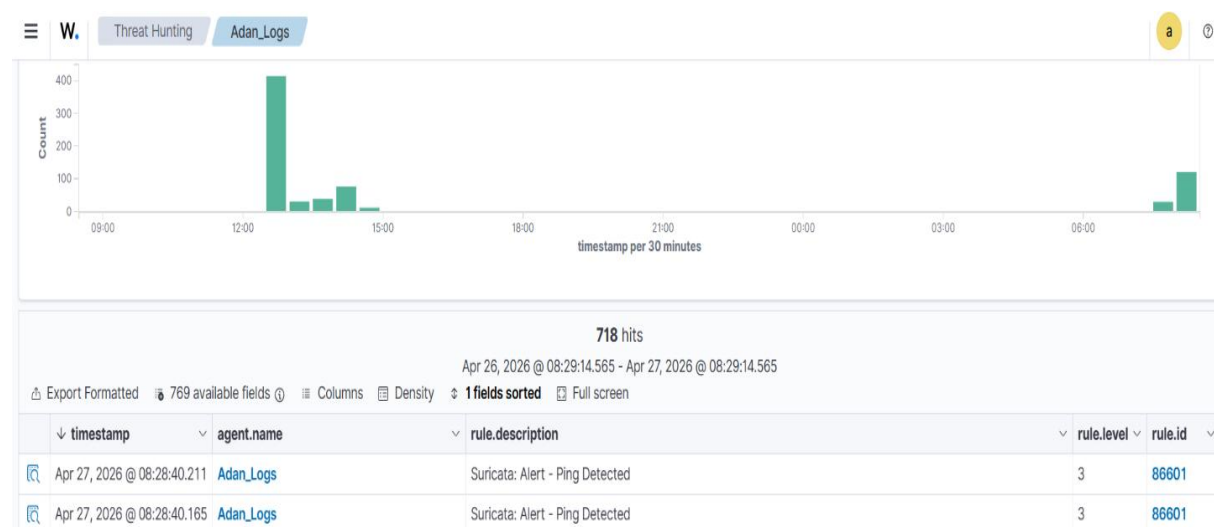
This command sends ICMP packets to test connectivity with the target IP address.

```
(kali㉿kali)-[~]
$ curl http://192.168.178.128
curl: (7) Failed to connect to 192.168.178.128 port 80 after 0 ms: Could not connect to server

(kali㉿kali)-[~]
$ ping -c 5 192.168.178.130
PING 192.168.178.130 (192.168.178.130) 56(84) bytes of data.
64 bytes from 192.168.178.130: icmp_seq=1 ttl=127 time=0.738 ms
64 bytes from 192.168.178.130: icmp_seq=2 ttl=127 time=0.874 ms
64 bytes from 192.168.178.130: icmp_seq=3 ttl=127 time=1.49 ms
64 bytes from 192.168.178.130: icmp_seq=4 ttl=127 time=1.11 ms
64 bytes from 192.168.178.130: icmp_seq=5 ttl=127 time=0.901 ms

— 192.168.178.130 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 4082ms
rtt min/avg/max/mdev = 0.738/1.022/1.488/0.261 ms
```

An ICMP echo request (ping) was executed, and the corresponding network activity was successfully detected and logged in Wazuh as a potential host discovery attempt



nmap -sS -p 1-1000 192.168.178.130

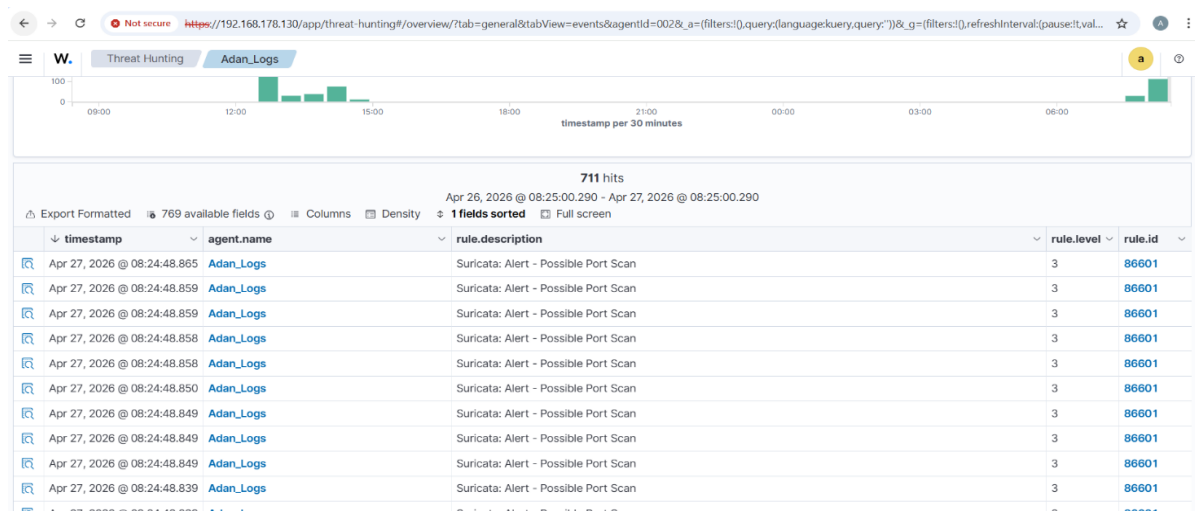
This command performs a TCP SYN stealth scan on ports 1–1000 of the target IP to identify open ports and services.

```
(kali㉿kali)-[~]
└─$ nmap -sS -p 1-1000 192.168.178.130
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-27 11:24 -0400
Nmap scan report for 192.168.178.130
Host is up (0.00033s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
443/tcp   open  https
MAC Address: 00:0C:29:21:57:6B (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.89 seconds

(kali㉿kali)-[~]
└─$
```

A TCP SYN scan was performed using Nmap (`nmap -sS -p 1-1000 192.168.178.130`), and the generated network activity was successfully detected and logged in Wazuh as a potential port scanning attempt.



9.Shuffle (SOAR):

Shuffle SOAR is a Security Orchestration, Automation, and Response (SOAR) platform used to automate cybersecurity workflows by connecting tools like Wazuh, TheHive, and Cortex, where it receives alerts, processes them through predefined workflows, and automatically performs actions such as alert enrichment, case creation, and notifications.

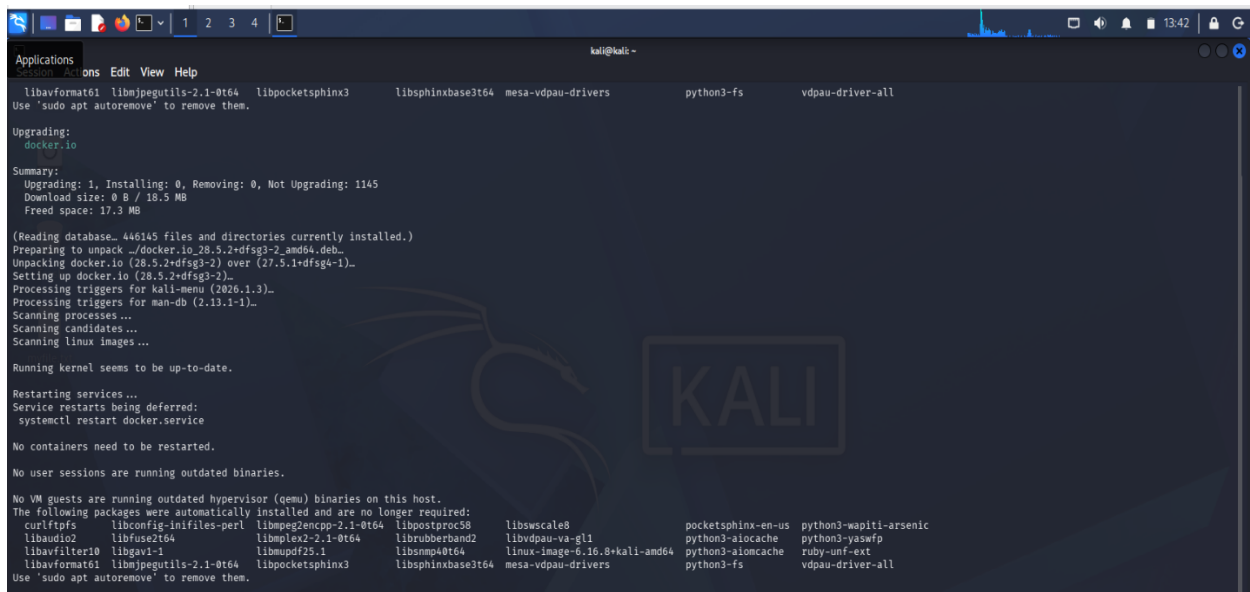
Installing Docker

`sudo apt install docker.io`

Installs Docker to run and manage containers for applications.

`sudo apt install docker-compose`

Installs Docker Compose to define and run multi-container applications using a YAML file.

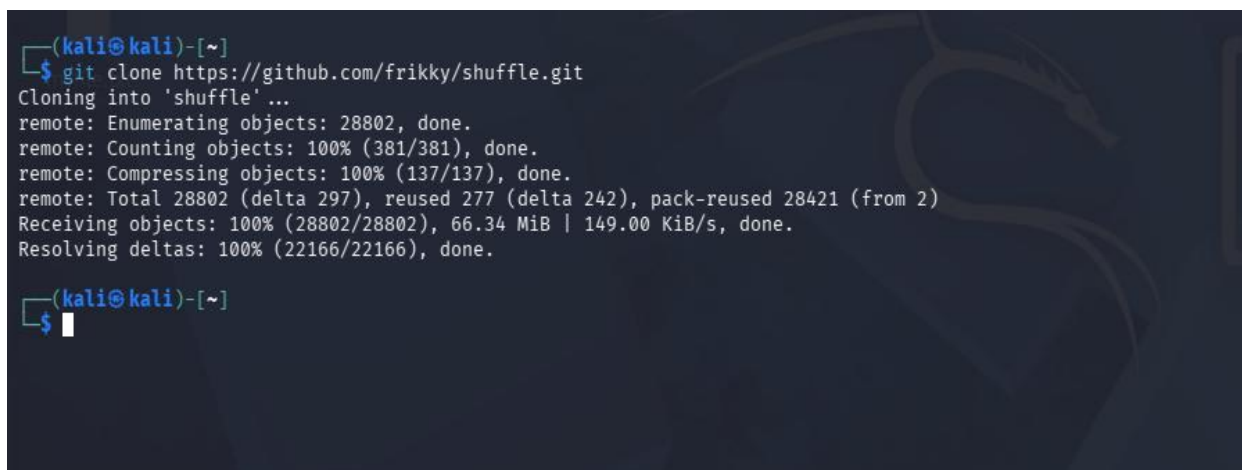
A terminal window on a Kali Linux system showing the command 'sudo apt install docker.io' being executed. The output displays the package list, the upgrade process for docker.io, and the installation summary. The summary indicates that 1 package is being installed, with a download size of 0 B / 18.5 MB and a freed space of 17.3 MB. The installation process includes unpacking the package, setting up the docker.io service, and processing triggers for kali-menu and man-db. The terminal also shows the status of the kernel, services, and containers, indicating that the installation was successful and no containers need to be restarted.

```
kali@kali: ~  
Applications  
libavformat61 libjpegutils-2.1-0t64 libpocketsphinx3 libspinxbase3t64 mesa-vdpau-drivers python3-fs vdpau-driver-all  
Use 'sudo apt autoremove' to remove them.  
Upgrading:  
  docker.io  
Summary:  
  Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 1145  
  Download size: 0 B / 18.5 MB  
  Freed space: 17.3 MB  
(Reading database, 446145 files and directories currently installed.)  
Preparing to unpack ./docker.io_28.5.2+dfsg3-2_amd64.deb...  
Unpacking docker.io (28.5.2+dfsg3-2) over (27.5.1+dfsg4-1)...  
Setting up docker.io (28.5.2+dfsg3-2)...  
Processing triggers for kali-menu (2026.1.3)...  
Processing triggers for man-db (2.13.1-1)...  
Scanning processes...  
Scanning candidates...  
Scanning linux images...  
Running kernel seems to be up-to-date.  
Restarting services...  
Service restarts being deferred:  
  systemctl restart docker.service  
No containers need to be restarted.  
No user sessions are running outdated binaries.  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
The following packages were automatically installed and are no longer required:  
  curlftpfs libconfig-inifiles-perl libmpg2encpp-2.1-0t64 libpostproc58 libswscale8 pocketsphinx-en-us python3-wapiti-arsenic  
  libaudio2 libfuse2t64 libmpeg2-2.1-0t64 librubberband2 libvdpau-va-gl1 python3-aiocache python3-yaswfo  
  libavfilter10 libgavl-1 libmupdf25.1 libsnmp40t64 linux-image-6.16.8+kali-amd64 python3-aiomcache python3-unf-ext  
  libavformat61 libjpegutils-2.1-0t64 libpocketsphinx3 libspinxbase3t64 mesa-vdpau-drivers python3-fs vdpau-driver-all  
Use 'sudo apt autoremove' to remove them.
```

Installing Shuffle(SOAR)

Clone Shuffle Repository

`git clone https://github.com/frikky/shuffle.git`

A terminal window on a Kali Linux system showing the command 'git clone https://github.com/frikky/shuffle.git' being executed. The output displays the cloning process, including enumerating objects, counting objects, compressing objects, and receiving objects. The terminal also shows the status of the repository, indicating that the cloning was successful and the repository is now ready for use.

```
(kali@kali)-[~]  
$ git clone https://github.com/frikky/shuffle.git  
Cloning into 'shuffle' ...  
remote: Enumerating objects: 28802, done.  
remote: Counting objects: 100% (381/381), done.  
remote: Compressing objects: 100% (137/137), done.  
remote: Total 28802 (delta 297), reused 277 (delta 242), pack-reused 28421 (from 2)  
Receiving objects: 100% (28802/28802), 66.34 MiB | 149.00 KiB/s, done.  
Resolving deltas: 100% (22166/22166), done.  
(kali@kali)-[~]  
$
```

cd shuffle

This command changes the directory to the Shuffle project folder.

```
(kali㉿kali)-[~]  
$ cd shuffle  
  
(kali㉿kali)-[~/shuffle]  
$
```

docker-compose up -d

This command starts and runs all services defined in the Docker Compose file in detached mode (background).

```
(kali㉿kali)-[~/shuffle]  
$ sudo docker-compose up -d  
[sudo] password for kali:  
[+] Running 36/38  
✓ orborus Pulled  
✓ 9824c27679d3 Pull complete  
✓ 065f90fabca4 Pull complete  
✓ 13b8d7420e7c Pull complete  
⌄ opensearch [=====] 165.1MB / 1.005GB Pulling  
✓ b9bd08b1e98f Pull complete  
✓ f8aa3279ea9 Pull complete  
✓ 02e053413b83 Pull complete  
⌄ fcd6b6fbb206 Downloading [=====] 106.9MB/947MB  
✓ e53b047a18c3 Download complete  
✓ f2a93b8e507c Download complete  
✓ frontend Pulled  
✓ 1733a4cd5924 Pull complete  
✓ aa8a9db0de30 Pull complete  
✓ 7f0da823c3c6 Pull complete  
✓ 1907fe38b1cf Pull complete  
✓ b25e0944b727 Pull complete  
✓ 8e5bb6a6b364 Pull complete  
✓ 7cc422329a3e Pull complete  
✓ f7e94fe79017 Pull complete  
✓ c5ffdd5d9e7f Pull complete  
✓ 8d9788875553 Pull complete  
✓ 01a9a0ab745e Pull complete  
✓ 8fc72a021fb1 Pull complete  
✓ 7b9109ef08a2 Pull complete  
✓ a0693ce983ac Pull complete  
✓ c6ae2810e4cb Pull complete  
✓ 40856057d16f Pull complete  
✓ 19c3440ff4c7 Pull complete  
✓ backend Pulled  
✓ 2d35ebdb57d9 Pull complete  
✓ 9b028d8bf166 Pull complete  
✓ 761eb53feb3f Pull complete  
✓ c9f34794a612 Pull complete  
✓ 8166b243b6a5 Pull complete  
✓ 7897a9235ce3 Pull complete  
✓ 46f5a7301183 Pull complete  
✓ 4f4fb700ef54 Download complete
```

This will automatically pull and run:

- Shuffle backend
- Frontend UI
- Database (usually MongoDB)

Check Running Containers

docker ps

This command lists all currently running Docker containers.

```
(kali@kali) ~/shuffle
$ sudo docker ps
[sudo] password for kali:
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
MES
632c673154ab   ghcr.io/shuffle/shuffle-frontend:latest "/entrypoint.sh nginx..." About a minute ago Up About a minute   0.0.0.0:3001→80/tcp, [::]:3001→80/tcp, 0.0.0.0:3443→443/tcp, [::]:3443→443/tcp   sh
uffle-frontend
27dc7092475e   opensearchproject/opensearch:3.2.0   "/opensearch-docker..." About a minute ago Up About a minute   9300/tcp, 9600/tcp, 0.0.0.0:9200→9200/tcp, :::9200→9200/tcp, 9650/tcp           sh
uffle-opensearch
ad937d8235bf   ghcr.io/shuffle/shuffle-backend:latest "./shufflebackend"        About a minute ago Up About a minute   0.0.0.0:5001→5001/tcp, :::5001→5001/tcp           sh
uffle-backend
0e9dff027475   ghcr.io/shuffle/shuffle-orborus:latest "./orborus"              About a minute ago Up About a minute                               sh
uffle-orborus

(kali@kali) ~/shuffle
$
```

Start deployment

Run:

```
sudo docker-compose down -v
```

stops and removes containers, networks, and volumes of the project.

```
sudo docker-compose up -d
```

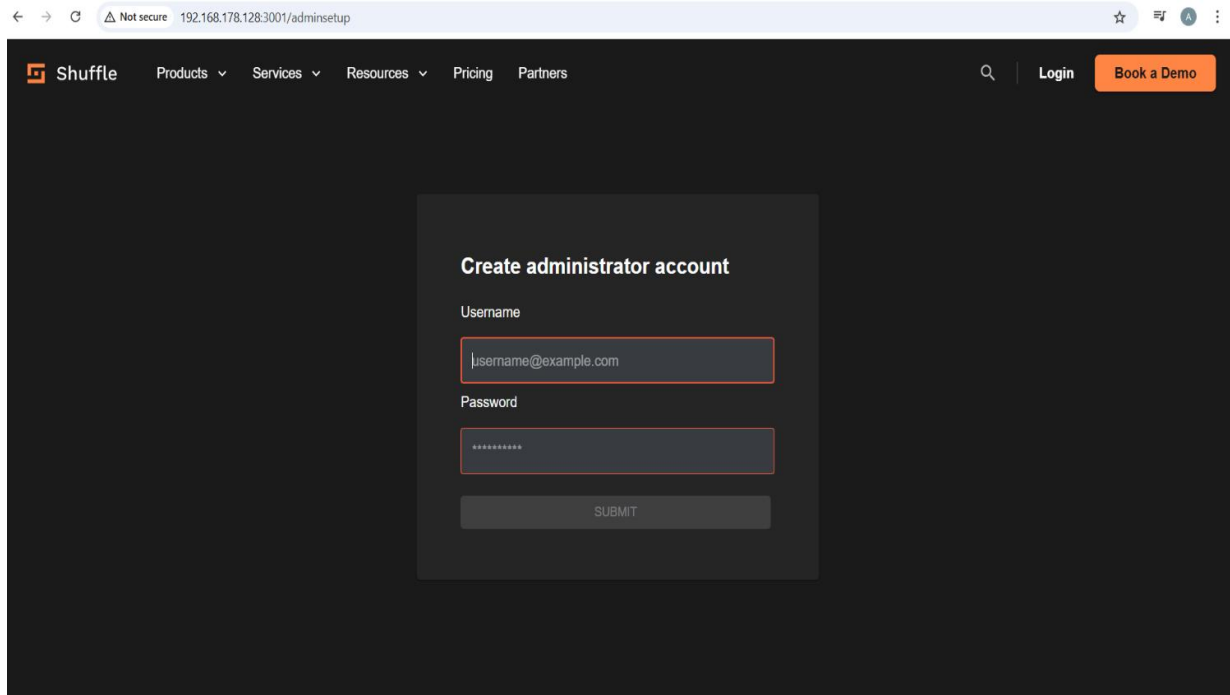
Rebuilds and starts all services in the background.

```
kali@kali: ~/Shuffle
Session Actions Edit View Help

(kali@kali) ~/Shuffle
$ sudo docker-compose down -v
sudo docker-compose up -d
WARN[0000] /home/kali/Shuffle/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
WARN[0000] /home/kali/Shuffle/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 6/95
  * redis [ ] Pulling
  * thehive [ ] Pulling
  * cassandra [ ] Pulling
  * misp_mysql [89.23MB / 141.2MB Pulling
  * misp-modules [ ] Pulling
  * cortex.local [ ] Pulling
  * minio [ ] Pulling
  * misp.local [ ] Pulling
```

Open your browser and go to: <http://localhost:3001>

Create a new user account and set credentials (username and password) for accessing the application.



10. TheHive Installation

Install dependencies

Run:

```
sudo apt update  
sudo apt install wget curl gnupg coreutils apt-transport-https git ca-certificates ca-certificates-  
java software-properties-common python3-pip lsb-release unzip -y
```

```
---(kali@kali)-[~]  
_-$ sudo apt install wget curl gnupg coreutils apt-transport-https git ca-certificates ca-certificates-java software-properties-common python3-pip lsb-release unzip -y  
[sudo] password for kali:  
wget is already the newest version (1.25.0-2).  
curl is already the newest version (8.19.0-3).  
gnupg is already the newest version (2.4.9-4).  
gnupg set to manually installed.  
coreutils is already the newest version (9.10-1).  
git is already the newest version (1:2.53.0-1).  
ca-certificates is already the newest version (20260223).  
ca-certificates-java is already the newest version (20260311).  
ca-certificates-java set to manually installed.  
software-properties-common is already the newest version (0.111-1).  
python3-pip is already the newest version (26.0.1+dfsg-1).  
lsb-release is already the newest version (12.1-2).  
unzip is already the newest version (6.0-29).  
unzip set to manually installed.  
Installing:  
  apt-transport-https  
Summary:  
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 58  
  Download size: 26.5 kB  
  Space needed: 37.9 kB / 52.6 GB available  
Get:1 http://http.kali.org/kali kali-rolling/main amd64 apt-transport-https all 3.2.0+kali1 [26.5 kB]  
Fetched 26.5 kB in 2s (13.3 kB/s)  
Selecting previously unselected package apt-transport-https.  
(Reading database... 442315 files and directories currently installed.)  
Preparing to unpack ./apt-transport-https_3.2.0+kali1_all.deb...  
Unpacking apt-transport-https (3.2.0+kali1)...  
Setting up apt-transport-https (3.2.0+kali1)...  
Scanning processes...  
Scanning linux images...  
Running kernel seems to be up-to-date.  
No services need to be restarted.  
No containers need to be restarted.  
No user sessions are running outdated binaries.  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
---(kali@kali)-[~]  
_-$
```

Install Java 11

```
wget -qO- https://apt.corretto.aws/corretto.key | sudo gpg --dearmor -o /usr/share/keyrings/corretto.gpg
```

```
echo "deb [signed-by=/usr/share/keyrings/corretto.gpg] https://apt.corretto.aws stable main" | sudo tee /etc/apt/sources.list.d/corretto.sources.list
```

```
sudo apt update  
sudo apt install java-11-amazon-corretto-jdk -y
```

This set of commands is used to add the Amazon Corretto repository, import its signing key, update package lists, and install Java 11 (Amazon Corretto JDK) on a Linux system.

```
(kali@kali)-[~]  
$ wget -qO- https://apt.corretto.aws/corretto.key | sudo gpg --dearmor -o /usr/share/keyrings/corretto.gpg  
echo "deb [signed-by=/usr/share/keyrings/corretto.gpg] https://apt.corretto.aws stable main" | sudo tee /etc/apt/sources.list.d/corretto.sources.list  
sudo apt update  
sudo apt install java-11-amazon-corretto-jdk -y  
deb [signed-by=/usr/share/keyrings/corretto.gpg] https://apt.corretto.aws stable main  
get:2 https://download.docker.com/linux/debian bookworm InRelease [46.6 kB]  
get:3 https://apt.corretto.aws stable InRelease [10.7 kB]  
get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]  
get:4 https://apt.corretto.aws stable/main amd64 Packages [26.1 kB]  
get:5 http://kali.download/kali kali-rolling/main amd64 Packages [21.1 MB]  
get:6 https://apt.corretto.aws stable/main amd64 Contents (deb) [33.5 kB]  
get:7 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.4 MB]  
get:8 http://kali.download/kali kali-rolling/contrib amd64 Packages [118 kB]  
get:9 http://kali.download/kali kali-rolling/non-free amd64 Packages [189 kB]  
get:10 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [908 kB]  
Fetched 75.8 MB in 2min 33s (496 kB/s)  
129 packages can be upgraded. Run 'apt list --upgradable' to see them.  
Installing:  
  java-11-amazon-corretto-jdk  
Summary:  
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 129  
  Download size: 196 MB  
  Space needed: 327 MB / 52.5 GB available  
get:1 https://apt.corretto.aws stable/main amd64 java-11-amazon-corretto-jdk amd64 1:11.0.31.11-1 [196 MB]  
Ign:1 https://apt.corretto.aws stable/main amd64 java-11-amazon-corretto-jdk amd64 1:11.0.31.11-1  
get:1 https://apt.corretto.aws stable/main amd64 java-11-amazon-corretto-jdk amd64 1:11.0.31.11-1 [196 MB]  
Fetched 90.2 MB in 20min 16s (74.1 kB/s)  
Selecting previously unselected package java-11-amazon-corretto-jdk:amd64.  
(Reading database... 442319 files and directories currently installed.)  
Preparing to unpack ./java-11-amazon-corretto-jdk_1%3a11.0.31.11-1_amd64.deb..  
Unpacking java-11-amazon-corretto-jdk:amd64 (1:11.0.31.11-1).  
Setting up java-11-amazon-corretto-jdk:amd64 (1:11.0.31.11-1).  
update-alternatives: using /usr/lib/jvm/java-11-amazon-corretto/bin/java to provide /usr/bin/java (java) in auto mode  
update-alternatives: using /usr/lib/jvm/java-11-amazon-corretto/bin/keytool to provide /usr/bin/keytool (keytool) in auto mode  
update-alternatives: using /usr/lib/jvm/java-11-amazon-corretto/bin/rmid to provide /usr/bin/rmid (rmid) in auto mode  
update-alternatives: using /usr/lib/jvm/java-11-amazon-corretto/bin/rmiregistry to provide /usr/bin/rmiregistry (rmiregistry) in auto mode  
update-alternatives: using /usr/lib/jvm/java-11-amazon-corretto/bin/jjs to provide /usr/bin/jjs (jjs) in auto mode  
update-alternatives: using /usr/lib/jvm/java-11-amazon-corretto/bin/pack200 to provide /usr/bin/pack200 (pack200) in auto mode  
update-alternatives: using /usr/lib/jvm/java-11-amazon-corretto/bin/unpack200 to provide /usr/bin/unpack200 (unpack200) in auto mode  
update-alternatives: using /usr/lib/jvm/java-11-amazon-corretto/bin/javac to provide /usr/bin/javac (javac) in auto mode  
update-alternatives: using /usr/lib/jvm/java-11-amazon-corretto/bin/jaotc to provide /usr/bin/jaotc (jaotc) in auto mode  
update-alternatives: using /usr/lib/jvm/java-11-amazon-corretto/bin/jlink to provide /usr/bin/jlink (jlink) in auto mode
```

Set JAVA_HOME:

```
echo 'JAVA_HOME="/usr/lib/jvm/java-11-amazon-corretto"' | sudo tee -a /etc/environment  
export JAVA_HOME="/usr/lib/jvm/java-11-amazon-corretto"
```

This command is used to set and configure the JAVA_HOME environment variable for Amazon Corretto JDK 11, making Java available system-wide for applications and services.

```
(kali@kali)-[~]
$ echo 'JAVA_HOME="/usr/lib/jvm/java-11-amazon-corretto"' | sudo tee -a /etc/environment
export JAVA_HOME="/usr/lib/jvm/java-11-amazon-corretto"
[sudo] password for kali:
JAVA_HOME="/usr/lib/jvm/java-11-amazon-corretto"

(kali@kali)-[~]
$
```

Verify:

java -version

```
(kali@kali)-[~]
$ java -version
openjdk version "11.0.31" 2026-04-21 LTS
OpenJDK Runtime Environment Corretto-11.0.31.11.1 (build 11.0.31+11-LTS)
OpenJDK 64-Bit Server VM Corretto-11.0.31.11.1 (build 11.0.31+11-LTS, mixed mode)

(kali@kali)-[~]
$
```

Install Cassandra (DATABASE)

Add repo:

```
wget -qO - https://downloads.apache.org/cassandra/KEYS | sudo gpg --dearmor -o
/usr/share/keyrings/cassandra.gpg
```

```
echo "deb [signed-by=/usr/share/keyrings/cassandra.gpg] https://deb.debian.org/debian/cassandra
41x main" | sudo tee /etc/apt/sources.list.d/cassandra.sources.list
```

This command is used to add the Apache Cassandra repository and import its GPG key so the system can securely install Cassandra packages via apt.

```
(kali@kali)-[~]
└─$ wget -qO - https://downloads.apache.org/cassandra/KEYS | sudo gpg --dearmor -o /usr/share/keyrings/cassandra.gpg
echo "deb [signed-by=/usr/share/keyrings/cassandra.gpg] https://debian.cassandra.apache.org 41x main" | sudo tee /etc/apt/sources.list.d/cassandra.sources.list
deb [signed-by=/usr/share/keyrings/cassandra.gpg] https://debian.cassandra.apache.org 41x main
(kali@kali)-[~]
└─$
```

Cassandra Installation:

sudo apt update

sudo apt install cassandra -y

```
(kali@kali)-[~]
└─$ sudo apt update
sudo apt install cassandra -y
Hit:1 http://http.kali.org/kali kali-rolling InRelease
Get:2 https://download.docker.com/linux/debian bookworm InRelease [46.6 kB]
Hit:3 https://apt.corretto.aws stable InRelease
Hit:4 https://apache.jfrog.io/artifactory/cassandra-deb 41x InRelease
Fetched 46.6 kB in 17s (2,756 B/s)
129 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
 cassandra

Suggested packages:
 cassandra-tools

Summary:
 Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 129
 Download size: 52.0 MB
 Space needed: 62.7 MB / 52.1 GB available

Get:1 https://apache.jfrog.io/artifactory/cassandra-deb 41x/main amd64 cassandra all 4.1.11 [52.0 MB]
Fetched 51.5 MB in 5min 40s (152 kB/s)
Selecting previously unselected package cassandra.
(Reading database... 442880 files and directories currently installed.)
Preparing to unpack .../cassandra_4.1.11_all.deb...
Unpacking cassandra (4.1.11)...
Setting up cassandra (4.1.11)...
vm.max_map_count = 1048575
net.ipv4.tcp_keepalive_time = 300
update-rc.d: We have no instructions for the cassandra init script.
update-rc.d: It looks like a network service, we disable it.
update-rc.d: warning: start and stop actions are no longer supported; falling back to defaults
Failed to disable unit: Unit cassandra.service does not exist
Failed to disable unit: Unit cassandra.service does not exist
Processing triggers for kali-menu (2026.2.2)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

ps aux | grep Cassandra

This command is used to check whether Apache Cassandra is running by listing active processes and filtering results for cassandra

```
(kali@kali):~$ sudo cassandra -R
(kali@kali):~$ ps aux | grep cassandra
root      37566 64.9 27.9 3274464 1767196 ?        Ssl   11:14   0:08 /usr/bin/java -ea -d32net.openhft... -XX:+UseThreadPriorities -XX:+HeapDumpOnOutOfMemoryError -Xss256k -XX:+AlwaysPreTouch -XX:-UseBiasedLockin
g -XX:+UseTLAB -XX:+ResizeTLAB -XX:+UseNUMA -XX:+PerfDisableSharedMem -Djava.net.preferIPv4Stack=true -XX:+UseConcMarkSweepGC -XX:+CMSParallelInitEnabled -XX:SurvivorRatio=8 -XX:MaxTenuringThreshold=1 -XX:CMS
InitiatingOccupancyFraction=75 -XX:+UseCMSInitiatingOccupancyOnly -XX:CMSInitDuration=10000 -XX:+CMSParallelInitialMarkEnabled -XX:+CMS EdenChunksRecordAlways -XX:+CMSClassUnloadingEnabled -Djdk.attach.allowAtta
chSelf=true --add-exports java.base/jdk.internal.misc=ALL-UNNAMED --add-exports java.base/jdk.internal.ref=ALL-UNNAMED --add-exports java.base/sun.nio.ch=ALL-UNNAMED --add-exports java.management.rmi/com.sun.jm
x.remote.internal.rmi=ALL-UNNAMED --add-exports java.rmi/sun.rmi.registry=ALL-UNNAMED --add-exports java.rmi/sun.rmi.server=ALL-UNNAMED --add-exports java.sql/java.sql=ALL-UNNAMED --add-opens java.base/java.lan
g.module=ALL-UNNAMED --add-opens java.base/jdk.internal.loader=ALL-UNNAMED --add-opens java.base/jdk.internal.ref=ALL-UNNAMED --add-opens java.base/jdk.internal.reflect=ALL-UNNAMED --add-opens java.base/jdk.int
ernal.math=ALL-UNNAMED --add-opens java.base/jdk.internal.module=ALL-UNNAMED --add-opens java.base/jdk.internal.util.jar=ALL-UNNAMED --add-opens jdk.management/com.sun.management.internal=ALL-UNNAMED -Dio.netty
.tryReflectionGetAccessible=true -Xlog:gc-info,heap=trace,age=debug,safepoint=info,promotion+trace:files=/var/log/cassandra/gc-logs:time,uptime,pid,tid,level,file:count=10,file:size=10485760 -Xms1545M -Xmx1545M
-Xmn386M -XX:+UseConcGCDMark -XX:CompileCommandFile=/etc/cassandra/hotspot_compiler -jvagent:/usr/share/cassandra/lib/jamm-0.3.2.jar -Dcassandra.jmx.local.port=7199 -Dcom.sun.management.jmxremote.authenticat
e=false -Dcom.sun.management.jmxremote.password.file=/etc/cassandra/jmxremote.password -Djava.library.path=/usr/share/cassandra/lib/sigar-bin -Dcassandra.libjemalloc=/usr/lib/x86_64-linux-gnu/libjemalloc.so.2 -
XX:OnOutOfMemoryError=kill -9 %p -Dlogback.configurationFile=logback.xml -Dcassandra.logdir=/var/log/cassandra -Dcassandra.storagedir=/var/lib/cassandra -cp /etc/cassandra:/usr/share/cassandra/lib/HdrHistogram-
2.1.12.jar:/usr/share/cassandra/lib/ST4-4.0.8.jar:/usr/share/cassandra/lib/airline-0.8.jar:/usr/share/cassandra/lib/antlr-runtime-3.5.2.jar:/usr/share/cassandra/lib/asm-9.1.jar:/usr/share/cassandra/lib/caffeine
-2.9.2.jar:/usr/share/cassandra/lib/cassandra-driver-core-3.11.5-shaded.jar:/usr/share/cassandra/lib/checker-qual-3.10.0.jar:/usr/share/cassandra/lib/chronicle-bytes-2.20.111.jar:/usr/share/cassandra/lib/chroni
cle-core-2.20.116.jar:/usr/share/cassandra/lib/chronicle-queue-5.20.112.jar:/usr/share/cassandra/lib/chronicle-threads-2.20.111.jar:/usr/share/cassandra/lib/chronicle-wire-2.20.117.jar:/usr/share/cassandra/lib/
commons-cli-1.1.jar:/usr/share/cassandra/lib/commons-codec-1.9.jar:/usr/share/cassandra/lib/commons-lang3-3.10.0.jar:/usr/share/cassandra/lib/commons-math3-3.2.jar:/usr/share/cassandra/lib/concurrent-trees-2.4.
0.jar:/usr/share/cassandra/lib/ecj-4.6.1.jar:/usr/share/cassandra/lib/error_prone_annotations-2.5.1.jar:/usr/share/cassandra/lib/guava-27.0-jre.jar:/usr/share/cassandra/lib/high-scale-lib-1.0.6.jar:/usr/share/c
assandra/lib/hppc-0.8.1.jar:/usr/share/cassandra/lib/ipaddress-5.3.3.jar:/usr/share/cassandra/lib/j2objc-annotations-1.3.jar:/usr/share/cassandra/lib/jackson-annotations-2.19.2.jar:/usr/share/cassandra/lib/jack
son-core-2.19.2.jar:/usr/share/cassandra/lib/jackson-databind-2.19.2.jar:/usr/share/cassandra/lib/jackson-datatype-jsr310-2.19.2.jar:/usr/share/cassandra/lib/jamm-0.3.2.jar:/usr/share/cassandra/lib/javax.inject
-1.jar:/usr/share/cassandra/lib/jbcrypt-0.4.jar:/usr/share/cassandra/lib/jcl-over-slf4j-1.7.25.jar:/usr/share/cassandra/lib/jcommander-1.20.jar:/usr/share/cassandra/lib/jctools-core-3.1.0.jar:/usr/share/cassand
ra/lib/jna-5.9.0.jar:/usr/share/cassandra/lib/json-simple-1.1.jar:/usr/share/cassandra/lib/jsr305-2.0.2.jar:/usr/share/cassandra/lib/jvm-attach-api-1.5.jar:/usr/share/cassandra/lib/log4j-over-slf4j-1.7.25.jar:/
usr/share/cassandra/lib/logback-classic-1.2.9.jar:/usr/share/cassandra/lib/logback-core-1.2.9.jar:/usr/share/cassandra/lib/lz4-java-1.10.1.jar:/usr/share/cassandra/lib/metrics-core-3.1.5.jar:/usr/share/cassandra
4/lib/metrics-jvm-3.1.5.jar:/usr/share/cassandra/lib/metrics-logback-3.1.5.jar:/usr/share/cassandra/lib/mxdump-0.14.jar:/usr/share/cassandra/lib/netty-all-4.1.58.Final.jar:/usr/share/cassandra/lib/netty-tcnativ
e-boringssl-static-2.0.36.Final.jar:/usr/share/cassandra/lib/ohc-core-0.5.1.jar:/usr/share/cassandra/lib/ohc-core-j8-0.5.1.jar:/usr/share/cassandra/lib/psjava-0.1.19.jar:/usr/share/cassandra/lib/reporter-config
-base-3.0.3.jar:/usr/share/cassandra/lib/reporter-config-3.0.3.jar:/usr/share/cassandra/lib/sigar-1.6.4.jar:/usr/share/cassandra/lib/sjk-cli-0.14.jar:/usr/share/cassandra/lib/sjk-core-0.14.jar:/usr/share/cassa
ndra/lib/sjk-json-0.14.jar:/usr/share/cassandra/lib/sjk-stacktrace-0.14.jar:/usr/share/cassandra/lib/slf4j-api-1.7.25.jar:/usr/share/cassandra/lib/snakeyaml-1.26.jar:/usr/share/cassandra/lib/snappy-java-1.1.10.
4.jar:/usr/share/cassandra/lib/snowball-stemmer-1.3.0.S01.1.jar:/usr/share/cassandra/lib/stream-2.5.2.jar:/usr/share/cassandra/lib/zstd-jni-1.5.7-2.jar:/usr/share/cassandra/lib/jsr223/*-*.jar:/usr/share/cassand
ra/apache-cassandra-4.1.11.jar:/usr/share/cassandra/apache-cassandra.jar:/usr/share/cassandra/fqltool.jar:/usr/share/cassandra/stress.jar: org.apache.cassandra.service.CassandraDaemon
kali      37912  0.0  0.0   6544   2440 pts/1    S+   11:15   0:00 grep --color=auto cassandra
(kali@kali):~$
```

Create service file

sudo nano /etc/systemd/system/cassandra.service

Step 2: Paste this

[Unit]

Description=Apache Cassandra

After=network.target

[Service]

Type=forking

ExecStart=/usr/sbin/cassandra -R

ExecStop=/bin/kill -15 \$MAINPID

User=cassandra

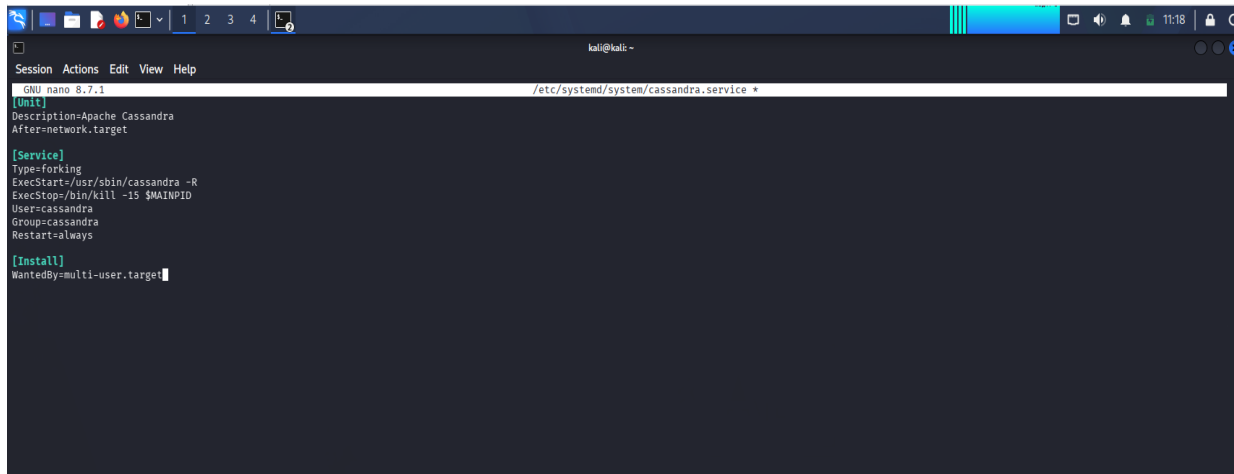
Group=cassandra

Restart=always

[Install]

WantedBy=multi-user.target

This configuration is used to create a systemd service file for Apache Cassandra, defining how the Cassandra service should start, stop, and run automatically on system boot.



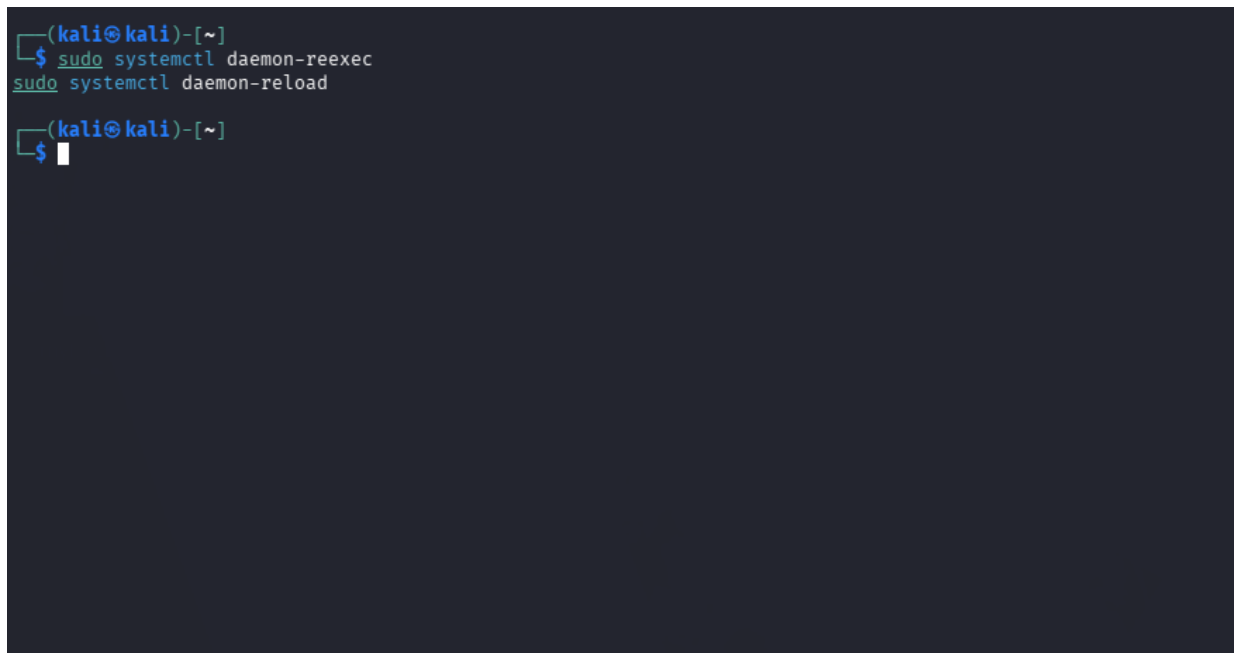
```
GNU nano 8.7.1 /etc/systemd/system/cassandra.service *
[Unit]
Description=Apache Cassandra
After=network.target

[Service]
Type=forking
ExecStart=/usr/sbin/cassandra -R
ExecStop=/bin/kill -15 $MAINPID
User=cassandra
Group=cassandra
Restart=always

[Install]
WantedBy=multi-user.target
```

Reload system

```
sudo systemctl daemon-reexec
sudo systemctl daemon-reload
```



```
(kali㉿kali)-[~]
$ sudo systemctl daemon-reexec
sudo systemctl daemon-reload

(kali㉿kali)-[~]
$
```

Run Cassandra:

```
sudo systemctl enable cassandra
sudo systemctl start cassandra
sudo systemctl status Cassandra
```

```
(kali@kali)-[~]
└─$ sudo systemctl enable cassandra
sudo systemctl start cassandra
sudo systemctl status cassandra
Created symlink '/etc/systemd/system/multi-user.target.wants/cassandra.service' → '/etc/systemd/system/cassandra.service'.
● cassandra.service - Apache Cassandra
   Loaded: loaded (/etc/systemd/system/cassandra.service; enabled; preset: disabled)
   Active: active (running) since Thu 2025-04-30 11:22:16 EDT; 304ms ago
     Invocation: 2df04369969c47a58dc736565a6c3556
   Process: 65508 ExecStart=/usr/sbin/cassandra -R (code=exited, status=0/SUCCESS)
    Main PID: 69636 (java)
       Tasks: 3 (limit: 7167)
      Memory: 416.4M (peak: 1.2G)
         CPU: 10.782s
    CGroup: /system.slice/cassandra.service
            └─69636 /usr/bin/java -ea -d:net.openhft... -XX:+UseThreadPriorities -XX:+HeapDumpOnOutOfMemoryError -Xss256k -XX:+AlwaysPreTouch -XX:-UseBiasedLocking -XX:+UseTLAB -XX:+ResizeTLAB -XX:+UseNUMA

Apr 30 11:21:23 kali systemd[1]: Starting cassandra.service - Apache Cassandra...
Apr 30 11:22:16 kali systemd[1]: Started cassandra.service - Apache Cassandra.
Apr 30 11:22:16 kali cassandra[69636]: OpenJDK 64-Bit Server VM warning: Option UseConcMarkSweepGC was deprecated in version 9.0 and will likely be removed in a future release.
[lines 1-15/15 (END)]
```

Important Cassandra fix

sudo chown -R cassandra:cassandra /var/lib/cassandra

```
(kali@kali)-[~]
└─$ sudo chown -R cassandra:cassandra /var/lib/cassandra

(kali@kali)-[~]
└─$
```

Enable authentication

Edit:

sudo nano /etc/cassandra/cassandra.yaml

Set:

authenticator: PasswordAuthenticator
authorizer: CassandraAuthorizer

This step is used to configure Apache Cassandra security settings by enabling authentication and authorization, allowing username/password login and controlling user access permissions within the database.

```
Session Actions Edit View Help
GNU nano 8.7.1 /etc/cassandra/cassandra.yaml *

# - dynamic_remote
# Using DynamicEndpointSnitch to select batchlog storage endpoints, prevents the
# local rack, if possible. This strategy offers the same availability guarantees
# as random_remote but selects the fastest endpoints according to the DynamicEndpointSnitch.
# (DynamicEndpointSnitch currently only tracks reads and not writes - i.e. write-only
# (or mostly-write) workloads might not benefit from this strategy.)
# Note: this strategy will fall back to random_remote, if dynamic_snitch is not enabled.
#
# - dynamic
# Mostly the same as dynamic_remote, except that local rack is not excluded, which mean it offers lower
# availability guarantee than random_remote or dynamic_remote.
# Note: this strategy will fall back to random_remote, if dynamic_snitch is not enabled.
#
batchlog_endpoint_strategy: random_remote

# Authentication backend, implementing IAuthenticator; used to identify users
# Out of the box, Cassandra provides org.apache.cassandra.auth.{AllowAllAuthenticator,
# PasswordAuthenticator}.
#
# - AllowAllAuthenticator performs no checks - set it to disable authentication.
# - PasswordAuthenticator relies on username/password pairs to authenticate
# users. It keeps usernames and hashed passwords in system_auth.roles table.
# Please increase system_auth keyspace replication factor if you use this authenticator.
# If using PasswordAuthenticator, CassandraRoleManager must also be used (see below)
authenticator: PasswordAuthenticator

# Authorization backend, implementing IAuthorizer; used to limit access/provide permissions
# Out of the box, Cassandra provides org.apache.cassandra.auth.{AllowAllAuthorizer,
# CassandraAuthorizer}.
#
# - AllowAllAuthorizer allows any action to any user - set it to disable authorization.
# - CassandraAuthorizer stores permissions in system_auth.role_permissions table. Please
# increase system_auth keyspace replication factor if you use this authorizer.
authorizer: CassandraAuthorizer

# Part of the Authentication & Authorization backend, implementing IRoleManager; used
# to maintain grants and memberships between roles.
# Out of the box, Cassandra provides org.apache.cassandra.auth.CassandraRoleManager,
# which stores role information in the system_auth keyspace. Most functions of the
# IRoleManager require an authenticated login, so unless the configured IAuthenticator
# actually implements authentication, most of this functionality will be unavailable.
#
# Example configuration for the CassandraRoleManager, if the system_auth keyspace is placed
```

Restart Cassandra:

sudo systemctl restart Cassandra

```
(kali㉿kali)-[~]
$ sudo systemctl restart cassandra

(kali㉿kali)-[~]
$
```

Elasticsearch Installation:

Add repo:

```
wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo gpg --dearmor -o /usr/share/keyrings/elasticsearch.gpg
```

```
echo "deb [signed-by=/usr/share/keyrings/elasticsearch.gpg] https://artifacts.elastic.co/packages/8.x/apt stable main" | sudo tee /etc/apt/sources.list.d/elasticsearch.list
```

This command is used to add the Elasticsearch repository and import its GPG key so the system can securely install Elasticsearch packages via apt.

```
(kali@kali)-[~]
$ wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo gpg --dearmor -o /usr/share/keyrings/elasticsearch.gpg
echo "deb [signed-by=/usr/share/keyrings/elasticsearch.gpg] https://artifacts.elastic.co/packages/8.x/apt stable main" | sudo tee /etc/apt/sources.list.d/elasticsearch.list
deb [signed-by=/usr/share/keyrings/elasticsearch.gpg] https://artifacts.elastic.co/packages/8.x/apt stable main
(kali@kali)-[~]
$
```

Install:

```
sudo apt update
sudo apt install elasticsearch -y
```

```
Session Actions Edit View Help
Upgrading:
  elasticsearch

Summary:
Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 129
1 not fully installed or removed.
Download size: 679 MB
Space needed: 1.308 MB / 50.6 GB available

Continue? [Y/n] Y
Get:1 https://artifacts.elastic.co/packages/8.x/apt stable/main amd64 elasticsearch amd64 8.19.15 [679 MB]
Fetched 679 MB in 18min 46s (603 kB/s)
(Reading database... 443053 files and directories currently installed.)
Preparing to unpack ./elasticsearch_8.19.15_amd64.deb...
Unpacking elasticsearch (8.19.15) over (8.19.15)...
Setting up elasticsearch (8.19.15)...

----- Security autoconfiguration information -----

Authentication and authorization are enabled.
TLS for the transport and HTTP layers is enabled and configured.

The generated password for the elastic built-in superuser is : PzslKHAKx_WaNjbc008E

If this node should join an existing cluster, you can reconfigure this with
'/usr/share/elasticsearch/bin/elasticsearch-reconfigure-node --enrollment-token <token-here>'
after creating an enrollment token on your existing cluster.

You can complete the following actions at any time:

Reset the password of the elastic built-in superuser with
'/usr/share/elasticsearch/bin/elasticsearch-reset-password -u elastic'.

Generate an enrollment token for Kibana instances with
'/usr/share/elasticsearch/bin/elasticsearch-create-enrollment-token -s kibana'.

Generate an enrollment token for Elasticsearch nodes with
'/usr/share/elasticsearch/bin/elasticsearch-create-enrollment-token -s node'.

### NOT starting on installation, please execute the following statements to configure elasticsearch service to start automatically using systemd
sudo systemctl daemon-reload
sudo systemctl enable elasticsearch.service
### You can start elasticsearch service by executing
sudo systemctl start elasticsearch.service
Processing triggers for procps (2:4.0.4-9+b2)...
Processing triggers for systemd (260.1-1)...
Scanning processes...
Scanning linux images...
```

Configure Elasticsearch:

`sudo nano /etc/elasticsearch/elasticsearch.yml`

Add:

`cluster.name: hive`

`xpack.security.enabled: true`

```
GNU nano 8.7.1 /etc/elasticsearch/e
#
# Cluster
# Use a descriptive name for your cluster:
#
#cluster.name: hive
#
# Node
# Use a descriptive name for the node:
#
#node.name: node-1
# Add custom attributes to the node:
#
#node.attr.rack: r1
#
# Paths
# Path to directory where to store the data (separate multiple locations by comma):
#
#path.data: /var/lib/elasticsearch
# Path to log files:
#
#path.logs: /var/log/elasticsearch
#
# Memory
# Lock the memory on startup:
#
#bootstrap.memory_lock: true
#
# Make sure that the heap size is set to about half the memory available
# on the system and that the owner of the process is allowed to use this
# limit.
#
# Elasticsearch performs poorly when the system is swapping the memory.
#
# Network
# By default Elasticsearch is only accessible on localhost. Set a different
# address here to expose this node on the network:
```

Set memory

`sudo nano /etc/elasticsearch/jvm.options.d/jvm.options`

Set:

`-Xms4g`

`-Xmx4g`

```
kali@kali: ~
Session Actions Edit View Help
GNU nano 8.7.1 /etc/elasticsearch/jvm.options.d/jvm.options *
-Xms4g
-Xmx4g
```

Start Elasticsearch:

```
sudo systemctl enable elasticsearch
sudo systemctl start elasticsearch
sudo systemctl status elasticsearch
```

```

(kali@kali)~$ sudo systemctl enable elasticsearch
(kali@kali)~$ sudo systemctl start elasticsearch
(kali@kali)~$ sudo systemctl status elasticsearch
Created symlink '/etc/systemd/system/multi-user.target.wants/elasticsearch.service' → '/usr/lib/systemd/system/elasticsearch.service'.
elasticsearch.service - Elasticsearch
   Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-04-30 15:13:35 EDT; 1s ago
     Invocation: 4a17e8ae512f4728b5f8ea4a7fc07861
       Docs: https://www.elastic.co
    Main PID: 23147 (java)
      Tasks: 109 (limit: 9271)
    Memory: 1.4G (peak: 1.5G, swap: 7.6M, swap peak: 7.6M)
       CPU: 3min 20.530s
    CGroup: /system.slice/elasticsearch.service
            └─23147 /usr/share/elasticsearch/jdk/bin/java -Xms4m -Xmx64m -XX:+UseSerialGC -Dcli.name=server -Dcli.script=/usr/share/elasticsearch/bin/elasticsearch
            └─23260 /usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.cache.ttl=60 -Des.networkaddress.cache.negative.ttl=10 -XX:+AlwaysPreTouch -Xss1m -D
            └─23864 /usr/share/elasticsearch/modules/x-pack-ml/platform/linux-x86_64/bin/controller

pr 30 15:04:04 kali systemd[1]: Starting elasticsearch.service - Elasticsearch ...
pr 30 15:13:35 kali systemd[1]: Started elasticsearch.service - Elasticsearch.
lines 1-16/16 (END)
```

Download the TheHive Debian package (version 5.7.1) from the official repository for installation on the system.

wget -O thehive.deb https://thehive.download.strangebee.com/5.7/deb/thehive_5.7.1-1_all.deb

```

(kali@kali)~$ wget -O thehive.deb https://thehive.download.strangebee.com/5.7/deb/thehive_5.7.1-1_all.deb
--2026-04-30 14:08:18-- https://thehive.download.strangebee.com/5.7/deb/thehive_5.7.1-1_all.deb
Resolving thehive.download.strangebee.com (thehive.download.strangebee.com)... 52.222.136.10, 52.222.136.100, 52.222.136.34, ...
Connecting to thehive.download.strangebee.com (thehive.download.strangebee.com)|52.222.136.10|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 342968976 (327M) [application/vnd.debian.binary-package]
Saving to: 'thehive.deb'

thehive.deb
100%[=====] 327.08M 4.65MB/s in 72s

2026-04-30 14:09:33 (4.57 MB/s) - 'thehive.deb' saved [342968976/342968976]
```

Install the downloaded TheHive Debian package on the system using apt.

sudo apt install ./thehive.deb -y

```

(kali@kali)-[~]
└─$ sudo apt install ./thehive.deb -y
Note, selecting 'thehive' instead of './thehive.deb'
Installing:
  thehive

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 129
  Download size: 0 B / 343 MB
  Space needed: 425 MB / 47.0 GB available

Get:1 /home/kali/thehive.deb thehive all 5.7.1-1 [343 MB]
Selecting previously unselected package thehive.
(Reading database... 443066 files and directories currently installed.)
Preparing to unpack /home/kali/thehive.deb...
Unpacking thehive (5.7.1-1)...
Setting up thehive (5.7.1-1)...
Creating system group: thehive
Creating system user: thehive in thehive with thehive daemon-user and shell /bin/false
1+0 records in
1+0 records out
1024 bytes (1.0 kB, 1.0 KiB) copied, 4.613e-05 s, 22.2 MB/s
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

```

Configure TheHive

Edit config:

```
sudo nano /etc/thehive/application.conf
```

Change base URL:

```
application.baseUrl = "http://localhost:9000"
```

```
play.http.context = "/"
```

```

GNU nano 8.7.1 /etc/thehive/application.conf
db.janusgraph {
  storage {
    backend = cql
    hostname = ["127.0.0.1"]
    # Cassandra authentication (if configured)
    # username = "thehive"
    # password = "password"
    cql {
      cluster-name = thp
      keyspace = thehive
    }
  }
}
index.search {
  backend = elasticsearch
  hostname = ["127.0.0.1"]
  index-name = thehive
}
}

# Attachment storage configuration
# By default, TheHive is configured to store files locally in the folder.
# The path can be updated and should belong to the user/group running thehive service. (by default: thehive:thehive)
storage {
  provider = localfs
  localfs.location = /opt/thp/thehive/files
}

# Define the maximum size for an attachment accepted by TheHive
play.http.parser.maxDiskBuffer = 1GB
# Define maximum size of http request (except attachment)
play.http.parser.maxMemoryBuffer = 10M

# Service configuration
application.baseUrl = "http://localhost:9000"
play.http.context = "/"

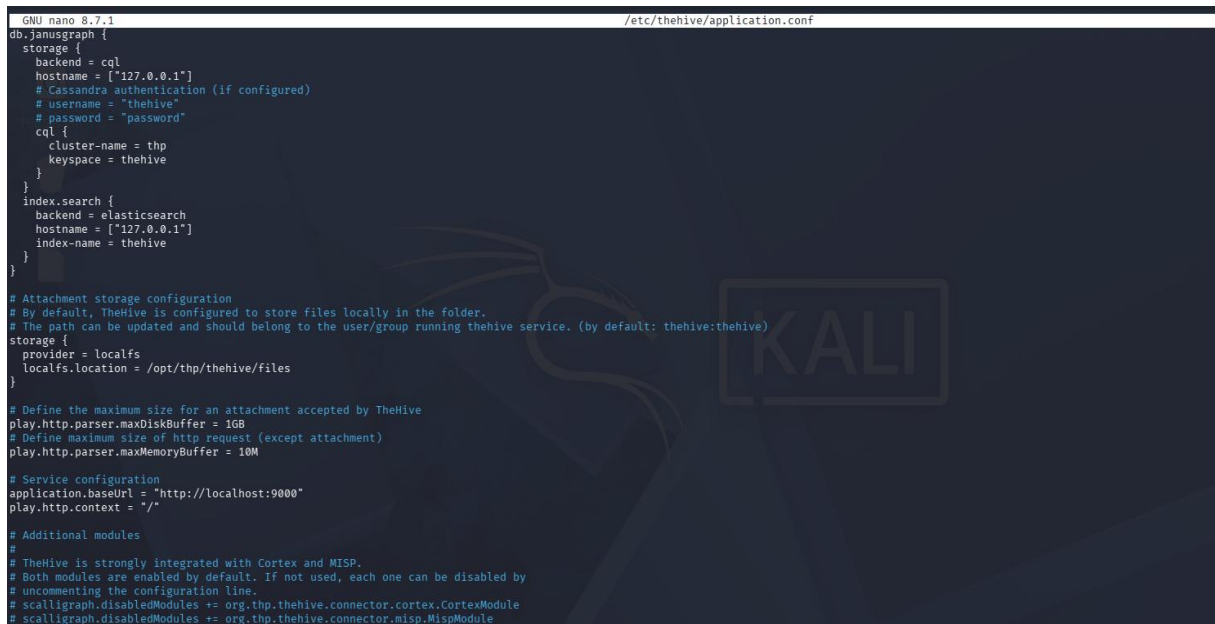
# Additional modules
#
# TheHive is strongly integrated with Cortex and MISP.
# Both modules are enabled by default. If not used, each one can be disabled by
# uncommenting the configuration line.
# scalligraph.disabledModules += org.thp.thehive.connector.cortex.CortexModule
# scalligraph.disabledModules += org.thp.thehive.connector.misp.MispModule

```

Add database config:

```
db.janusgraph {
  storage {
    backend = cql
    hostname = ["127.0.0.1"]
    cql {
      keyspace = thehive
    }
  }

  index.search {
    backend = elasticsearch
    hostname = ["127.0.0.1"]
    index-name = thehive
  }
}
```



```
GNU nano 8.7.1 /etc/thehive/application.conf
db.janusgraph {
  storage {
    backend = cql
    hostname = ["127.0.0.1"]
    # Cassandra authentication (if configured)
    # username = "thehive"
    # password = "password"
    cql {
      cluster-name = thp
      keyspace = thehive
    }
  }
  index.search {
    backend = elasticsearch
    hostname = ["127.0.0.1"]
    index-name = thehive
  }
}

# Attachment storage configuration
# By default, TheHive is configured to store files locally in the folder.
# The path can be updated and should belong to the user/group running thehive service. (by default: thehive:thehive)
storage {
  provider = localfs
  localfs.location = /opt/thp/thehive/files
}

# Define the maximum size for an attachment accepted by TheHive
play.http.parser.maxDiskBuffer = 1GB
# Define maximum size of http request (except attachment)
play.http.parser.maxMemoryBuffer = 10M

# Service configuration
application.baseUrl = "http://localhost:9000"
play.http.context = "/"

# Additional modules
#
# TheHive is strongly integrated with Cortex and MISP.
# Both modules are enabled by default. If not used, each one can be disabled by
# uncommenting the configuration line.
# scalligraph.disabledModules += org.thp.thehive.connector.cortex.CortexModule
# scalligraph.disabledModules += org.thp.thehive.connector.misp.MispModule
```

Create file storage

```
sudo mkdir -p /opt/thp/thehive/files
sudo chown -R thehive:thehive /opt/thp/thehive/files
sudo chmod 700 /opt/thp/thehive/files
```

```
(kali@kali)-[~]
$ sudo mkdir -p /opt/thp/thehive/files
$ sudo chown -R thehive:thehive /opt/thp/thehive/files
$ sudo chmod 700 /opt/thp/thehive/files
```

```
(kali@kali)-[~]
$
```

Start TheHive

`sudo systemctl enable thehive`

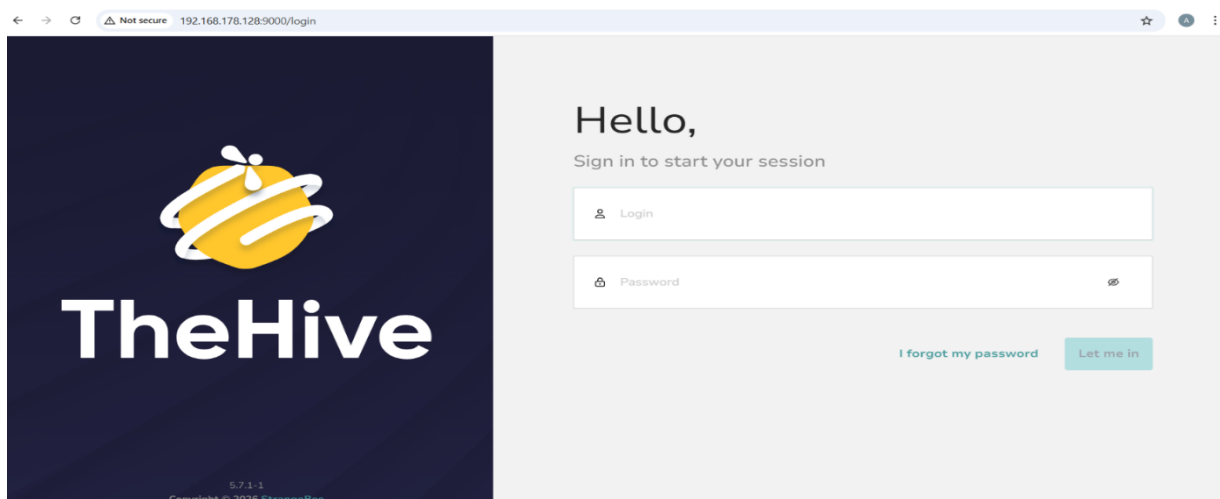
`sudo systemctl start thehive`

`sudo systemctl status thehive`

```
(kali@kali)-[~]
$ sudo systemctl enable thehive
$ sudo systemctl start thehive
$ sudo systemctl status thehive
Created symlink '/etc/systemd/system/multi-user.target.wants/thehive.service' → '/usr/lib/systemd/system/thehive.service'.
● thehive.service - Scalable, Open Source and Free Security Incident Response Solutions
   Loaded: loaded (/usr/lib/systemd/system/thehive.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-04-30 14:22:56 EDT; 31ms ago
 Invocation: 13061b721ae446cab8a3a5dfdaa637ef
    Docs: https://strangebee.com
   Main PID: 72578 ((thehive))
     Tasks: 1 (limit: 9271)
    Memory: 2.3M (peak: 2.3M)
       CPU: 7ms
    CGroup: /system.slice/thehive.service
            └─72578 "(thehive)"

Apr 30 14:22:56 kali systemd[1]: Started thehive.service - Scalable, Open Source and Free Security Incident Response Solutions.
```

Write `http://<ip>:9000` in browser to access



The default administrative credentials (username: admin@thehive.local) were used to log in to TheHive for initial configuration and access.

Username = admin@thehive.local

Password = secret

This instance uses a Platinum License for Trial purpose, and will expire in 15 days. [Register now.](#)

1

Users

?

D

+

Export list

Default

+ Add filter

		Name	Login	Type	Organisations	MFA	Created by	Dates	C.	U.	
☐	Active	D	Default admin user admin@thehive.local	Normal	A		T TheHive system user	C. 02/05/2026 00:01			

Cases can be created and managed in TheHive to investigate and track security incidents efficiently.

Adding a Case Template

Prefix
Case template title prefix...

* Name
Case template name...

Display name
Case template display name...

TLP

PAP

Severity

Tags
Tags

* Description
[Rich text editor with toolbar and preview]

Tasks
Custom fields

[Cancel](#) [Confirm case template creation](#)

Cortex Insatllation

Install Dependencies

Cortex requires **Java + ElasticSearch**.

1. Install Java

sudo apt update

sudo apt install openjdk-11-jdk -y

```
(kali@kali)-[~]
└─$ sudo apt update
sudo apt install openjdk-11-jdk -y
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.1 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.4 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [118 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [262 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [189 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [908 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [15.3 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [38.7 kB]
Fetched 76.0 MB in 4min 47s (265 kB/s)
1973 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  openjdk-11-jdk
Installing dependencies:
  openjdk-11-jdk-headless  openjdk-11-jre  openjdk-11-jre-headless
Suggested packages:
  openjdk-11-demo  openjdk-11-source  visualvm  fonts-ipafont-gothic  fonts-ipafont-mincho  fonts-wqy-microhei | fonts-wqy-zenhei  fonts-indic
Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 1973
  Download size: 117 MB
  Space needed: 264 MB / 63.1 GB available
Get:1 http://http.kali.org/kali kali-rolling/main amd64 openjdk-11-jre-headless amd64 11.0.25-5ea-1 [38.4 MB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 openjdk-11-jre amd64 11.0.25-5ea-1 [195 kB]
Get:3 http://http.kali.org/kali kali-rolling/main amd64 openjdk-11-jdk-headless amd64 11.0.25-5ea-1 [73.9 MB]
Get:4 http://http.kali.org/kali kali-rolling/main amd64 openjdk-11-jdk amd64 11.0.25-5ea-1 [4,485 kB]
Fetched 117 MB in 4min 44s (412 kB/s)
Selecting previously unselected package openjdk-11-jre-headless:amd64.
(Reading database ... 422308 files and directories currently installed.)
Preparing to unpack .../openjdk-11-jre-headless_11.0.25-5ea-1_amd64.deb ...
Unpacking openjdk-11-jre-headless:amd64 (11.0.25-5ea-1) ...
Selecting previously unselected package openjdk-11-jre:amd64.
Preparing to unpack .../openjdk-11-jre_11.0.25-5ea-1_amd64.deb ...
Unpacking openjdk-11-jre:amd64 (11.0.25-5ea-1) ...
Selecting previously unselected package openjdk-11-jdk-headless:amd64.
Preparing to unpack .../openjdk-11-jdk-headless_11.0.25-5ea-1_amd64.deb ...
Unpacking openjdk-11-jdk-headless:amd64 (11.0.25-5ea-1) ...
Selecting previously unselected package openjdk-11-jdk:amd64.
Preparing to unpack .../openjdk-11-jdk_11.0.25-5ea-1_amd64.deb ...
Unpacking openjdk-11-jdk:amd64 (11.0.25-5ea-1) ...
Setting up openjdk-11-jre-headless:amd64 (11.0.25-5ea-1) ...
```

java -version

```
(kali@kali)-[~]
└─$ java -version
openjdk version "21.0.9" 2025-10-21
OpenJDK Runtime Environment (build 21.0.9+10-Debian-1)
OpenJDK 64-Bit Server VM (build 21.0.9+10-Debian-1, mixed mode, sharing)

(kali@kali)-[~]
└─$
```

Install Elasticsearch

sudo apt update

sudo apt install elasticsearch -y

```
Session Actions Edit View Help
Upgrading:
  elasticsearch
Summary:
  Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 129
  1 not fully installed or removed.
  Download size: 679 MB
  Space needed: 1,308 MB / 50.6 GB available

Continues? [Y/n] y
Get:1 https://artifacts.elastic.co/packages/8.x/apt stable/main amd64 elasticsearch amd64 8.19.15 [679 MB]
Fetched 679 MB in 18min 46s (603 kB/s)
(Reading database.. 443053 files and directories currently installed.)
Preparing to unpack ./elasticsearch-8.19.15_amd64.deb...
Unpacking elasticsearch (8.19.15) over (8.19.15)...
Setting up elasticsearch (8.19.15)...

----- Security autoconfiguration information -----

Authentication and authorization are enabled.
TLS for the transport and HTTP layers is enabled and configured.

The generated password for the elastic built-in superuser is : PzslKHAKx_WaNjbc008E

If this node should join an existing cluster, you can reconfigure this with
'/usr/share/elasticsearch/bin/elasticsearch-reconfigure-node --enrollment-token <token-here>'
after creating an enrollment token on your existing cluster.

You can complete the following actions at any time:

Reset the password of the elastic built-in superuser with
'/usr/share/elasticsearch/bin/elasticsearch-reset-password -u elastic'.

Generate an enrollment token for Kibana instances with
'/usr/share/elasticsearch/bin/elasticsearch-create-enrollment-token -s kibana'.

Generate an enrollment token for Elasticsearch nodes with
'/usr/share/elasticsearch/bin/elasticsearch-create-enrollment-token -s node'.

### NOT starting on installation, please execute the following statements to configure elasticsearch service to start automatically using systemd
sudo systemctl daemon-reload
sudo systemctl enable elasticsearch.service
### You can start elasticsearch service by executing
sudo systemctl start elasticsearch.service
Processing triggers for procs (2:4.0.4-9+b2)...
Processing triggers for systemd (260.1-1)...
Scanning processes...
Scanning linux images...
```

Configure Elasticsearch:

sudo nano /etc/elasticsearch/elasticsearch.yml

Add:

cluster.name: hive

xpack.security.enabled: true

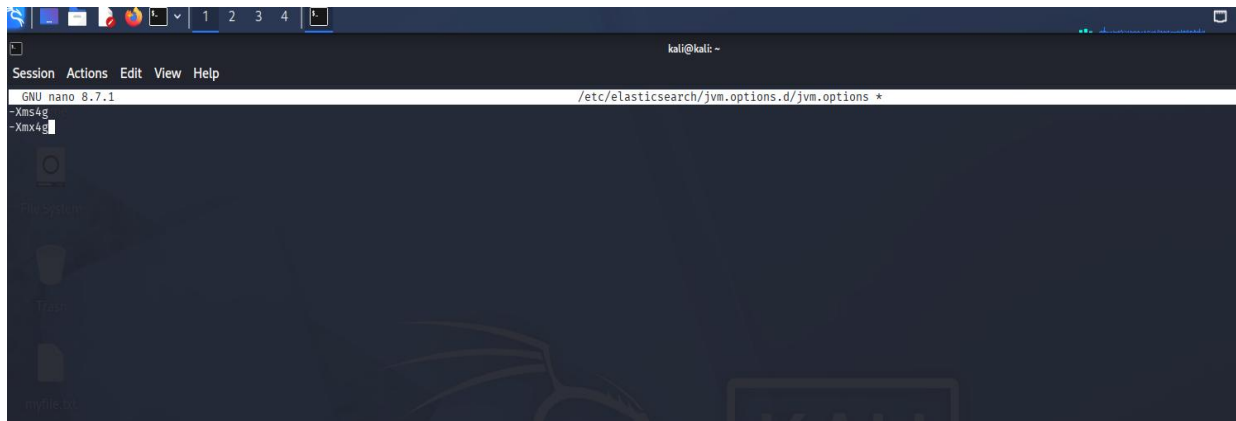
```
GNU nano 8.7.1 /etc/elasticsearch/e
# ----- Cluster -----
# Use a descriptive name for your cluster:
#cluster.name: hive
#
# ----- Node -----
# Use a descriptive name for the node:
#node.name: node-1
# Add custom attributes to the node:
#node.attr.rack: r1
#
# ----- Paths -----
# Path to directory where to store the data (separate multiple locations by comma):
#path.data: /var/lib/elasticsearch
# Path to log files:
#path.logs: /var/log/elasticsearch
#
# ----- Memory -----
# Lock the memory on startup:
#bootstrap.memory_lock: true
# Make sure that the heap size is set to about half the memory available
# on the system and that the owner of the process is allowed to use this
# limit.
# Elasticsearch performs poorly when the system is swapping the memory.
#
# ----- Network -----
# By default Elasticsearch is only accessible on localhost. Set a different
# address here to expose this node on the network:
```

Set memory

```
sudo nano /etc/elasticsearch/jvm.options.d/jvm.options
```

Set:

```
-Xms4g  
-Xmx4g
```



Start Elasticsearch:

```
sudo systemctl enable elasticsearch  
sudo systemctl start elasticsearch  
sudo systemctl status elasticsearch
```

```
—(kali@kali)~  
$ sudo systemctl enable elasticsearch  
$ sudo systemctl start elasticsearch  
$ sudo systemctl status elasticsearch  
Created symlink '/etc/systemd/system/multi-user.target.wants/elasticsearch.service' → '/usr/lib/systemd/system/elasticsearch.service'.  
elasticsearch.service - Elasticsearch  
   Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; enabled; preset: disabled)  
   Active: active (running) since Thu 2026-04-30 15:13:35 EDT; 1s ago  
 Invocation: 4a17e8ae512f4728b5f8ea4a7fc07861  
    Docs: https://www.elastic.co  
   Main PID: 23147 (java)  
     Tasks: 109 (limit: 9271)  
  Memory: 1.4G (peak: 1.5G, swap: 7.6M, swap peak: 7.6M)  
    CPU: 3min 20.530s  
   CGroup: /system.slice/elasticsearch.service  
           └─23147 /usr/share/elasticsearch/jdk/bin/java -Xms4m -Xmx64m -XX:+UseSerialGC -Dcli.name=server -Dcli.script=/usr/share/elasticsearch/bin/elasticsearch  
           └─23260 /usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.cache.ttl=60 -Des.networkaddress.cache.negative.ttl=10 -XX:+AlwaysPreTouch -Xss1m -D  
           └─23864 /usr/share/elasticsearch/modules/x-pack-ml/platform/linux-x86_64/bin/controller  
  
pr 30 15:04:04 kali systemd[1]: Starting elasticsearch.service - Elasticsearch ...  
pr 30 15:13:35 kali systemd[1]: Started elasticsearch.service - Elasticsearch.  
ines 1-16/16 (END)
```

Install Docker (if not installed)

```
sudo apt update
sudo apt install docker.io docker-compose -y
sudo systemctl enable docker
sudo systemctl start docker
```

```
(kali@kali)-[/opt]
└─$ sudo apt update
sudo apt install docker.io docker-compose -y
sudo systemctl enable docker
sudo systemctl start docker
Hit:1 http://http.kali.org/kali kali-rolling InRelease
1973 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  docker-compose docker.io

Installing dependencies:
  containerd criu docker-buildx docker-cli libcompel1 python3-protobuf python3-pycrui runc tini-static

Suggested packages:
  containernetworking-plugins docker-doc btrfs-progs debootstrap rinse rootlesskit xfsprogs zfs-fuse | zfsutils-linux

Recommended packages:
  needrestart

Summary:
  Upgrading: 0, Installing: 11, Removing: 0, Not Upgrading: 1973
  Download size: 90.0 MB
  Space needed: 393 MB / 62.9 GB available

Get:2 http://http.kali.org/kali kali-rolling/main amd64 containerd amd64 2.1.6+ds1-1 [27.8 MB]
Get:1 http://http.kali.org/kali kali-rolling/main amd64 runc amd64 1.3.5+ds1-1 [6,726 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 libcompel1 amd64 4.2-3 [63.1 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 criu amd64 4.2-3 [556 kB]
Get:5 http://http.kali.org/kali kali-rolling/main amd64 docker-buildx amd64 0.29.1+ds1-3 [14.1 MB]
Get:6 http://http.kali.org/kali kali-rolling/main amd64 docker-cli amd64 28.5.2+dfsg3-2 [7,084 kB]
Get:7 http://kali.download/kali kali-rolling/main amd64 docker-compose amd64 2.40.3-3 [13.9 MB]
Get:8 http://http.kali.org/kali kali-rolling/main amd64 tini-static amd64 0.19.0-6+b1 [281 kB]
Get:9 http://http.kali.org/kali kali-rolling/main amd64 docker.io amd64 28.5.2+dfsg3-2 [18.5 MB]
Get:10 http://http.kali.org/kali kali-rolling/main amd64 python3-protobuf amd64 3.21.12-15+b1 [156 kB]
Get:11 http://kali.download/kali kali-rolling/main amd64 python3-pycrui all 4.2-3 [43.2 kB]
Fetched 90.0 MB in 6min 12s (242 kB/s)
Selecting previously unselected package runc.
(Reading database ... 422815 files and directories currently installed.)
Preparing to unpack .../00-runc_1.3.5+ds1-1_amd64.deb ...
Unpacking runc (1.3.5+ds1-1) ...
Selecting previously unselected package containerd.
Preparing to unpack .../01-containerd_2.1.6+ds1-1_amd64.deb ...
Unpacking containerd (2.1.6+ds1-1) ...
Selecting previously unselected package libcompel1:amd64.
Preparing to unpack .../02-libcompel1_4.2-3_amd64.deb ...
Unpacking libcompel1:amd64 (4.2-3) ...
Selecting previously unselected package criu.
```

Create Cortex Docker Setup

```
mkdir/opt/cortex-docker
cd/opt/cortex-docker
nano docker-compose.yml
```

Paste this configuration

version: "3.8"

services:

cortex:

image: thehiveproject/cortex:3.1.8

container_name: cortex

ports:

- "9001:9001"

environment:

- job_directory=/tmp/cortex-jobs

depends_on:

- elasticsearch

elasticsearch:

image: docker.elastic.co/elasticsearch/elasticsearch:7.17.9

container_name: cortex-es

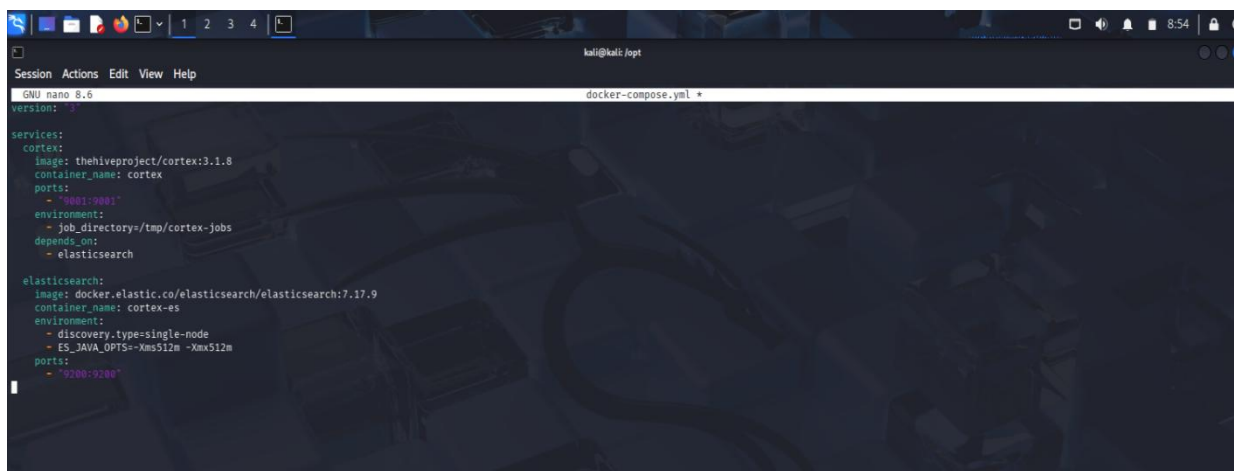
environment:

- discovery.type=single-node
- ES_JAVA_OPTS=-Xms512m -Xmx512m

ports:

- "9200:9200"

A Docker Compose configuration was created to deploy Cortex along with Elasticsearch, enabling automated analysis capabilities and data storage required for integration with TheHive.



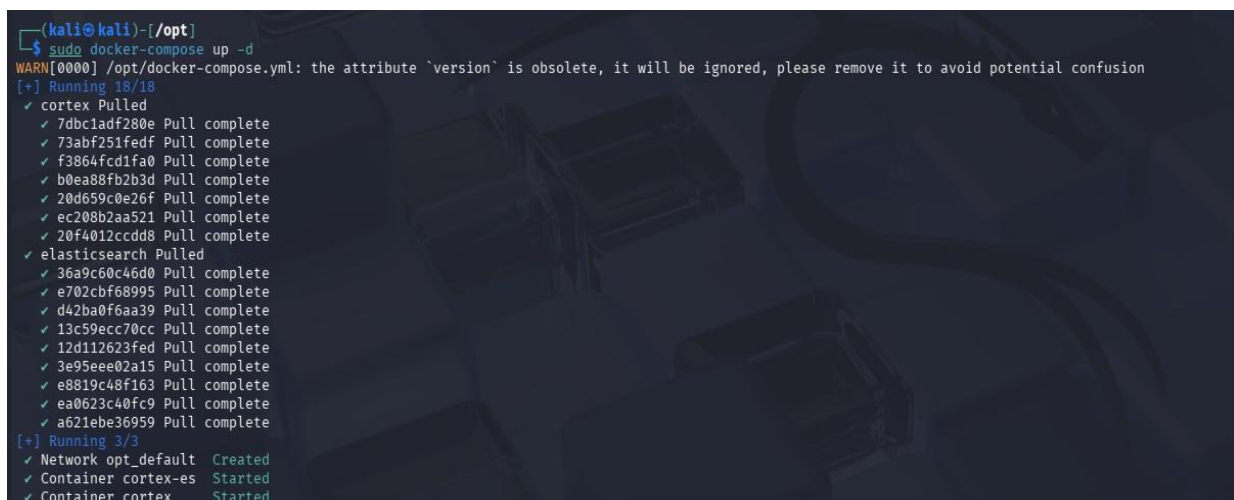
```
Session Actions Edit View Help
GNU nano 8.6 docker-compose.yml
version: 3

services:
  cortex:
    image: thehiveproject/cortex:3.1.8
    container_name: cortex
    ports:
      - 9082:9082
    environment:
      - job_directory=/tmp/cortex-jobs
    depends_on:
      - elasticsearch

  elasticsearch:
    image: docker.elastic.co/elasticsearch/elasticsearch:7.17.9
    container_name: cortex-es
    environment:
      - discovery.type=single-node
      - ES_JAVA_OPTS=-Xms512m -Xmx512m
    ports:
      - 9200:9200
```

sudo docker-compose up -d

The Cortex service was started using Docker Compose (docker-compose up -d), running the containers in detached mode for background execution



```
(kali@kali)-[/opt]
$ sudo docker-compose up -d
WARN[0000] /opt/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 18/18
✓ cortex Pulled
  7dbc1adf280e Pull complete
  73abf251fedf Pull complete
  f3864fcd1fa0 Pull complete
  b0ea88fb2b3d Pull complete
  20d659c0e26f Pull complete
  ec208b2aa521 Pull complete
  20f4012ccdd8 Pull complete
✓ elasticsearch Pulled
  36a9c60c46d0 Pull complete
  e702cbf68995 Pull complete
  d42ba0f6aa39 Pull complete
  13c59ecc70cc Pull complete
  12d112623fed Pull complete
  3e95ee02a15 Pull complete
  e8819c48f163 Pull complete
  ea0623c40fc9 Pull complete
  a621ebe36959 Pull complete
[+] Running 3/3
✓ Network opt_default Created
✓ Container cortex-es Started
✓ Container cortex Started
```

Configure Cortex

```
(kali@kali)~/Shuffle  
$ mkdir -p cortex  
  
(kali@kali)~/Shuffle  
$ nano cortex/application.conf
```

Now create config file:

The Cortex configuration file was edited to define the application secret key, enable local authentication, and specify paths for analyzers and responders in Cortex

nano cortex/application.conf

Paste this:

```
play.http.secret.key="cortexlabsecret"
```

```
auth {
```

```
  provider = local
```

```
}
```

```
analyzer {
```

```
  path = ["/opt/cortex/analyzers"]
```

```
}
```

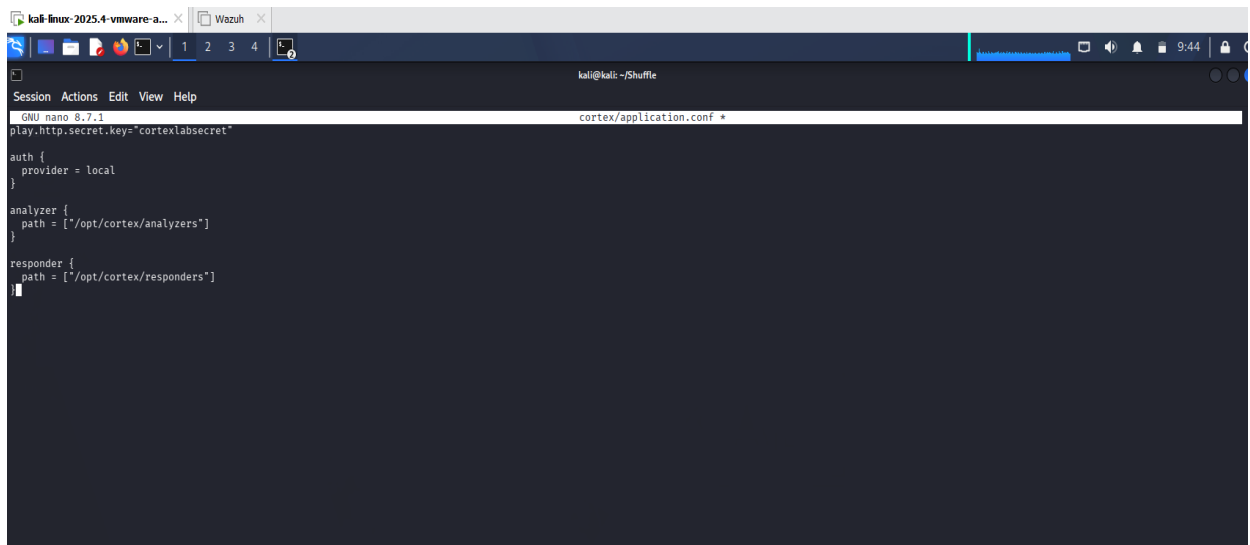
```
responder {
```

```
  path = ["/opt/cortex/responders"]
```

```
}
```

Save:

- CTRL + X
- Y
- Enter



```
kali@kali: ~/Shuffle
cortex/application.conf *
play.http.secret.key="cortexlabsecret"

auth {
  provider = local
}

analyzer {
  path = ["/opt/cortex/analyzers"]
}

responder {
  path = ["/opt/cortex/responders"]
}
```

Now open your compose file:

nano docker-compose.yml

Then Add:

version: "3.7"

services:

=====

THEHIVE

=====

thehive:

image: strangebee/thehive:5.2

container_name: thehive

restart: unless-stopped

depends_on:

cassandra:

condition: service_started

elasticsearch:

condition: service_started

minio:

condition: service_started

cortex.local:

condition: service_started

mem_limit: 2g

ports:

- "9000:9000"

environment:

- JVM_OPTS=-Xms1g -Xmx1g

command:

- --secret

- "lab123456789"

- "--cql-hostnames"

- "cassandra"

- "--index-backend"

- "elasticsearch"

- "--es-hostnames"

- "elasticsearch"
- "--s3-endpoint"
- "http://minio:9002"
- "--s3-access-key"
- "minioadmin"
- "--s3-secret-key"
- "minioadmin"
- "--s3-use-path-access-style"

volumes:

- thehivedata:/etc/thehive/application.conf

networks:

- SOC_NET

=====

CASSANDRA

=====

cassandra:

image: cassandra:4

container_name: cassandra

restart: unless-stopped

ports:

- "9042:9042"

environment:

- CASSANDRA_CLUSTER_NAME=TheHive

volumes:

- cassandradata:/var/lib/cassandra

networks:

- SOC_NET

=====

ELASTICSEARCH

=====

elasticsearch:

image: docker.elastic.co/elasticsearch/elasticsearch:7.17.9

container_name: elasticsearch

restart: unless-stopped

mem_limit: 1g

ports:

- "9200:9200"

environment:

- discovery.type=single-node
- xpack.security.enabled=false
- cluster.name=hive
- http.host=0.0.0.0
- ES_JAVA_OPTS=-Xms512m -Xmx512m

volumes:

- elasticsearchdata:/usr/share/elasticsearch/data

networks:

- SOC_NET

=====

MINIO (S3 STORAGE)

=====

minio:

image: quay.io/minio/minio

container_name: minio

restart: unless-stopped

command: server /data --console-address ":9002"

environment:

- MINIO_ROOT_USER=minioadmin
- MINIO_ROOT_PASSWORD=minioadmin

ports:

- "9002:9002"

volumes:

- miniodata:/data

networks:

- SOC_NET

=====

CORTEX

=====

cortex.local:

image: thehiveproject/cortex:3.1.7

container_name: cortex

restart: unless-stopped

environment:

- PLAY_HTTP_SECRET_KEY=cortexlabsecret
- JOB_DIRECTORY=/tmp/cortex-jobs
- JVM_OPTS=-Xms512m -Xmx512m

ports:

- "9001:9001"

volumes:

- /var/run/docker.sock:/var/run/docker.sock
- /tmp/cortex-jobs:/tmp/cortex-jobs
- ./cortex/application.conf:/etc/cortex/application.conf

depends_on:

- elasticsearch

networks:

- SOC_NET

=====

MISP

=====

misp.local:

image: coolacid/misp-docker:core-latest

container_name: misp

restart: unless-stopped

depends_on:

- misp_mysql

ports:

- "80:80"
- "443:443"

volumes:

- "./server-configs:/var/www/MISP/app/Config/"
- "./logs:/var/www/MISP/app/tmp/logs/"
- "./files:/var/www/MISP/app/files"
- "./ssl:/etc/nginx/certs"

environment:

- MYSQL_HOST=misp_mysql
- MYSQL_DATABASE=mispdb
- MYSQL_USER=mispuser
- MYSQL_PASSWORD=misppass
- MISP_ADMIN_EMAIL=mispadmin@lab.local
- MISP_ADMIN_PASSPHRASE=mispadminpass

- MISP_BASEURL=localhost
- TIMEZONE=Europe/London
- INIT=true
- CRON_USER_ID=1
- REDIS_FQDN=redis
- HOSTNAME=https://10.0.2.10

networks:

- SOC_NET

misp_mysql:

image: mysql/mysql-server:5.7

container_name: misp_mysql

restart: unless-stopped

environment:

- MYSQL_DATABASE=mispdb
- MYSQL_USER=mispuser
- MYSQL_PASSWORD=misppass
- MYSQL_ROOT_PASSWORD=mispass

volumes:

- mispsqldata:/var/lib/mysql

networks:

- SOC_NET

redis:

image: redis:latest

container_name: redis

networks:

- SOC_NET

misp-modules:

image: coolacid/misp-docker:modules-latest

container_name: misp-modules

restart: unless-stopped

environment:

- REDIS_BACKEND=redis

depends_on:

- redis
- misp_mysql

networks:

- SOC_NET

=====

VOLUMES

=====

volumes:

miniodata:

cassandradata:

elasticsearchdata:

thehivedata:

mispsqldata:

=====

NETWORK

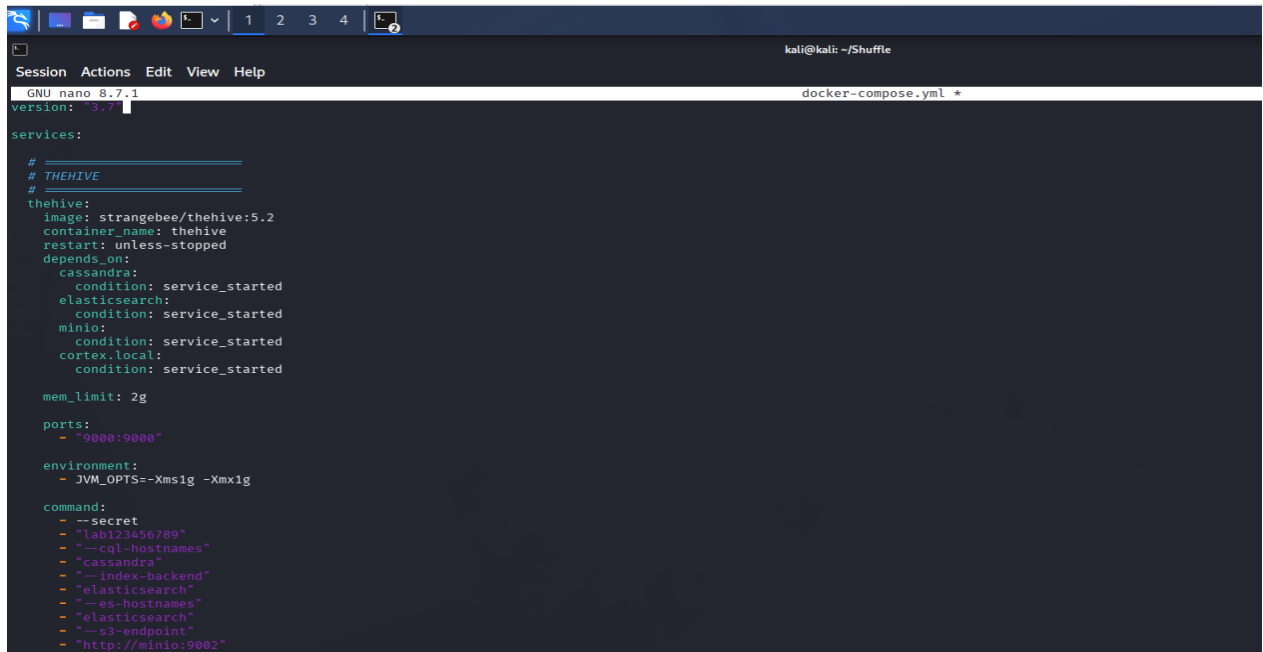
=====

networks:

SOC_NET:

driver: bridge

This Docker Compose configuration defines and deploys a complete SOC environment including TheHive, Cortex, MISP, Elasticsearch, and supporting services using containers.



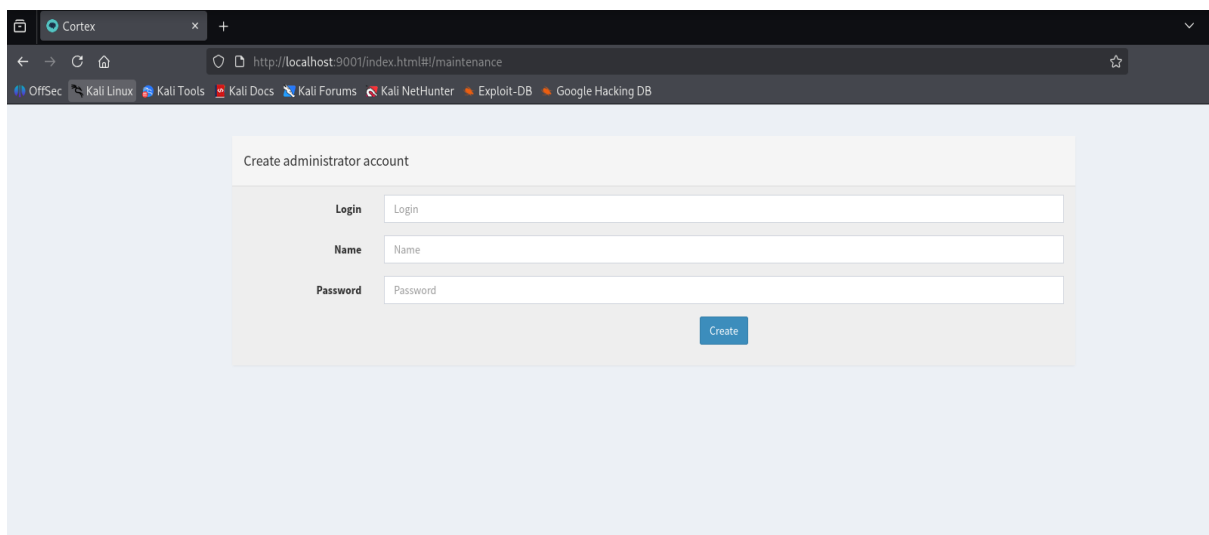
```
GNU nano 8.7.1 docker-compose.yml
version: '3.7'

services:
  #
  # THEHIVE
  #
  thehive:
    image: strangebee/thehive:5.2
    container_name: thehive
    restart: unless-stopped
    depends_on:
      cassandra:
        condition: service_started
      elasticsearch:
        condition: service_started
      minio:
        condition: service_started
      cortex.local:
        condition: service_started
    mem_limit: 2g
    ports:
      - "9000:9000"
    environment:
      - JVM_OPTS=-Xms1g -Xmx1g
    command:
      - --secret
      - "lab123456789"
      - "--cql-hostnames"
      - "cassandra"
      - "--index-backend"
      - "elasticsearch"
      - "--es-hostnames"
      - "elasticsearch"
      - "--s3-endpoint"
      - "http://minio:9002"
```

Access Cortex

Open browser:

<http://localhost:9001>



Cortex

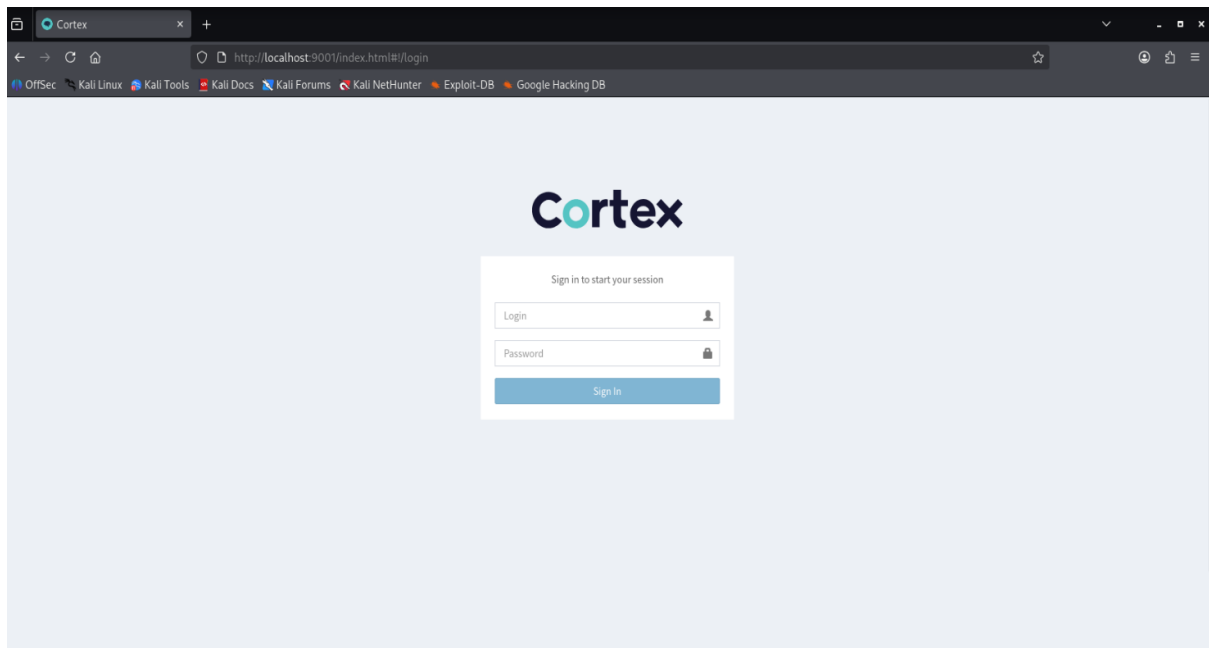
http://localhost:9001/index.html#/maintenance

Create administrator account

Login	Login
Name	Name
Password	Password

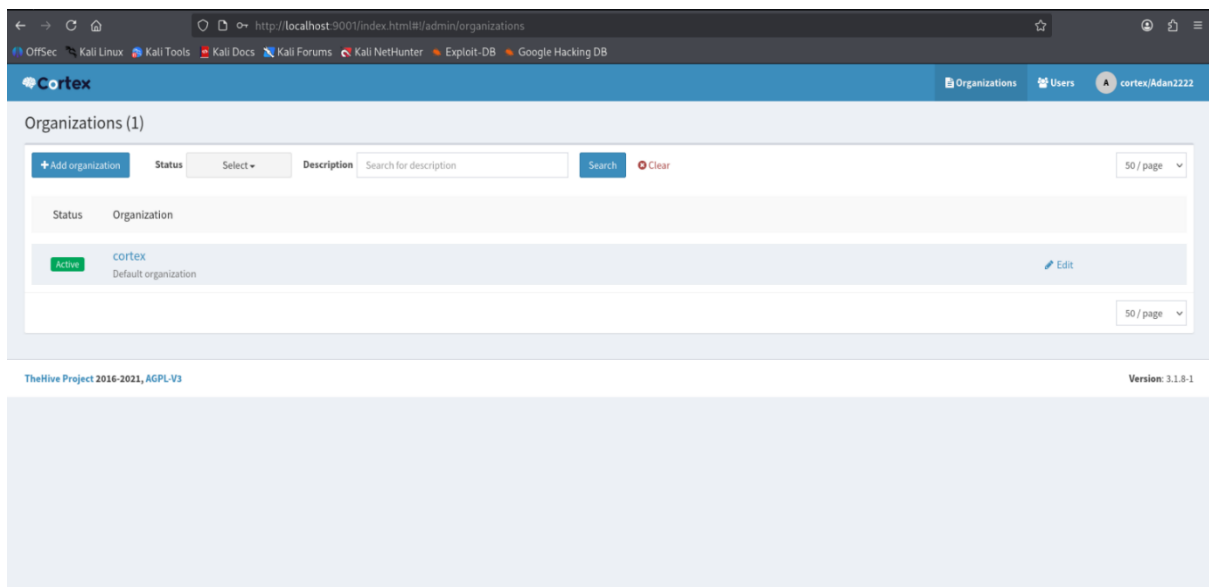
Create

Add credentials



Login:

- **user:** adan22
- **pass:** _____



Generate API Key

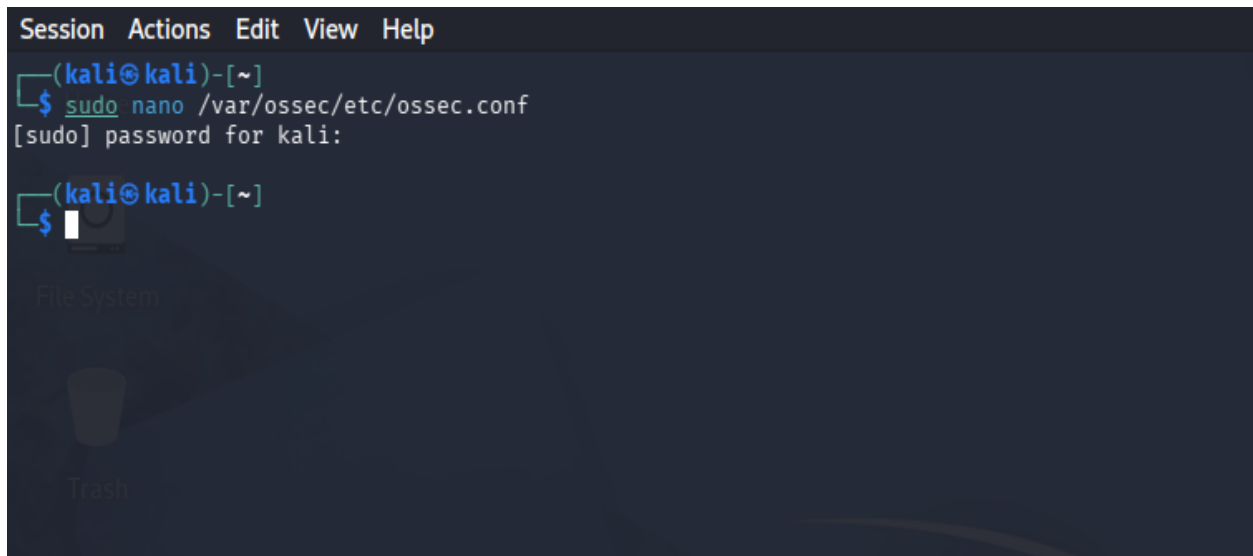
- Open Cortex dashboard
- Go to **Organization Settings**
- Generate API key

Connect Wazuh to Shuffle

On your Wazuh system:

Open config:

```
sudo nano /var/ossec/etc/ossec.conf
```

A terminal window with a dark background and light blue text. The window has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The prompt is '(kali@kali)-[~]'. The user enters the command '\$ sudo nano /var/ossec/etc/ossec.conf'. The terminal shows '[sudo] password for kali:' followed by a blank line. The prompt returns to '(kali@kali)-[~]'. The user enters a dollar sign '\$' followed by a cursor. In the background, there is a faint desktop environment with icons for 'File System' and 'Trash'.

```
<integration>
```

```
<name>custom-webhook</name>
```

```
<hook_url>http://192.168.1.20:3001/api/v1/hooks/wazuh</hook_url>
```

```
</integration>
```

A custom webhook integration was configured in Wazuh to forward alerts to an external service via HTTP, enabling automated incident response and integration with other security tools.

```
</localfile>

  <localfile>
    <log_format>json</log_format>
    <location>/var/log/suricata/eve.json</location>
  </localfile>

<integration>
  <name>custom-webhook</name>
  <hook_url>http://192.168.1.20:3001/api/v1/hooks/wazuh</hook_url>
</integration>

</ossec_config>
```

Restart Wazuh:

```
sudo systemctl restart wazuh-manager
```

[illegible]

Open and edit the main configuration file of Wazuh (ossec.conf) using the nano text editor for customizing rules, integrations, and system settings

```
sudo nano /var/ossec/etc/ossec.conf
```

```
Session Actions Edit View Help
(kali㉿kali)-[~]
$ sudo nano /var/ossec/etc/ossec.conf
[sudo] password for kali:

(kali㉿kali)-[~]
$
```

A custom webhook integration was added in Wazuh configuration to forward security alerts to an external endpoint (<http://192.168.1.20:3001/api/v1/hooks/wazuh>) for automated processing and incident response.

```
<integration>
```

```
<name>custom-webhook</name>
```

```
<hook_url>http://192.168.1.20:3001/api/v1/hooks/wazuh</hook_url>
```

```
</integration>
```

```
<location>/var/log/suricata/log <location>
</localfile>

<localfile>
<log_format>json</log_format>
<location>/var/log/suricata/eve.json</location>
</localfile>

<integration>
<name>custom-webhook</name>
<hook_url>http://192.168.1.20:3001/api/v1/hooks/wazuh</hook_url>
</integration>

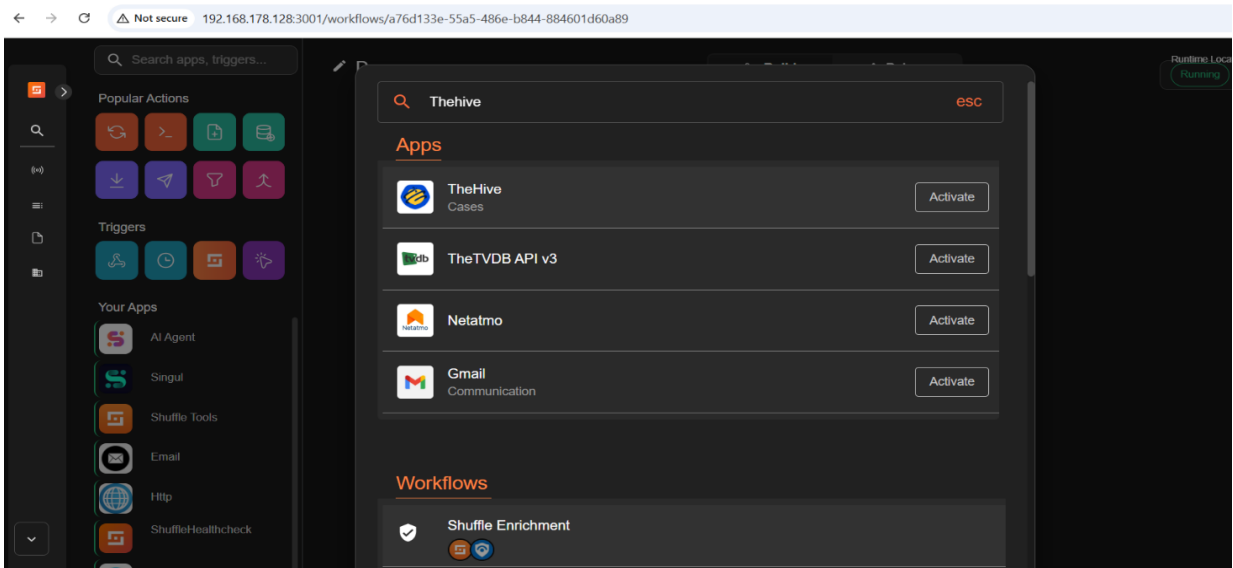
</ossec_config>
```

Restart Wazuh:

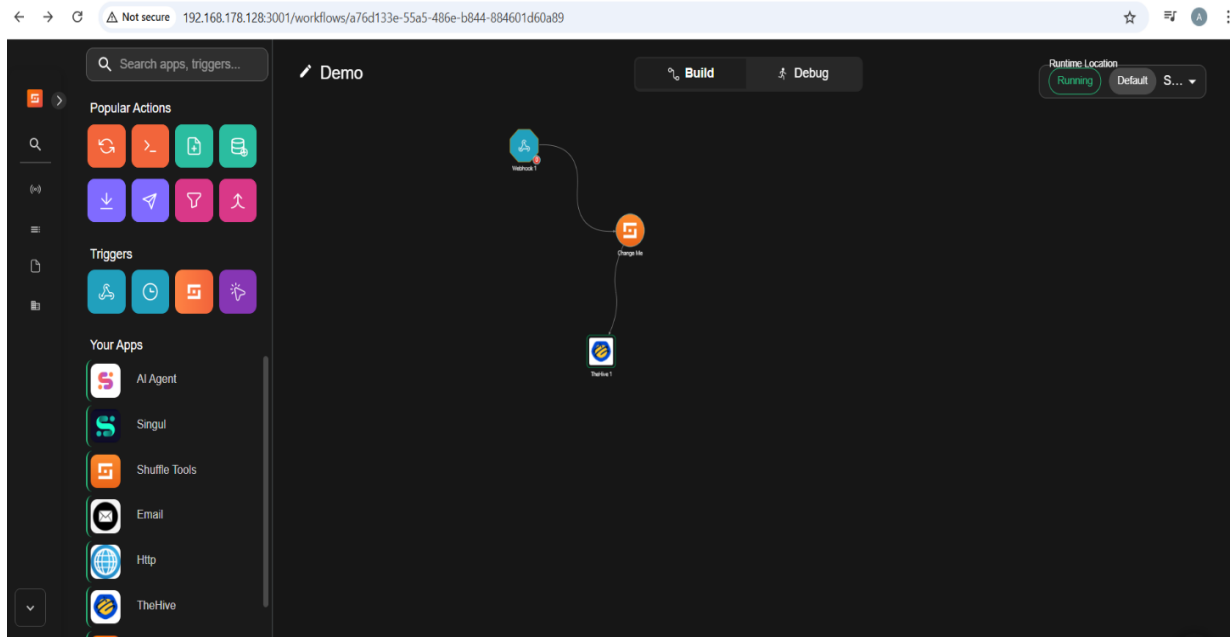
```
sudo systemctl restart wazuh-manager
```

[illegible]

The TheHive service was activated to enable incident management and case handling functionalities before integrating it with other security tools



Add Action → TheHive (Create Case)



Create Workflow in Shuffle

1. Open Shuffle & Create Workflow

Go to Shuffle

Click Workflows

Click New Workflow

Name it:

Wazuh SIEM Flow

2. Add Webhook (Receive Wazuh Alert)

Drag Webhook node

Rename: Receive Wazuh Alert

Copy the Webhook URL

Configure in Wazuh

Go to Wazuh → Integrations

Add webhook:

JSON

```
{  
  "url": "YOUR_SHUFFLE_WEBHOOK_URL",  
  "level": 10  
}
```

Now Wazuh will send alerts to Shuffle

3. Add “Change Me” (Data Processing)

Add Shuffle Tools → “Change Me”

Connect: Webhook → Change Me

Configure:

Extract fields from JSON:

JSON

```
{  
  "source_ip": "$.data.srcip",  
  "rule": "$.rule.description",  
  "level": "$.rule.level"  
}
```

4. Add “Create Alert” (TheHive)

Drag TheHive → Create Alert

Connect: Change Me → Create Alert

Configure:

Title: Wazuh Alert: Port Scan

Description: Use data from Change Me

Severity: {{level}}

Now alert is stored in TheHive

5. Add “Add Observables”

Drag TheHive → Add Observable

Connect: Create Alert → Add Observable

Configure:

Data Type: ip

Value: {{source_ip}}

👉 This adds IP for analysis

6. Add Cortex Analysis

Drag Cortex → Analyze

Connect: Add Observable → Cortex

Configure:

Analyzer: VirusTotal / AbuseIPDB

Observable: {{source_ip}}

Checks if IP is malicious

7. Add “Get Analysis”

Drag Cortex → Get Analysis

Connect: Cortex → Get Analysis

Fetch results from analyzers

8. Add “Create Case” (TheHive)

Drag TheHive → Create Case

Connect: Get Analysis → Create Case

Configure:

Title: Investigation: Port Scan

Description: include analysis results

Full incident case created

9. Add Gmail

 Gmail Setup

Drag Gmail node

Add credentials (OAuth or App Password)

 Gmail_1 (Initial Alert)

Connect: Change Me → Gmail_1

Subject:

Wazuh Alert: Port Scan Detected

Source IP: {{source_ip}}

Rule: {{rule}}

Severity: {{level}}

 Gmail_2 (After Observables)

Connect: Add Observable → Gmail_2

Sends enriched alert

📧 Gmail_3 (Final Report)

Connect: Create Case → Gmail_3

Sends final investigation result

10. Test Workflow

Click Run

Trigger test alert from Wazuh:



Receive Alert (Webhook – Wazuh)

First node: “Receive Wazuh Alert” (webhook)

Wazuh sends alert → Shuffle receives it via webhook

Example alert:

Port scan from 192.168.178.128

Change Me (Data Processing Node)

This node formats or extracts useful data from alert

Example:

Extract IP

Extract alert type (port scan)

Clean JSON data

Basically: prepare data for next steps

Create Alert (TheHive / Alert System)

Creates a security alert in incident platform

Stores:

Source IP

Severity

Description

Now alert is officially recorded

Add Observables

Adds important data (IOCs):

IP: 192.168.178.128

These observables are used for threat analysis

SEVERITY	STATUS	TITLE	# CASE	SOURCE	TYPE	DATE	AGE	OWNER
Medium	New	Port scan (rule id: 86601) detected. Rule: 86601 Source IP: 192.168.52.152 Destination IP: 192.168.52.10 Destination Port: 51516	-	wazuh_alert	86601	02/05/2026 19:12	a few seconds	Adan22
Medium	New	Port scan (rule id: 86601) detected. Rule: 86601 Source IP: 192.168.52.152 Destination IP: 192.168.52.10 Destination Port: 51523	-	wazuh_alert	86601	02/05/2026 19:12	a few seconds	Adan22
Medium	New	Port scan (rule id: 86601) detected. Rule: 86601 Source IP: 192.168.52.152 Destination IP: 192.168.52.10 Destination Port: 51527	-	wazuh_alert	86601	02/05/2026 19:12	a few seconds	Adan22
Medium	New	Port scan (rule id: 86601) detected. Rule: 86601 Source IP: 192.168.52.152 Destination IP: 192.168.52.10 Destination Port: 51533	-	wazuh_alert	86601	02/05/2026 19:12	a few seconds	Adan22
Medium	New	Port scan (rule id: 86601) detected. Rule: 86601 Source IP: 192.168.52.152 Destination IP: 192.168.52.10 Destination Port: 51539	-	wazuh_alert	86601	02/05/2026 19:12	a few seconds	Adan22
Medium	New	Port scan (rule id: 86601) detected. Rule: 86601 Source IP: 192.168.52.152 Destination IP: 192.168.52.10 Destination Port: 51546	-	wazuh_alert	86601	02/05/2026 19:12	a few seconds	Adan22
Medium	New	Port scan (rule id: 86601) detected. Rule: 86601 Source IP: 192.168.52.152 Destination IP: 192.168.52.10 Destination Port: 51552	-	wazuh_alert	86601	02/05/2026 19:12	a few seconds	Adan22
Medium	New	Port scan (rule id: 86601) detected. Rule: 86601 Source IP: 10.0.0.5 Destination IP: 192.168.52.10 Destination Port: 51516	-	wazuh_alert	86601	02/05/2026 19:12	a few seconds	Adan22

0 - 30 of 129 Next Show 30

Cortex Analysis

Sends observables to analyzers (Cortex)

Example checks:

IP reputation

Threat intelligence lookup

Finds if IP is malicious or known attacker

6. Get Analysis

Retrieves results from Cortex

Example result:

IP flagged as suspicious

Known scanning activity

Cortex

+ New Analysis

Jobs History

Analyzers

Responders

Organization

AD Adan22

Jobs History (17)

Data Types (14)

Select ▾

Job Type (2)

Select ▾

Analyzers (7)

Select ▾

Observable

Search for observable data

Search

Clear

Pagination

50 / page ▾

Status	Job details	TLP	PAP
Success	[ip] 192.168.52.1 Analyzer: PortScan_86601	Date: 2 May 2026 19:12	User: Adan22
		TLP:GREEN	PAP:GREEN
		View	Delete
Success	[domain] example.com Analyzer: PortScan_86601	Date: 2 May 2026 19:12	User: Adan22
		TLP:GREEN	PAP:GREEN
		View	Delete
Success	[ip] 10.0.0.5 Analyzer: PortScan_86601	Date: 2 May 2026 19:12	User: Adan22
		TLP:AMBER	PAP:AMBER
		View	Delete
Success	[domain] test.local Analyzer: PortScan_86601	Date: 2 May 2026 19:12	User: Adan22
		TLP:WHITE	PAP:WHITE
		View	Delete
Success	[ip] 172.16.1.11 Analyzer: PortScan_86601	Date: 2 May 2026 19:12	User: Adan22
		TLP:WHITE	PAP:WHITE
		View	Delete
Success	[domain] internal.company.local Analyzer: PortScan_86601	Date: 2 May 2026 19:12	User: Adan22
		TLP:WHITE	PAP:WHITE
		View	Delete
Success	[ip] 8.8.8.8 Analyzer: PortScan_86601	Date: 2 May 2026 19:12	User: Adan22
		TLP:WHITE	PAP:WHITE
		View	Delete
Success	[ip] 192.168.1.100 Analyzer: PortScan_86601	Date: 2 May 2026 19:12	User: Adan22
		TLP:GREEN	PAP:GREEN
		View	Delete

0 - 8 of 17

Next

Show

50 ▾

Create Case

Creates a full incident case (TheHive)

Includes:

Alert details

Analysis results

This is where investigation happens

8. Send Email (Gmail instead of Teams)

now:

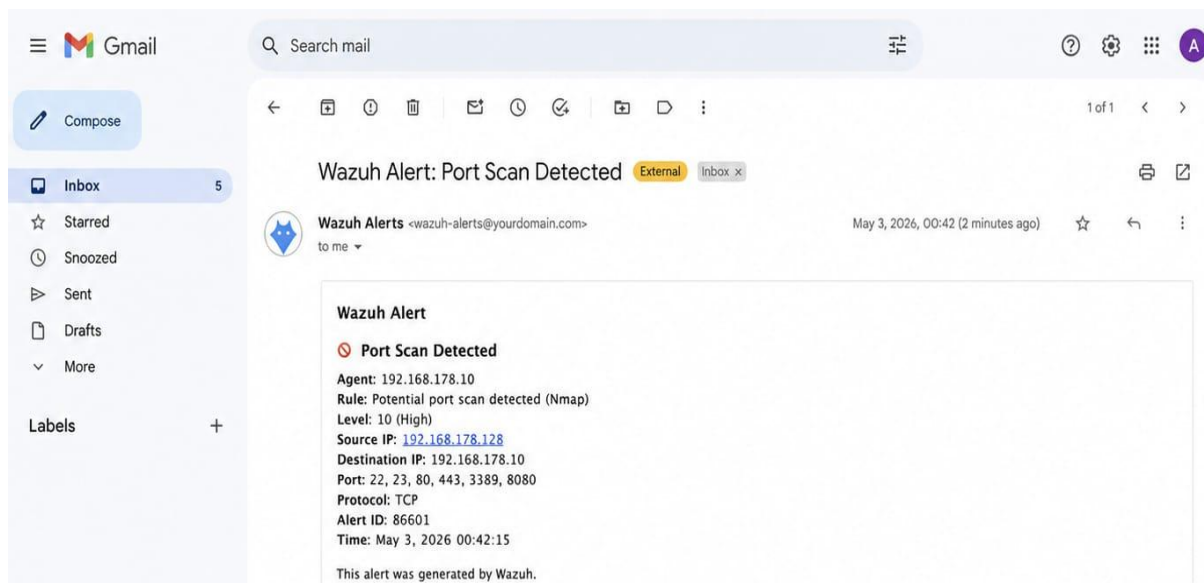
Gmail_1 → Gmail_2 → Gmail_3

 Emails are sent at different stages:

Initial alert

After analysis

Final case update



Conclusion:

This project successfully demonstrates the implementation of a complete network security monitoring and incident response system using Suricata, Wazuh, TheHive, Cortex, and Shuffle. The system is capable of capturing and analyzing network traffic in real time, detecting malicious activities, and generating alerts for further investigation.

Suricata functions as the primary intrusion detection tool by monitoring network packets and generating logs in eve.json format. Wazuh processes these logs to provide centralized monitoring, visualization, and alert management through its dashboard. The deployment of Wazuh agents ensures continuous data collection from monitored systems.

The integration of TheHive enhances incident response by allowing structured case management, while Cortex provides automated threat intelligence and analysis of observables. Shuffle adds automation to the workflow by connecting all tools and enabling a streamlined security response pipeline.

Overall, this integrated environment simulates a real-world Security Operations Center (SOC), improving threat detection efficiency, reducing manual workload, and enabling faster incident response.